Digital Communication Systems

Solutions - Section A (Multiplexing & Line Coding)

Question 1(a)

What is line coding? Why is it important to minimize DC components in line coding, and which coding schemes help achieve this? (10)

Solution

1. What is Line Coding?

Line coding is the process of converting digital data (a sequence of discrete bits, 0s and 1s) into a digital signal (a sequence of voltage or current pulses) that can be transmitted over a baseband channel (like a wire or cable). It represents the digital data in a format optimized for physical transmission.

2. Why is it important to minimize DC components?

- **AC Coupling / Transformers:** Many communication links (like telephone lines) use transformers or capacitors for isolation and AC coupling. These components block DC signals (0 Hz). If a line code has a significant DC component, that part of the signal gets blocked, causing the received signal to be distorted (a phenomenon known as baseline wandering).
- **Power Efficiency:** A DC component means extra power is being transmitted that carries no useful information, which wastes energy.
- **Interference:** DC bias can interfere with the proper operation of certain electronic amplifiers and repeaters in the transmission path.

3. Which coding schemes help achieve this?

Schemes that maintain a balanced number of positive and negative voltage levels over time effectively eliminate the DC component.

- **Manchester Encoding:** Every bit has a transition in the middle (High-to-Low or Low-to-High). This guarantees exactly 50% positive and 50% negative voltage duration, perfectly eliminating the DC component.
- **Differential Manchester:** Also has a mid-bit transition for every bit, yielding zero DC component.
- **Bipolar AMI (Alternate Mark Inversion):** Binary '0' is 0V, but binary '1's alternate between $+V$ and $-V$. Over time, the positive and negative pulses cancel out, resulting in zero DC bias.
- **Block Coding (e.g., 8B/10B):** Maps 8-bit blocks to 10-bit blocks designed specifically to balance the number of 1s and 0s.

Question 1(b)

In a joint TDM and FDM multiplexing system, 20 sources are multiplexed in each TDM block. The bandwidth of each source is 4 kHz. The Nyquist sampling rate is used... [Details omitted for brevity]. Determine: (i) the data rate of each TDM output and (ii) the bandwidth of the

FDM output signal. (15)

Solution

Given Parameters:

- Number of sources per TDM block = 20
- Bandwidth of each source = 4 kHz
- Sampling rate = Nyquist rate
- Encoding = 8 bits per sample
- Interleaving = 1 sample (8 bits) per source per frame
- Synchronization = 1 bit per frame
- FDM = Multiplexes 4 TDM blocks using ASK modulation
- Guard band = 2 kHz

(i) Determine the data rate of each TDM output:

1. **Sampling Rate (f_s):** According to Nyquist, $f_s = 2 \times \text{Bandwidth} = 2 \times 4 \text{ kHz} = 8 \text{ kHz}$ (or 8000 samples/sec).
2. **Frame Size:** Each frame contains 1 sample (8 bits) from each of the 20 sources, plus 1 framing/sync bit.
$$\text{Frame Size} = (20 \text{ sources} \times 8 \text{ bits/source}) + 1 \text{ sync bit} = 160 + 1 = \mathbf{161} \text{ bits/frame}$$
3. **Frame Rate:** Since we take 1 sample per source per frame, the frame rate equals the sampling rate.
$$\text{Frame Rate} = f_s = \mathbf{8000} \text{ frames/sec}$$
4. **TDM Data Rate (R_{TDM}):**
$$R_{\text{TDM}} = \text{Frame Size} \times \text{Frame Rate} = 161 \text{ bits/frame} \times 8000 \text{ frames/sec}$$
$$R_{\text{TDM}} = 1,288,000 \text{ bps} = \mathbf{1.288} \text{ Mbps}$$

(ii) Determine the bandwidth of the FDM output signal:

1. **Bandwidth of each ASK-modulated TDM signal (B_{ASK}):**
For ASK (Amplitude Shift Keying), the transmission bandwidth is equal to the bit rate of the modulating signal (assuming standard first-null bandwidth for a baseband NRZ signal shifted to a carrier frequency).
$$B_{ASK} = R_{\text{TDM}} = \mathbf{1.288} \text{ MHz} = \mathbf{1288} \text{ kHz}$$
2. **Total FDM Bandwidth (B_{FDM}):**
The FDM system multiplexes 4 such channels. To prevent interference, guard bands are placed between the channels. For 4 channels, there are exactly $(4 - 1) = 3$ guard bands between them.
$$\begin{aligned} B_{FDM} &= (4 \times B_{ASK}) + (3 \times B_{guard}) \\ B_{FDM} &= (4 \times 1288 \text{ kHz}) + (3 \times 2 \text{ kHz}) \\ B_{FDM} &= 5152 \text{ kHz} + 6 \text{ kHz} \\ B_{FDM} &= \mathbf{5158} \text{ kHz} = \mathbf{5.158} \text{ MHz} \end{aligned}$$

Answers: (i) 1.288 Mbps (ii) 5.158 MHz

Question 1(c)

The waveform shown in Fig. 1(c) represents the Multiline Transmission: MLT-3 encoding of a data stream. Decode the data stream. (10)

Solution

Decoding the MLT-3 Waveform

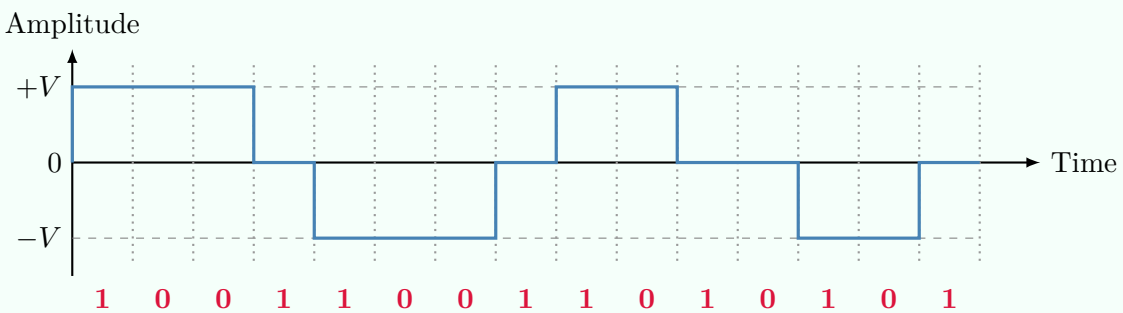
Rules of MLT-3 (Multiline Transmission, 3-level): MLT-3 uses three voltage levels: $+V, 0,$ and $-V$. The decoding rules are:

- **If the signal level does not change** from one bit period to the next, the bit is a **'0'**.
- **If the signal level changes** (transitions), the bit is a **'1'**.

The transitions for '1' follow a specific cycle: $+V \to 0 \to -V \to 0 \to +V \ldots$
By examining the given waveform at each dashed vertical interval (bit boundary):

- The signal starts at level 0 before the first bit.
- **Bit 1:** Transitions from 0 to $+V \implies$ **1**
- **Bit 2:** Stays at $+V \implies$ **0**
- **Bit 3:** Stays at $+V \implies$ **0**
- **Bit 4:** Transitions from $+V$ to $0 \implies$ **1**
- **Bit 5:** Transitions from 0 to $-V \implies$ **1**
- **Bit 6:** Stays at $-V \implies$ **0**
- **Bit 7:** Stays at $-V \implies$ **0**
- **Bit 8:** Transitions from $-V$ to $0 \implies$ **1**
- **Bit 9:** Transitions from 0 to $+V \implies$ **1**
- **Bit 10:** Stays at $+V \implies$ **0**
- **Bit 11:** Transitions from $+V$ to $0 \implies$ **1**
- **Bit 12:** Stays at $0 \implies$ **0**
- **Bit 13:** Transitions from 0 to $-V \implies$ **1**
- **Bit 14:** Stays at $-V \implies$ **0**
- **Bit 15:** Transitions from $-V$ to $0 \implies$ **1**

Visual Breakdown:



Final Decoded Data Stream:
The decoded binary sequence is: **1 0 0 1 1 0 0 1 1 0 1 0 1 0 1**

Communication Systems Engineering

Solutions - Thermal Noise, Probability & Networking

Question 4(a)

What is thermal noise in a communication system? Derive the equation of equivalent thermal noise voltage and current with necessary sketches. (10)

Solution

1. What is Thermal Noise?

Thermal noise (also known as Johnson-Nyquist noise) is the unavoidable electrical noise generated by the random thermal motion of electrons (charge carriers) inside an electrical conductor, regardless of any applied voltage. It occurs in all resistive electronic components and increases with temperature. Because its power spectral density is nearly flat across the frequency spectrum, it is typically modeled as White Noise.

2. Derivation of Equivalent Thermal Noise Voltage and Current:

From physics principles, the power spectral density (PSD) of thermal noise voltage across a resistor R is constant and given by:

$$S_v(f) = 2kTR \quad \text{(in Volts}^2\text{/Hz)}$$

Where:

- k = Boltzmann's constant (1.38×10^{-23} J/K)
- T = Absolute temperature in Kelvin (K)
- R = Resistance in Ohms (Ω)

Because this is a double-sided PSD, the total mean square noise voltage $\overline{v_n^2}$ measured over a specific system bandwidth B (in Hz) is calculated by integrating over positive frequencies (multiplying by 2):

$$\overline{v_n^2} = 2 \times S_v(f) \times B = 2 \times (2kTR) \times B = 4kTRB$$

Therefore, the **root-mean-square (RMS) equivalent thermal noise voltage (V_n)** is:

$$\mathbf{V_n = \sqrt{4kTRB}}$$

To find the **equivalent thermal noise current (I_n)**, we apply Norton's theorem (Ohm's law, $I = V/R$):

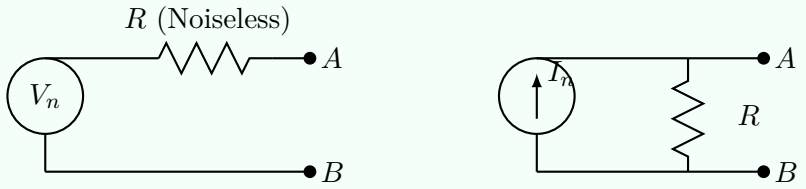
$$I_n = \frac{V_n}{R} = \frac{\sqrt{4kTRB}}{R} = \sqrt{\frac{4kTRB}{R^2}}$$

$$\mathbf{I_n = \sqrt{\frac{4kTB}{R}}}$$

3. Necessary Sketches (Equivalent Circuits):

A noisy resistor can be modeled as a noiseless resistor in series with a voltage source (Thévenin) or in parallel with a current source (Norton).

(a) Thévenin Equivalent (Voltage) **(b) Norton Equivalent (Current)**



Question 4(b)

An information packet contains 200 bits. This packet is transmitted over a communications channel where the probability of error for each bit is 10^{-3}. Suppose the packet includes an error-correcting code that can correct up to three errors located anywhere in the packet. What is the probability that a particular packet is received in error in this case? (10)

Solution

Given Parameters:

- Total number of bits in a packet, $n = 200$
- Probability of a single bit error, $p = 10^{-3} = 0.001$
- Error correction capability, $k = 3$ errors.

Since the system can successfully correct up to 3 errors, a packet will be received in error **only if it contains more than 3 errors** (i.e., 4, 5, ..., 200 errors).
Let X be the random variable representing the number of bit errors in a packet. This follows a Binomial Distribution:

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

The probability of a packet error (P_{error}) is the complement of the probability that the packet is received successfully (having 0, 1, 2, or 3 errors):

$$P_{\text{error}} = 1 - P(X \leq 3) = 1 - \left[P(X=0) + P(X=1) + P(X=2) + P(X=3)\right]$$

Now, we calculate the probability of each successful case:

$$P(X=0) = \binom{200}{0}(0.001)^0(0.999)^{200} \approx 1 \times 1 \times 0.8186488 \approx 0.8186488$$

$$P(X=1) = \binom{200}{1}(0.001)^1(0.999)^{199} \approx 200 \times 0.001 \times 0.8194683 \approx 0.1638936$$

$$P(X=2) = \binom{200}{2}(0.001)^2(0.999)^{198} = \frac{200 \times 199}{2} \times 10^{-6} \times 0.820288 \approx 0.0163237$$

$$P(X=3) = \binom{200}{3}(0.001)^3(0.999)^{197} = \frac{200 \times 199 \times 198}{6} \times 10^{-9} \times 0.821110 \approx 0.0010785$$

Summing these probabilities gives the total probability of successful reception:

$$P(X \leq 3) \approx 0.8186488 + 0.1638936 + 0.0163237 + 0.0010785 = 0.9999446$$

Finally, the probability that the packet is received in error is:

$$P_{\text{error}} = 1 - 0.9999446 = \mathbf{0.0000554}$$

Answer: The probability that a particular packet is received in error is approximately 5.54×10^{-5}.

Question 4(c)

A network system consists of four computers, two routers, and five LANs... [Topology description omitted for brevity]. You have to send a data packet from computer 2 to computer 4. Draw the network connection diagram, showing the data packet with physical and logical addresses at each hop. (15)

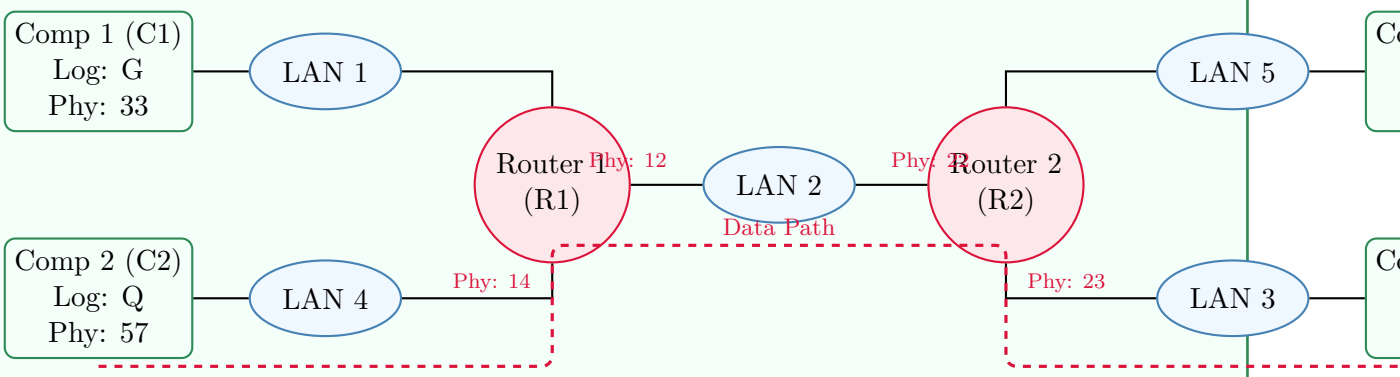
Solution

1. Network Connection Diagram:

Based on the given description:

- **Router 1 (R1)** connects to LAN 1, LAN 2, and LAN 4.
- **Router 2 (R2)** connects to LAN 2, LAN 3, and LAN 5.
- **Comp 1 (C1):** LAN 1 (Phy: 33, Log: G). **Comp 2 (C2):** LAN 4 (Phy: 57, Log: Q).
- **Comp 3 (C3):** LAN 5 (Phy: 65, Log: L). **Comp 4 (C4):** LAN 3 (Phy: 80, Log: W).

To send data from **Computer 2 to Computer 4**, the path is:
C2 → LAN 4 → R1 → LAN 2 → R2 → LAN 3 → C4.
Assumption: Let the physical (MAC) addresses of the router interfaces be as follows: R1 on LAN 4 = 14; R1 on LAN 2 = 12; R2 on LAN 2 = 22; R2 on LAN 3 = 23.



2. Data Packet at Each Hop:

Note: The logical addresses (IP) from source to destination remain unchanged across the entire journey. Only the physical addresses (MAC) change at each hop.

- **Hop 1: Computer 2 to Router 1 (across LAN 4)**

Src Phy (MAC)	Dest Phy (MAC)	Src Log (IP)	Dest Log (IP)	Data
57 (C2)	**14** (R1 Interface)	**Q** (C2)	**W** (C4)	[Payload]

- **Hop 2: Router 1 to Router 2 (across LAN 2)**

Src Phy (MAC)	Dest Phy (MAC)	Src Log (IP)	Dest Log (IP)	Data
12 (R1 Interface)	**22** (R2 Interface)	**Q** (C2)	**W** (C4)	[Payload]

- **Hop 3: Router 2 to Computer 4 (across LAN 3)**

Src Phy (MAC)	Dest Phy (MAC)	Src Log (IP)	Dest Log (IP)	Data
23 (R2 Interface)	**80** (C4)	**Q** (C2)	**W** (C4)	[Payload]

Communication Systems Engineering

Exam Solutions (Radar, Satellite & Networking)

Question 3(a)

Briefly discuss the different types of radar systems. Additionally, explain how radar is utilized in Air Traffic Control systems. (13)

Solution

1. Different Types of Radar Systems:

- **Continuous Wave (CW) Radar:** Transmits a continuous high-frequency signal. It relies on the Doppler effect to detect moving targets and their velocities but cannot easily determine the target's range (distance).
- **Pulsed Radar:** Transmits short, high-power pulses of radio waves and listens for the echoes during the rest period. It is excellent for determining the exact range of a target by measuring the time delay of the echo.
- **Doppler Radar:** Specifically designed to measure the velocity of moving objects by analyzing the frequency shift (Doppler shift) of the reflected signal. Commonly used in weather forecasting and speed enforcement.
- **Phased Array Radar:** Uses an array of many small antennas. Instead of mechanically rotating the dish, it steers the radar beam electronically by shifting the phase of the signals. It can track multiple targets simultaneously at incredible speeds.

2. Utilization in Air Traffic Control (ATC) Systems:
Radar is the backbone of modern ATC systems, ensuring the safe and orderly flow of air traffic. It is primarily utilized in two forms:

- **Primary Surveillance Radar (PSR):** This is a traditional pulsed radar. It bounces radio waves off the aircraft's body and measures the reflected echo to determine the aircraft's exact distance (range) and direction (bearing). It works even if the aircraft's internal systems are completely turned off.
- **Secondary Surveillance Radar (SSR):** Instead of just bouncing signals, the SSR transmits an interrogation signal. A device on the aircraft (called a *Transponder*) receives this signal and actively transmits a reply back to the ground. This reply contains crucial data such as the aircraft's identity (squawk code) and its precise altitude (barometric flight level).

By combining PSR and SSR, air traffic controllers get a complete 3D picture of the airspace, including who the aircraft is, where it is, and how high it is flying.

Question 3(b)

Calculate the maximum range of a radar system with an operating frequency of 12 GHz. The radar transmits a peak power of 300 kW, and the receiving antenna has an effective aperture of 5 m^2. The target has a radar cross section area (RCS) of 25 m^2, and the minimum detectable signal power is 1 nW. (10)

Solution

Given Parameters:

- Operating frequency, $f = 12$ GHz $= 12 \times 10^9$ Hz
- Peak transmitted power, $P_t = 300$ kW $= 3 \times 10^5$ W
- Effective aperture, $A_e = 5$ m^2
- Radar Cross Section (RCS), $\sigma = 25$ m^2
- Minimum detectable signal, $S_{min} = 1$ nW $= 1 \times 10^{-9}$ W
- Speed of light, $c = 3 \times 10^8$ m/s

Step 1: Calculate the Wavelength (λ)

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{12 \times 10^9} = \frac{1}{40} = 0.025 \text{ m}$$

Step 2: Radar Range Equation

Since we are given the effective aperture A_e instead of the antenna gain G_t, we can substitute the relationship $G_t = \frac{4\pi A_e}{\lambda^2}$ into the standard radar equation:

$$R_{max}^4 = \frac{P_t G_t \sigma A_e}{(4\pi)^2 S_{min}} = \frac{P_t \left(\frac{4\pi A_e}{\lambda^2}\right) \sigma A_e}{(4\pi)^2 S_{min}} = \frac{P_t A_e^2 \sigma}{4\pi \lambda^2 S_{min}}$$

Step 3: Calculation

$$\begin{aligned} R_{max}^4 &= \frac{(3 \times 10^5) \times (5)^2 \times 25}{4\pi \times (0.025)^2 \times 10^{-9}} \\ &= \frac{(3 \times 10^5) \times 25 \times 25}{4\pi \times 0.000625 \times 10^{-9}} \\ &= \frac{187500 \times 10^3}{2.5\pi \times 10^{-12}} \\ &= \frac{1.875 \times 10^8}{2.5\pi \times 10^{-12}} = \frac{1.875}{2.5\pi} \times 10^{20} \\ &= \frac{0.75}{\pi} \times 10^{20} \approx 0.23873 \times 10^{20} \text{ m}^4 \end{aligned}$$

Taking the fourth root of both sides:

$$\begin{aligned} R_{max} &= (0.23873)^{1/4} \times (10^{20})^{1/4} \\ &\approx 0.69897 \times 10^5 \text{ m} \\ &\approx 69,897 \text{ meters} = \mathbf{69.897} \text{ km} \end{aligned}$$

Answer: The maximum range of the radar system is approximately **69.9 km**.

Question 3(c)

Explain the function of a satellite transponder using a detailed diagram. (12)

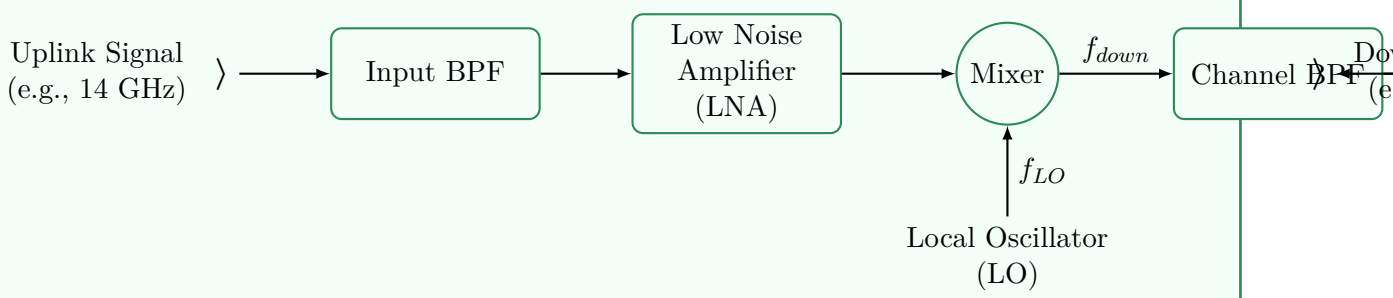
Solution

Function of a Satellite Transponder:

A transponder (Transmitter-Responder) is the primary payload of a communications satellite. Its core functions are to receive the weak uplink signal from Earth, clean it, change its frequency, amplify it heavily, and transmit it back to Earth as a downlink signal.

1. **Low Noise Amplification (LNA):** The signal arriving from Earth is extremely weak. The LNA amplifies this signal while introducing minimal electronic noise.
2. **Frequency Translation (Mixer):** The transponder uses a mixer and a Local Oscillator (LO) to step down the high uplink frequency (e.g., 14 GHz) to a lower downlink frequency (e.g., 11 GHz). This frequency shift is crucial to prevent the satellite's own powerful transmitted signal from interfering with its sensitive receiver.
3. **Filtering (BPF):** Bandpass filters remove any unwanted noise or out-of-band signals generated during the mixing process, isolating the specific channel.
4. **High Power Amplification (HPA):** Finally, a High Power Amplifier (typically a Traveling Wave Tube Amplifier - TWTA) blasts the signal's power to a very high level so it can survive the long journey back to Earth.

Detailed Block Diagram of a Satellite Transponder:



Question 4(a)

Explain the need for satellite launching methods. Describe any one method of satellite launching. (12)

Solution

1. Need for Satellite Launching Methods:

Satellites cannot simply be shot straight up into space; they require precise trajectory and immense speed to stay in orbit without falling back to Earth due to gravity. Launching methods are needed to:

- Overcome Earth's gravitational pull and atmospheric drag.
- Impart the required *orbital velocity* (e.g., 7.8 km/s for LEO) to achieve a stable orbit.
- Place the satellite in specific target orbits (like Geostationary Orbit - GEO, 35,786 km above Earth) with minimal fuel consumption. Directly lifting a heavy satellite to GEO in one vertical burn would require an impossibly massive and expensive rocket. Multi-stage transfer methods save fuel and maximize the satellite's payload.

2. Method of Satellite Launching (Hohmann Transfer Orbit to GEO):
The most common and energy-efficient method to place a satellite into a Geostationary Orbit is via a **Geostationary Transfer Orbit (GTO)**.

- **Stage 1 (Parking Orbit):** The launch vehicle lifts off and places the satellite into a low, circular orbit around the Earth called the Low Earth Orbit (LEO) or "Parking Orbit" (around 200-500 km altitude).
- **Stage 2 (Transfer Orbit):** While in LEO, at a specific point (typically over the equator), the rocket's upper stage engine fires. This provides a massive burst of speed, converting the circular LEO into a highly stretched elliptical orbit called the GTO. The closest point of this ellipse (Perigee) remains at LEO altitude, while the farthest point (Apogee) reaches out to the target GEO altitude of 35,786 km.
- **Stage 3 (Circularization into GEO):** The satellite coasts along the ellipse until it reaches the Apogee. At this exact moment, an onboard rocket engine (Apogee Kick Motor - AKM) fires. This burn circularizes the orbit and corrects any inclination, permanently parking the satellite in a circular geostationary orbit.

Question 4(b)

What is meant by a protocol in networking? Describe in detail the function of each layer in the OSI model. (13)

Solution

1. What is meant by a Protocol?
In networking, a **protocol** is a standard set of rules and conventions that govern how data is formatted, transmitted, and received across a network. It allows devices from different manufacturers to communicate seamlessly. Protocols define three key elements: *Syntax* (data format/structure), *Semantics* (meaning of control information), and *Timing* (speed of transmission).
2. Functions of each layer in the OSI Model:
The Open Systems Interconnection (OSI) model consists of 7 layers (from bottom to top):

- **Layer 1: Physical Layer:** Responsible for transmitting raw bit streams (0s and 1s) over a physical medium (cables, Wi-Fi). It defines hardware specifications, voltage levels, and pin layouts.
- **Layer 2: Data Link Layer:** Responsible for node-to-node delivery within the same local network. It packages data into *Frames*, adds physical (MAC) addresses, and handles error detection (using CRC) and flow control.
- **Layer 3: Network Layer:** Responsible for routing data across multiple distinct networks from the original source to the final destination. It packages data into *Packets*, adds logical (IP) addresses, and determines the best route.
- **Layer 4: Transport Layer:** Ensures reliable, end-to-end message delivery. It breaks large data into *Segments*, handles error recovery, flow control, and uses Port numbers to direct data to specific applications (e.g., TCP or UDP).
- **Layer 5: Session Layer:** Establishes, maintains, and terminates communication sessions (dialogues) between two computers. It manages synchronization and checkpointing.

- **Layer 6: Presentation Layer:** Acts as the network's translator. It formats, compresses, and encrypts/decrypts data so that the application layer can understand it (e.g., converting ASCII to EBCDIC, JPEG encoding).
- **Layer 7: Application Layer:** The topmost layer that directly interacts with the user's software applications. It provides high-level network services like web browsing (HTTP), email (SMTP), and file transfer (FTP).

Question 4(c)

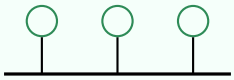
Define data communication. Describe various LAN topologies with the necessary diagram. (10)

Solution

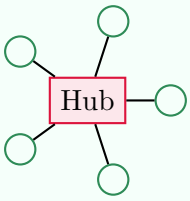
1. Definition of Data Communication:
Data communication is the exchange of data (in the form of 1s and 0s) between two or more devices via some form of transmission medium (such as a wire cable or wireless signal). For data communication to occur, the communicating devices must be part of a communication system made up of hardware and software, adhering to common protocols.

2. Various LAN Topologies:
Topology refers to the physical or logical geometric arrangement of devices (nodes) on a network.

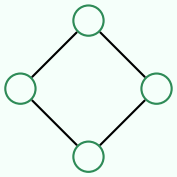
(a) Bus Topology:
All devices are connected to a single central cable, called the backbone. Data is broadcasted to all nodes, but only the intended recipient accepts it. It requires terminators at both ends to prevent signal reflection. *Advantage:* Cheap and easy to install.



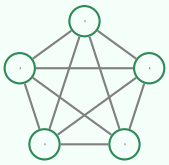
(b) Star Topology:
Every device has a dedicated point-to-point connection to a central controller, usually a Hub or a Switch. Devices communicate only through this central node. *Advantage:* Failure of one cable only affects one device.



(c) Ring Topology:
Each device is connected to exactly two other devices, forming a closed continuous loop. Data travels in one direction, from node to node, until it reaches its destination (often using token passing). *Disadvantage:* If one node fails, the entire ring breaks.



(d) Mesh Topology:
Every device is connected to every other device via dedicated links. For N devices, $N(N-1)/2$ links are required. *Advantage:* Extremely robust and secure; no single point of failure. *Disadvantage:* Expensive and difficult to wire.



Communication Systems Engineering

Solutions - Radar & Satellite Earth Station

Question (b)

Determine the maximum range of a radar system, provided that the peak transmitted power is 250 kW, minimum detectable signal power is 10 pW, the radar cross section (RCS) of the target is 20 m^2, the gain of the transmitting antenna is 4000, and the effective aperture of the receiver antenna is 4.5 m^2. (10)

Solution

Given Parameters:

- Peak transmitted power, $P_t = 250 \text{ kW} = 250 \times 10^3 \text{ W} = 2.5 \times 10^5 \text{ W}$
- Minimum detectable signal power, $S_{min} = 10 \text{ pW} = 10 \times 10^{-12} \text{ W} = 10^{-11} \text{ W}$
- Radar Cross Section (RCS), $\sigma = 20 \text{ m}^2$
- Gain of transmitting antenna, $G_t = 4000$
- Effective aperture of receiver antenna, $A_e = 4.5 \text{ m}^2$

Formula:

The maximum range of a radar system (R_{max}) is given by the Free-Space Radar Range Equation:

$$R_{max} = \left[\frac{P_t G_t \sigma A_e}{(4\pi)^2 S_{min}}\right]^{1/4}$$

Calculation:

Step 1: Calculate the numerator.

$$\begin{aligned}\text{Numerator} &= P_t \times G_t \times \sigma \times A_e \\ &= (250 \times 10^3) \times 4000 \times 20 \times 4.5 \\ &= 10^9 \times 90 \\ &= 90 \times 10^9 = 9 \times 10^{10}\end{aligned}$$

Step 2: Calculate the denominator.

$$\begin{aligned}\text{Denominator} &= (4\pi)^2 \times S_{min} \\ &= 16\pi^2 \times 10^{-11} \\ &\approx 157.9137 \times 10^{-11} \\ &= 1.579137 \times 10^{-9}\end{aligned}$$

Step 3: Divide and find the fourth root.

$$\begin{aligned}R_{max}^4 &= \frac{9 \times 10^{10}}{1.579137 \times 10^{-9}} \\ &\approx 5.6993 \times 10^{19} \\ &= 5699.3 \times 10^{16} \text{ m}^4\end{aligned}$$

Taking the fourth root of both sides:

$$\begin{aligned} R_{max} &= (5699.3)^{1/4} \times (10^{16})^{1/4} \\ &\approx 8.6915 \times 10^4 \text{ meters} \\ &= \mathbf{86.915} \text{ km} \end{aligned}$$

Answer: The maximum range of the radar system is approximately **86.9 km**.

Question (c)

What are the main tasks of an earth station? Describe with block diagram how the terrestrial components communicate through a satellite. (12)

Solution

1. Main Tasks of an Earth Station:

An Earth station acts as the critical bridge (gateway) between the ground-based terrestrial networks (like internet, telephone networks) and the satellite in space. Its main tasks are:

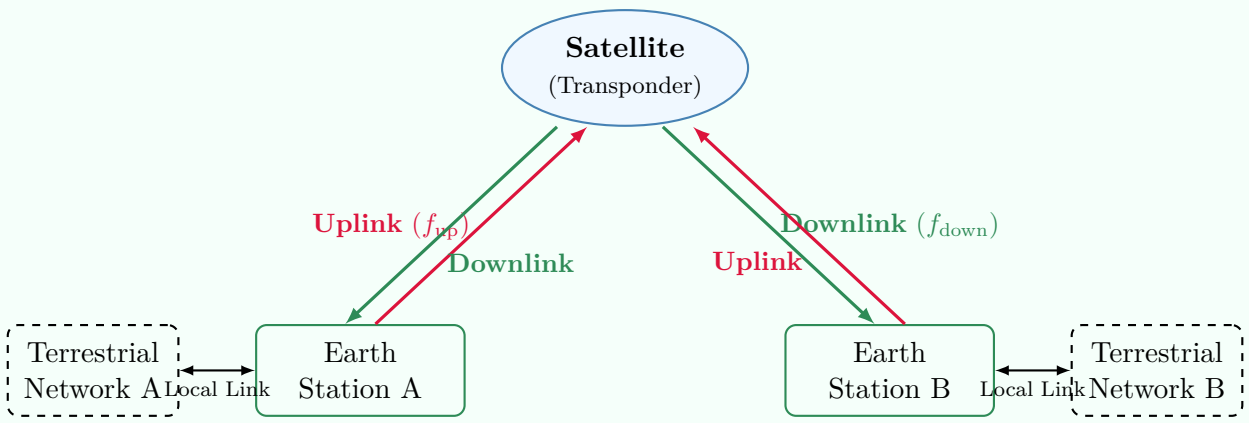
- **Modulation & Demodulation:** Modulating the user data onto high-frequency carrier waves for transmission and demodulating the received signals back into usable data.
- **Frequency Translation (Up/Down Conversion):** Converting the intermediate frequency (IF) to a high radio frequency (RF) for the uplink (e.g., 14 GHz), and stepping down the received RF downlink (e.g., 11 GHz) back to IF.
- **Amplification:** Using High Power Amplifiers (HPAs) to blast the signal to the satellite, and Low Noise Amplifiers (LNAs) to boost the extremely weak signals received from space without adding noise.
- **Tracking and Telemetry:** Continuously tracking the satellite's position in the sky to keep the antenna perfectly aligned, and monitoring the satellite's health (TT&C).

2. Communication of Terrestrial Components through a Satellite:

When a user (Terrestrial Component A) wants to communicate with another user (Terrestrial Component B) situated far away, they cannot do it directly due to the curvature of the earth. They use the satellite as a microwave relay station in the sky.

- **Uplink Process:** Data from Terrestrial Network A goes to Earth Station A. The Earth Station processes, heavily amplifies, and transmits this signal up to the satellite at a specific frequency (f_{up}).
- **Transponder Process:** The satellite receives this weak signal, amplifies it, translates its frequency to a different, lower frequency (f_{down}) to avoid interference, and broadcasts it back down to Earth.
- **Downlink Process:** Earth Station B receives this signal, amplifies it (using LNA), demodulates it, and passes the original data to Terrestrial Network B.

Block Diagram:



In the diagram, Earth Station A uplinks data to the satellite, which then downlinks it to Earth Station B, allowing the two separated terrestrial networks to communicate seamlessly.

Communication Systems Engineering

Solutions - Question 4

Question 4(a)

Explain the operation of a Doppler radar.

Solution

Operation of a Doppler Radar:

A Doppler radar operates on the principle of the **Doppler Effect**, which observes that the frequency of a wave changes when there is relative motion between the source of the wave and the observer (or target).

1. **Transmission:** The radar continuously (or in pulses) transmits a highly stable Radio Frequency (RF) signal at a known frequency, f_t.
2. **Reflection:** When this RF signal strikes a moving target (like a car or an aircraft), it reflects back to the radar antenna as an echo.
3. **Frequency Shift (Doppler Shift):**
   - If the target is moving **towards** the radar, the reflected waves are compressed, causing the received frequency to be higher than the transmitted one ($f_r > f_t$).
   - If the target is moving **away**, the reflected waves are stretched, making the received frequency lower ($f_r < f_t$).
4. **Doppler Frequency Calculation:** The difference between the received frequency (f_r) and the transmitted frequency (f_t) is the Doppler shift (f_d):

   $$f_d = f_r - f_t = \frac{2v_r}{\lambda}$$

   Where v_r is the radial velocity of the target and λ is the wavelength of the radar signal.
5. **Detection & Velocity Measurement:** The receiver mixes the incoming echo with a sample of the transmitted signal. A low-pass filter extracts the difference frequency (f_d). By directly measuring f_d, the radar can instantly and accurately calculate the velocity v_r of the moving target. Stationary targets yield $f_d = 0$ and are easily filtered out (clutter rejection).

Question 4(b)

What is meant by protocol? Describe the function of each layer of TCP/IP model.

Solution

1. What is meant by Protocol?

In data communications, a **protocol** is a formal set of rules, conventions, and data structures that governs how devices on a network exchange information. It defines *what* is communicated, *how* it is communicated, and *when* it is communicated. The key elements of a protocol are syntax (data format), semantics (meaning of control info), and timing (speed and sequencing).

2. Functions of each layer of the TCP/IP Model:
The TCP/IP reference model is a practical, 4-layer networking architecture that forms the basis of the Internet.

- **Layer 1: Network Access Layer (or Link Layer)**
  - **Function:** This layer deals with the physical transmission of data over the network medium (cables, optical fibers, wireless). It handles the conversion of digital data into physical signals and manages MAC (hardware) addressing for node-to-node frame delivery on the same local network.
  - *Examples:* Ethernet, Wi-Fi, ARP.
- **Layer 2: Internet Layer**
  - **Function:** Responsible for logical addressing and routing. It takes data packets, assigns them an IP address, and determines the best optimal path to route them across multiple interconnected networks to reach the final destination.
  - *Examples:* IP (IPv4/IPv6), ICMP (Ping).
- **Layer 3: Transport Layer**
  - **Function:** Provides end-to-end communication and process-to-process delivery. It breaks data into segments, handles error control, flow control, and uses port numbers to ensure data reaches the correct software application on the destination device.
  - *Examples:* **TCP** (reliable, connection-oriented) and **UDP** (fast, connectionless).
- **Layer 4: Application Layer**
  - **Function:** The topmost layer that directly provides network services to user applications. It handles data formatting, encryption, and the specific rules for high-level network services like web browsing or email.
  - *Examples:* HTTP (Web), SMTP (Email), FTP (File transfer), DNS.

Question 4(c)

Briefly explain the frame transmission technique in a ring LAN.

Solution

Frame Transmission Technique in a Ring LAN (Token Passing):
In a Ring Local Area Network (LAN), computers are connected in a closed physical or logical loop. The most common and standardized technique used for frame transmission in such networks is **Token Passing** (e.g., IEEE 802.5 Token Ring).
Here is the step-by-step explanation of the transmission process:

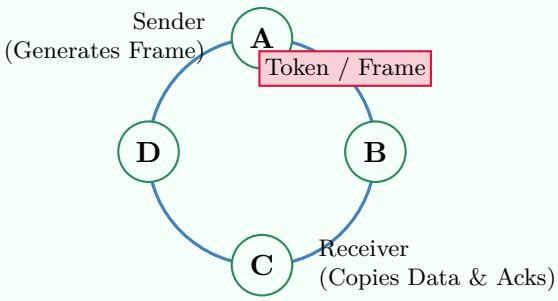
1. **The Token:** A small, special control frame called a "Token" continuously circulates around the ring in one direction when no station is transmitting. A token can be in two states: *Free* or *Busy*.
2. **Capturing the Token (Accessing the Medium):** When a station (e.g., Station A) wants to transmit data, it must wait for the Free Token to arrive. Once it receives the Free Token, it seizes it and changes its status to *Busy*.

3. **Data Transmission:** Station A attaches its data payload, the destination address (e.g., Station C), and its own source address to the busy token, transforming it into a **Data Frame**. It then transmits this frame onto the ring.

4. **Circulation and Copying:** The data frame travels around the ring, passing through each intermediate station. Every station examines the destination address on the frame.

   - If the address does not match, the station simply regenerates and forwards the frame to the next node.
   - When the frame reaches the destination (Station C), it recognizes its address, **copies the data** into its buffer, and modifies certain bits at the end of the frame (Frame Status field) to acknowledge that the data was received successfully.

5. **Frame Removal & Token Release:** The frame continues around the ring until it returns to the original sender (Station A). Station A checks the acknowledgment bits to ensure successful delivery. Finally, Station A **removes the data frame** from the ring and generates a new **Free Token**, passing it to the next station so others get a chance to transmit.



Communication Systems Engineering

Solutions - Radar, Satellite & Networking

Question 3(a)

Explain the operation of Doppler radar. (13)

Solution

Operation of a Doppler Radar:
A Doppler radar operates based on the **Doppler Effect**, which is the change in frequency of a wave in relation to an observer who is moving relative to the wave source.

1. **Signal Transmission:** The radar continuously transmits a highly stable Radio Frequency (RF) signal at a specific frequency, f_t.
2. **Target Reflection:** When this transmitted signal strikes a moving target, it is reflected back as an echo.
3. **Frequency Shift (Doppler Shift):**
   - If the target is moving **towards** the radar, the reflected waves are compressed, increasing the frequency ($f_r > f_t$).
   - If the target is moving **away** from the radar, the reflected waves are stretched, decreasing the frequency ($f_r < f_t$).
4. **Doppler Frequency Calculation:** The difference between the received frequency (f_r) and the transmitted frequency (f_t) is called the Doppler shift (f_d):

   $$f_d = f_r - f_t = \frac{2v_r}{\lambda}$$

   Where v_r is the radial velocity of the target relative to the radar, and λ is the wavelength of the transmitted signal.
5. **Signal Detection:** The receiver mixes (multiplies) the received echo signal with a sample of the transmitted signal. A low-pass filter extracts the difference frequency f_d. By measuring this Doppler shift, the radar precisely calculates the velocity of the target.
6. **Clutter Rejection:** Stationary targets (like buildings or trees) return an echo with $f_d = 0$. The radar system easily filters these out, making it highly effective at detecting only moving targets.

Question 3(b)

Derive the free-space radar range equation. Explain the factors on which radar range depend. (12)

Solution

1. Derivation of Free-Space Radar Range Equation:
Let P_t be the transmitted power. If the antenna radiates isotropically (equally in all directions), the power density at a distance R from the radar is $P_t/4\pi R^2$.

Radar antennas are highly directional. With a transmitting antenna gain G_t, the power density at the target at range R is:

$$\text{Power density at target} = \frac{P_t G_t}{4\pi R^2}$$

The target intercepts a portion of this power and scatters it back. The measure of how much power is scattered back is called the Radar Cross Section (RCS), denoted by σ. The power of the reflected signal is:

$$P_{\text{reflected}} = \frac{P_t G_t \sigma}{4\pi R^2}$$

This reflected power spreads out spherically as it returns to the radar. The power density of the echo at the receiving antenna is:

$$\text{Power density at receiver} = \frac{P_{\text{reflected}}}{4\pi R^2} = \frac{P_t G_t \sigma}{(4\pi R^2)^2}$$

The radar receiving antenna captures a portion of this power depending on its effective aperture area, A_e. The total received power P_r is:

$$P_r = \frac{P_t G_t \sigma A_e}{(4\pi R^2)^2} = \frac{P_t G_t \sigma A_e}{(4\pi)^2 R^4}$$

The maximum radar range (R_{max}) is the distance beyond which the target cannot be detected. This occurs when the received echo power P_r just equals the Minimum Detectable Signal (S_{min}) of the receiver. Setting $P_r = S_{min}$ and solving for R, we get the **Radar Range Equation**:

$$\mathbf{R_{max} = \left[\frac{P_t G_t \sigma A_e}{(4\pi)^2 S_{min}}\right]^{1/4}}$$

2. Factors on which Radar Range Depends: From the derived equation, the maximum range R_{max} depends on:

- **Transmitter Power (P_t):** Range is directly proportional to the fourth root of P_t. To double the range, transmitted power must be increased by 16 times (2^4).
- **Antenna Gain (G_t) & Aperture (A_e):** Larger, more directional antennas increase range.
- **Radar Cross Section (σ):** A measure of the target's reflectivity. Larger or highly reflective targets (like passenger planes) can be detected at greater ranges than stealthy or small targets.
- **Minimum Detectable Signal (S_{min}):** A highly sensitive receiver with a lower S_{min} can detect weaker echoes, thus increasing the maximum range. It is heavily dependent on receiver noise.

Question 3(c)

Describe the satellite launching method in a geo-stationary orbit. (10)

Solution

Satellite Launching Method in a Geo-Stationary Orbit (GEO):
Placing a satellite directly into a Geostationary Earth Orbit (GEO) at an altitude of approximately 35,786 km above the equator requires a multi-stage process, typically involving a transfer orbit to save fuel and maximize payload capacity.

1. **Launch into Low Earth Orbit (LEO) / Parking Orbit:** The launch vehicle (rocket) lifts off from the Earth. After the initial stages burn out, the satellite is placed into a circular Low Earth Orbit (usually between 200 km to 500 km altitude) known as a parking orbit.

2. **Geostationary Transfer Orbit (GTO):** While in the parking orbit, the rocket aligns itself optimally (usually at the equator crossing). The upper stage engine (Perigee Kick Motor) fires, providing a massive velocity boost. This elongates the orbit into a highly elliptical path called the Geostationary Transfer Orbit (GTO).

   - The lowest point (Perigee) of the GTO remains at the LEO altitude.
   - The highest point (Apogee) reaches the target geostationary altitude (35,786 km).

3. **Circularization into GEO:** The satellite coasts along the elliptical GTO until it reaches the Apogee (the farthest point). At this exact moment, an onboard engine called the **Apogee Kick Motor (AKM)** fires. This burn serves two critical purposes:

   - It circularizes the orbit, lifting the perigee up to 35,786 km.
   - It performs an inclination change, reducing the orbital angle to 0° so that the satellite orbits exactly along the Earth's equator.

4. **Final Positioning:** The satellite is now in a geostationary orbit. Thrusters are used to precisely drift the satellite to its assigned longitude (orbital slot) and stabilize it.

Question 4(a)

Explain the different sub-station of satellite earth station. (15)

Solution

Sub-systems (Sub-stations) of a Satellite Earth Station:
An Earth station acts as the terrestrial gateway for a satellite communication system. It is composed of several critical sub-systems:

1. **Antenna Sub-system:** This includes the large parabolic dish, the feed horn, and the mount. It provides high directive gain to transmit the uplink signal to the satellite and capture the weak downlink signal. It also includes an **Antenna Tracking System** (motors and sensors) to keep the dish precisely pointed at the satellite despite slight orbital drifts.

2. **Transmit Sub-system:** Responsible for sending the signal to the satellite.

   - *Up-converter:* Converts the intermediate frequency (IF) modulated signal to the high radio frequency (RF) uplink band (e.g., 6 GHz or 14 GHz).
   - *High Power Amplifier (HPA):* Boosts the RF signal to extremely high power levels (using Klystrons or Traveling Wave Tubes) to overcome free-space path loss.

3. **Receive Sub-system:** Responsible for processing the signal received from the satellite.
   - *Low Noise Amplifier (LNA):* The first component after the antenna. It amplifies the incredibly weak received signal while introducing almost zero thermal noise.
   - *Down-converter:* Steps down the high RF downlink frequency (e.g., 4 GHz or 11 GHz) to a lower, manageable IF for demodulation.
4. **Baseband & Modem Sub-system:** Handles the digital or analog information. It multiplexes user data, adds error correction codes, and uses Modems to modulate data onto the IF carrier (Tx) or demodulate data from the IF carrier (Rx).
5. **Telemetry, Tracking, and Command (TT&C) Sub-system:** A specialized unit used strictly for satellite control. It receives telemetry (health status) from the satellite, tracks its precise location, and sends commands (e.g., to fire thrusters or switch transponders).
6. **Power Supply Sub-system:** Provides highly reliable, uninterruptible power to the entire earth station using grid power, large battery banks, and backup diesel generators.

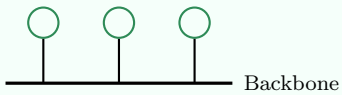
Question 4(b)

Describe different LAN topologies with necessary diagram. (10)

Solution

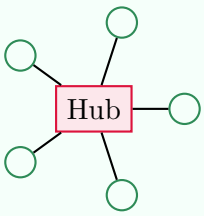
Local Area Network (LAN) Topologies:
Topology refers to the physical or logical layout of a network.

1. Bus Topology:
All devices share a single, central communication cable called the "backbone". Terminators are required at both ends.



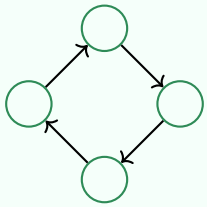
- *Pros:* Easy to install and requires less cable.
- *Cons:* If the main backbone cable breaks, the entire network fails. High collision rates with many devices.

2. Star Topology:
All devices are connected to a central device, such as a Hub or a Switch. Data from any device must pass through the central node.



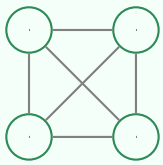
- *Pros:* If one cable fails, only that device is affected. Easy to troubleshoot and manage.
- *Cons:* If the central hub/switch fails, the entire network goes down.

3. Ring Topology:
Each device is connected to exactly two other devices, forming a closed loop. Data travels in one direction (unidirectional) around the ring.



- *Pros:* Orderly data transmission (often uses a Token), reducing collisions.
- *Cons:* If a single link or device fails, the token cannot pass, bringing down the whole network.

4. Mesh Topology:
Every device has a dedicated point-to-point link to every other device in the network.



- *Pros:* Highly robust and reliable. Failure of one link does not affect the rest of the network. High security and privacy.
- *Cons:* Extremely expensive and difficult to install due to the massive amount of cabling required.

Question 4(c)

Briefly describe different layers of TCP/IP model. (10)

Solution

Layers of the TCP/IP Model:
The TCP/IP model is a practical, 4-layer reference model that acts as the foundation of the modern Internet.

1. **Network Access Layer (or Link Layer):**
   - *Function:* This is the lowest layer, representing the physical network hardware. It is responsible for the physical transmission of data bits over the network medium (like Ethernet cables, Wi-Fi, optical fiber) and handling MAC addresses.
   - *Protocols:* Ethernet, Wi-Fi (802.11), ARP, PPP.
2. **Internet Layer (or Network Layer):**
   - *Function:* Responsible for logical transmission of data packets over multiple networks. It provides IP addressing, determines the best path (Routing) for the packet to reach its final destination, and handles packet fragmentation.
   - *Protocols:* IP (Internet Protocol - IPv4/IPv6), ICMP (Ping), ARP.
3. **Transport Layer:**
   - *Function:* Responsible for end-to-end communication and process-to-process delivery. It handles error control, flow control, and uses port numbers to direct data to the correct application.
   - *Protocols:*

  - **TCP (Transmission Control Protocol):** Provides reliable, connection-oriented delivery (ensures all packets arrive intact).
  - **UDP (User Datagram Protocol):** Provides fast, connectionless delivery (used for live video/audio where speed is prioritized over reliability).

4. **Application Layer:**
   - *Function:* The topmost layer that directly interacts with software applications. It provides the user interface and high-level network services. It encompasses the roles of the Application, Presentation, and Session layers of the OSI model.
   - *Protocols:* HTTP/HTTPS (Web browsing), SMTP (Email), FTP (File Transfer), DNS (Domain resolution).

Digital & Radar Communication Systems

Solutions - Question 3

Question 3(a)

Describe the operation of a Doppler radar and mention its application. (17)

Solution

1. Operation of a Doppler Radar:

A Doppler radar operates on the principle of the **Doppler Effect**, which states that the frequency of a wave changes when there is relative motion between the source and the observer (target).

- **Transmission:** The radar transmits a continuous wave (CW) or a series of pulsed Radio Frequency (RF) signals at a known, highly stable frequency (f_t).
- **Reflection:** When this signal strikes a moving target, the wave is reflected back to the radar. If the target is moving *towards* the radar, the reflected waves are compressed, leading to a higher frequency. If the target is moving *away*, the waves are stretched, resulting in a lower frequency.
- **Doppler Shift Calculation:** The received frequency (f_r) differs from the transmitted frequency (f_t). This difference is called the Doppler shift (f_d), given by:

  $$f_d = f_r - f_t = \frac{2v_r}{\lambda}$$

  Where v_r is the radial velocity of the target and λ is the wavelength of the transmitted signal.
- **Detection:** The radar receiver mixes the transmitted signal and the received echo signal. A low-pass filter is then used to extract the difference frequency (the Doppler shift). By measuring this shift f_d, the radar can precisely calculate the velocity of the moving target. It inherently filters out stationary objects (clutter) because they produce zero Doppler shift ($f_d = 0$).

2. Applications of Doppler Radar:

1. **Traffic Law Enforcement:** Used by police as "speed guns" to measure the speed of moving vehicles.
2. **Meteorology (Weather Radar):** Used to detect precipitation, calculate its motion, and identify severe weather conditions like tornadoes (by observing wind velocities).
3. **Aviation & Air Traffic Control:** Used to monitor the speed and trajectory of aircraft.
4. **Medical Diagnostics:** Used in echocardiography and fetal monitors to measure blood flow velocity and detect heartbeats.
5. **Military & Defense:** Used for tracking moving targets, missiles, and in proximity fuses for artillery.

Question 3(b)

What is speech coding? What are the objectives of speech coding? (08)

Solution

1. What is Speech Coding?

Speech coding is the process of converting an analog human speech signal into a digital format (digitization) and compressing it to use the minimum possible number of bits, while maintaining an acceptable level of speech quality. It is a specialized form of data compression tailored specifically for the characteristics of the human voice, removing redundancies and irrelevant information that the human ear cannot perceive. Examples include PCM, ADPCM, and LPC (Linear Predictive Coding).

2. Objectives of Speech Coding:

- **Minimize Bit Rate / Bandwidth:** The primary goal is to reduce the amount of data required to represent speech, thereby saving transmission bandwidth and storage space. This is crucial for cellular networks (like GSM, LTE) and VoIP.
- **Maintain Speech Quality:** To ensure that the decoded speech is highly intelligible (understandable) and natural-sounding to the listener, despite the heavy compression.
- **Minimize Algorithmic Delay:** To keep the processing time (encoding and decoding delay) as low as possible to prevent annoying lag during real-time, two-way conversations.
- **Reduce Computational Complexity:** To ensure that the speech coding algorithms can be executed quickly by the hardware (like smartphone processors) without draining the battery excessively.
- **Robustness to Channel Errors:** To design the code such that the speech quality does not degrade catastrophically if some bits are lost or corrupted during transmission over a noisy channel (like a weak mobile signal).

Question 3(c)

What is meant by MSK? Derive the expression of the phase in MSK for symbol 0 and 1. (10)

Solution

1. What is meant by MSK?

MSK (Minimum Shift Keying) is a special, highly bandwidth-efficient form of Continuous Phase Frequency Shift Keying (CPFSK). It is characterized by two key features:

- The modulation index is exactly $h = 0.5$.
- The phase of the carrier changes continuously without any abrupt jumps at the bit boundaries.

It is called "Minimum" because $h = 0.5$ provides the minimum possible frequency separation ($\Delta f = \frac{1}{2T_b}$) that allows the two frequencies representing '1' and '0' to be orthogonal for coherent detection. This continuous phase leads to a very compact power spectrum with rapidly decaying side-lobes.

2. Derivation of the Phase Expression in MSK for Symbol 1 and 0:

A general CPFSK signal can be represented as:

$$s(t) = \sqrt{\frac{2E_b}{T_b}} \cos[2\pi f_c t + \theta(t)]$$

Where $\theta(t)$ is the continuous phase angle, E_b is the bit energy, T_b is the bit duration, and f_c is the carrier frequency.
The phase $\theta(t)$ for the interval $0 \leq t \leq T_b$ is given by the integral of the modulating signal $m(t)$:

$$\theta(t) = \theta(0) + \frac{\pi h}{T_b} \int_0^t m(\tau) d\tau$$

Where:

- $\theta(0)$ is the initial phase at $t = 0$.
- h is the modulation index. For MSK, $\mathbf{h = 0.5}$.
- $m(t)$ is the baseband binary data sequence taking values $+1$ (for symbol 1) and -1 (for symbol 0).

Substituting $h = 0.5$ into the phase equation:

$$\theta(t) = \theta(0) + \frac{\pi(0.5)}{T_b} \int_0^t m(\tau) d\tau$$

$$\theta(t) = \theta(0) + \frac{\pi}{2T_b} \int_0^t m(\tau) d\tau$$

Since $m(\tau)$ is constant over a single bit interval $0 \leq t \leq T_b$, the integral $\int_0^t m(\tau) d\tau$ simply becomes $m \cdot t$. Thus, the phase expression simplifies to:

$$\theta(t) = \theta(0) + \frac{\pi t}{2T_b} m$$

Now, we derive the phase for each symbol:
Case 1: For Symbol 1 ($m = +1$) Substituting $m = +1$ into the phase equation:

$$\boldsymbol{\theta_1(\mathbf{t}) = \theta(\mathbf{0}) + \frac{\pi}{\mathbf{2T_b}} \mathbf{t}}$$

This means for symbol 1, the phase increases linearly by exactly $+\frac{\pi}{2}$ ($90°$) over the bit duration T_b.
Case 2: For Symbol 0 ($m = -1$) Substituting $m = -1$ into the phase equation:

$$\boldsymbol{\theta_0(\mathbf{t}) = \theta(\mathbf{0}) - \frac{\pi}{\mathbf{2T_b}} \mathbf{t}}$$

This means for symbol 0, the phase decreases linearly by exactly $-\frac{\pi}{2}$ ($-90°$) over the bit duration T_b.

Data Communication Systems

Solutions - Topologies & OSI Model

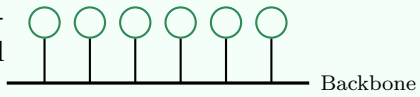
Question 3(a)

For each of the following four networks, discuss the consequences if a connection fails: (i) Six devices arranged in a bus topology (ii) Four devices arranged in a ring topology (iii) Five devices arranged in a mesh topology (iv) Seven devices arranged in a star topology (not counting the hub). (10)

Solution

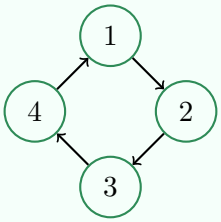
Consequences of a Connection Failure:

(i) Bus Topology (6 devices):
In a bus topology, all devices share a single main communication cable (the backbone).



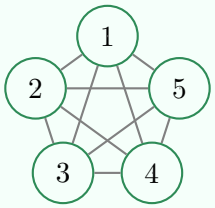
- **If the main backbone cable fails (breaks):** The entire network goes down. The break creates unterminated ends, causing signal reflection and destroying all communication.
- **If a drop-line (connection between a device and the backbone) fails:** Only that specific device is disconnected. The other 5 devices can continue to communicate normally.

(ii) Ring Topology (4 devices):
In a basic unidirectional ring topology, each device is connected to exactly two other devices, forming a closed loop. Data travels in one direction.



- **If a connection (link) fails:** The entire network fails. The closed loop is broken, meaning the token (or data) cannot circulate to reach its destination. *(Note: A dual-ring topology, like FDDI, could recover by wrapping the ring, but a standard ring will fail entirely).*

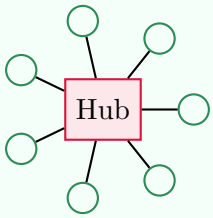
(iii) Mesh Topology (5 devices):
In a fully connected mesh, every device has a dedicated point-to-point link to every other device. For 5 devices, there are $n(n-1)/2 = 10$ links.



- **If a connection fails:** The network is highly robust. A single link failure only affects the direct communication between the two devices connected by that specific link. Even then, those two devices can still communicate indirectly by routing their data through the other 3 devices. The rest of the network remains completely unaffected.

(iv) Star Topology (7 devices + 1 Hub):
In a star topology, all devices are connected to a central controller (Hub or Switch).

- **If a connection (cable) fails:** Only the device connected to that specific failed cable is cut off from the network. The remaining 6 devices continue to communicate with each other through the hub without any issues. *(However, if the central hub itself fails, the entire network goes down).*



Question 3(b)

What is the purpose of a reference model in data communication system? How is the OSI reference model different from the TCP/IP reference model? Mention the responsibilities of the each layer of the OSI model. (15)

Solution

1. Purpose of a Reference Model:
A reference model provides a standardized, conceptual framework for how data communication should take place. Its main purposes are:

- **Modularity:** It breaks down the complex process of network communication into smaller, manageable, and simpler layers.
- **Interoperability:** It ensures that hardware and software from different vendors can communicate seamlessly.
- **Troubleshooting:** By dividing the network into layers, it becomes much easier to isolate and troubleshoot network problems.

2. Differences between OSI and TCP/IP Reference Models:

OSI Reference Model	TCP/IP Reference Model
It has **7 layers**.	It has **4 layers** (or 5, depending on the version).
Strict separation of Application, Presentation, and Session layers.	Combines Application, Presentation, and Session into a single **Application layer**.
It is a theoretical and conceptual model developed *before* the protocols were invented.	It is a practical model developed *based on* standard protocols (like TCP and IP).
The Network layer supports both connection-oriented and connectionless communication.	The Internet (Network) layer supports **only** connectionless communication (IP).
Strictly separates the Data Link and Physical layers.	Often combines them into a single **Network Access / Link layer**.

3. Responsibilities of Each Layer of the OSI Model:

1. **Physical Layer:** Responsible for the physical connection between devices and transmitting raw bit streams (0s and 1s) over the physical medium (cables, wireless).

2. **Data Link Layer:** Responsible for node-to-node frame delivery. It handles MAC (physical) addressing, framing, and error detection/control (like CSMA/CD).
3. **Network Layer:** Responsible for source-to-destination (end-to-end) delivery of packets across multiple networks. It handles logical (IP) addressing and routing.
4. **Transport Layer:** Responsible for process-to-process delivery of the entire message. It handles segmentation, reliable/unreliable delivery (TCP/UDP), and port addressing.
5. **Session Layer:** Responsible for establishing, maintaining, synchronizing, and terminating sessions (dialogs) between communicating applications.
6. **Presentation Layer:** Responsible for the translation, encryption/decryption, and compression/decompression of data so that it is readable by the application layer.
7. **Application Layer:** Provides network services directly to the user's applications (e.g., HTTP for web browsing, SMTP for email, FTP for file transfer).

Satellite Communication Systems

Solutions - Transponders & Earth Stations

Question 3(a)

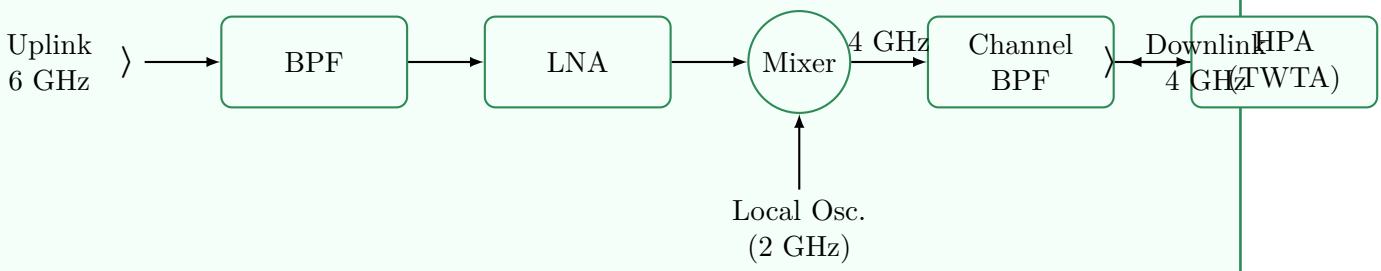
What is the main function of transponder in a satellite communication system? Illustrate the block diagrams of the single- and double-frequency conversion transponders for 6/4 GHz and 14/11 GHz bands respectively. (12)

Solution

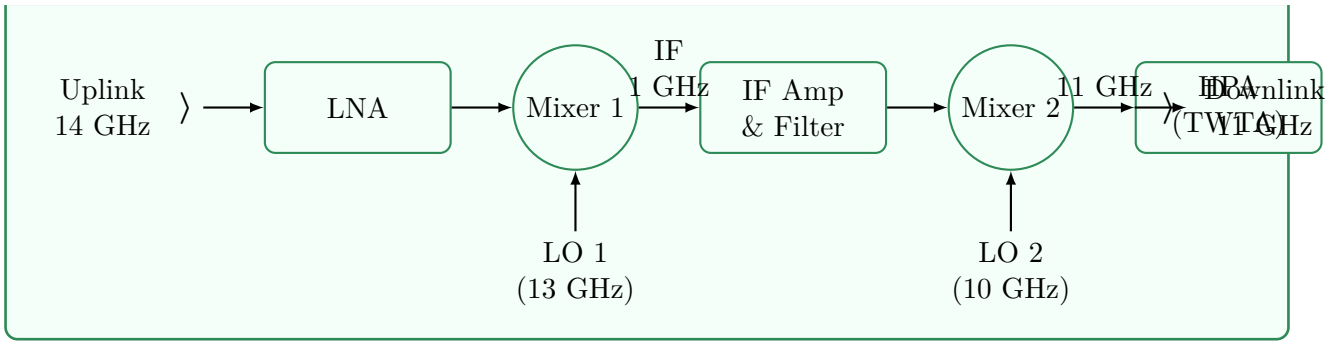
1. Main Function of a Transponder:
A transponder (Transmitter-Responder) is the most critical subsystem onboard a communications satellite. Its main functions are:

- **Reception:** It receives the weak uplink signal transmitted from the Earth station.
- **Amplification:** It heavily amplifies the weak received signal using Low Noise Amplifiers (LNAs) to minimize noise, and later High Power Amplifiers (HPAs/TWTAs) before retransmission.
- **Frequency Translation:** It converts (shifts) the high uplink frequency to a lower downlink frequency. This prevents the high-power transmitted signal from interfering with the weak incoming received signal on the satellite.

2. Single-Frequency Conversion Transponder (6/4 GHz Band):
In the C-band (6 GHz uplink / 4 GHz downlink), the frequency translation is usually done in a single step using one mixer because the frequency difference is small enough to avoid severe design complexities.



3. Double-Frequency Conversion Transponder (14/11 GHz Band):
In the Ku-band (14 GHz uplink / 11 GHz downlink) or higher bands, direct single conversion is difficult due to poor mixer performance and filtering issues at such high frequencies. Therefore, a double conversion technique is used: first down-converting to an Intermediate Frequency (IF, e.g., 1 GHz) for amplification and filtering, and then up-converting to the final downlink frequency.



Question 3(b)

With a neat sketch, explain the main tasks of an earth station in a satellite communication system. (11)

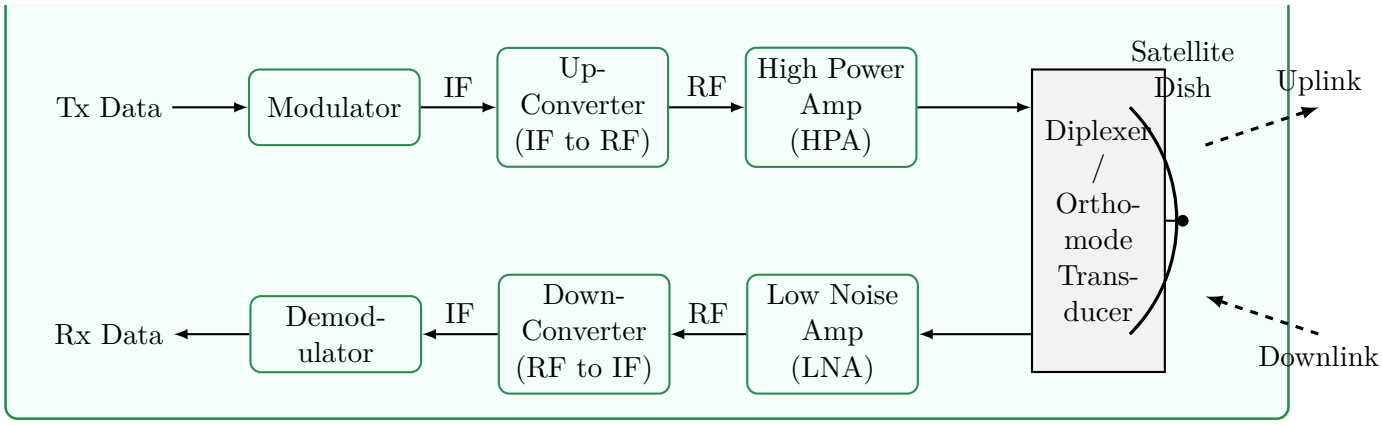
Solution

1. Main Tasks of an Earth Station:

The Earth station acts as the gateway between terrestrial communication networks and the space segment (satellite). Its primary tasks include:

- **Baseband Processing:** Multiplexing various user data streams (voice, video, internet) into a single composite baseband signal for transmission, and demultiplexing received signals.
- **Modulation & Demodulation:** Modulating the baseband signal onto an Intermediate Frequency (IF) carrier for transmission, and demodulating the received IF carrier back to baseband.
- **Frequency Translation:**
  - **Up-conversion:** Converting the modulated IF signal to a high Radio Frequency (RF) uplink frequency (e.g., 6 GHz or 14 GHz).
  - **Down-conversion:** Converting the received weak RF downlink signal (e.g., 4 GHz or 11 GHz) down to the IF band.
- **Power Amplification (HPA):** Boosting the RF uplink signal to a very high power level (using Klystrons or TWTAs) to overcome the massive free-space path loss before sending it to the antenna.
- **Low Noise Amplification (LNA):** Amplifying the extremely weak signal received from the satellite while adding minimal thermal noise.
- **Antenna Tracking:** Continuously adjusting the parabolic antenna's pointing direction to accurately track the satellite's position in orbit.

2. Block Diagram of an Earth Station:


Tx Data
Modulator
IF
Up-
Converter
(IF to RF)
RF
High Power
Amp
(HPA)
Satellite
Dish
Uplink
Diplexer
/
Ortho-
mode
Trans-
ducer
Downlink
Rx Data
Demod-
ulator
IF
Down-
Converter
(RF to IF)
RF
Low Noise
Amp
(LNA)

Data Communication Systems

Transmission Schemes & Network Types

Question

Explain simplex, half-duplex and full-duplex data transmission schemes with examples. Also, state the features of LANs, WANs and MANs. (10)

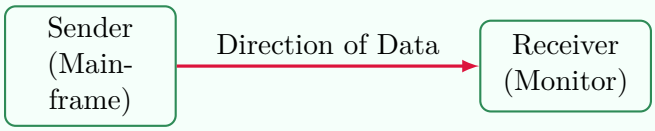
Solution

Part 1: Data Transmission Schemes

Data transmission modes (or schemes) define the direction of signal flow between two connected devices. There are three basic modes:

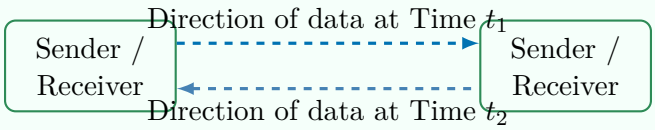
1. Simplex Mode:

- **Explanation:** Communication is strictly **unidirectional** (one-way), like a one-way street. Only one device on a link can transmit data, and the other can only receive it.
- **Example:** A Keyboard and a Monitor. The keyboard can only introduce input, and the monitor can only display output. Radio and TV broadcasting are also classic examples.



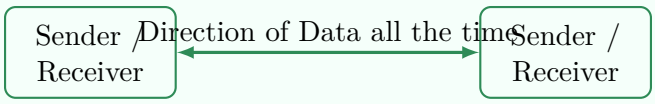
2. Half-Duplex Mode:

- **Explanation:** Communication is **bidirectional, but not simultaneously**. Each station can both transmit and receive, but when one device is sending, the other can only receive, and vice versa. It is like a one-lane bridge for two-way traffic.
- **Example: Walkie-Talkie**. A user must press a button to talk and release it to listen. Both cannot talk at the exact same time.



3. Full-Duplex Mode:

- **Explanation:** Communication is **bidirectional and simultaneous**. Both stations can transmit and receive at the exact same time. It is like a two-way street with traffic flowing in both directions simultaneously.
- **Example:** A **Telephone Network** or Mobile Phone conversation. Both people can talk and listen at the same time.



Part 2: Features of Networks (LAN, MAN, WAN)

1. Local Area Network (LAN):

- **Coverage:** Covers a very small geographical area, such as a single room, a building, or a small campus.
- **Speed:** Extremely high data transfer rates (typically 100 Mbps, 1 Gbps, or even 10 Gbps).
- **Error Rate:** Very low delay and negligible error rates.
- **Ownership:** Usually privately owned, installed, and maintained by a single organization.

2. Metropolitan Area Network (MAN):

- **Coverage:** Covers a larger area than a LAN, typically spanning a city, a large university campus, or a metropolitan area.
- **Speed:** Moderate to high data transfer rates.
- **Purpose:** Designed to connect multiple LANs together or provide internet access to a city.
- **Example:** A city's Cable TV network or WiMAX network. Ownership can be private or public.

3. Wide Area Network (WAN):

- **Coverage:** Covers vast geographical areas, spanning across cities, countries, or even continents.
- **Speed:** Generally lower data transfer rates compared to LANs (though optical fiber WANs are getting faster).
- **Error Rate:** Higher propagation delay and relatively higher error rates due to the long distances involved.
- **Infrastructure:** Relies heavily on public telecommunication lines, leased lines, or satellite links. The **Internet** is the largest example of a WAN.

Digital Communication Systems

Solutions - Question 2 (Final)

Question 2(a)

Explain the advantage of Gray encoding in multilevel modulation scheme with an example. Explain the relation between the bit error rate and the level of QAM scheme in noisy condition with the help of Fig. 2(a). Also, state the reason behind this sort of relationship. (10)

Solution

1. Advantage of Gray Encoding:
In multilevel modulation schemes (like M-ary PSK or M-ary QAM), symbols are represented by multiple bits. Because of channel noise, the most probable error at the receiver is confusing a transmitted symbol with one of its closest, adjacent neighbors in the constellation diagram.
Gray encoding dictates that adjacent symbols differ in exactly **one bit position**. **Advantage:** By using Gray coding, if a symbol is misinterpreted as an adjacent symbol due to noise, only a **single bit error** occurs. If standard binary encoding were used, an adjacent symbol error could result in multiple bit errors, significantly worsening the overall Bit Error Rate (BER).
Example (QPSK):
Consider a 4-PSK (QPSK) constellation.

- **Without Gray Coding:** Symbols might be mapped sequentially as $00 \to 01 \to 10 \to 11$ around the circle. If noise pushes '01' into the adjacent '10' region, **2 bits** are in error.
- **With Gray Coding:** Symbols are mapped as $00 \to 01 \to 11 \to 10$. If noise pushes '01' into the adjacent '11' region, only **1 bit** is in error.

2. Relation between BER and Level of QAM (M):
(Assuming standard behavior in lieu of Fig. 2(a)):
For a fixed average transmit power and a constant Signal-to-Noise Ratio (SNR), **as the level of the QAM scheme (M) increases (e.g., from 4-QAM to 16-QAM to 64-QAM), the Bit Error Rate (BER) also significantly increases.**
Reason behind this relationship:
In a constellation diagram, the average signal power is proportional to the average distance of the points from the origin. If the average transmit power is kept constant, packing more symbols (higher M) into the same signal space forces the points to be squeezed closer together.

- The minimum Euclidean distance (d_{min}) between adjacent symbols decreases.
- Because the points are closer, it takes much less noise energy to push a transmitted symbol across a decision boundary into an adjacent symbol's region.
- Therefore, a higher-level QAM system is much more susceptible to noise, leading to a higher BER for the same SNR. To maintain the same BER, a higher-level QAM requires a proportionately higher SNR.

Question 2(b)

Calculate the theoretical highest bit rate of a regular telephone line if the signal-to-noise ratio is 35 dB. A telephone line normally has a bandwidth of 3100 Hz (300 Hz to 3400 Hz). How many amplitude levels can be correctly detected in case of your calculated bit rate? (10)

Solution

1. Calculating the Theoretical Highest Bit Rate (Shannon Capacity)
Given Parameters:

- Bandwidth, $B = 3100$ Hz
- Signal-to-Noise Ratio (in dB), $SNR_{dB} = 35$ dB

First, we must convert the SNR from decibels to a linear ratio:

$$SNR_{dB} = 10\log_{10}(SNR)$$

$$35 = 10\log_{10}(SNR) \implies 3.5 = \log_{10}(SNR)$$

$$SNR = 10^{3.5} \approx 3162.28$$

According to the Shannon-Hartley theorem, the theoretical highest bit rate (Channel Capacity, C) is:

$$C = B\log_2(1 + SNR)$$

$$C = 3100 \times \log_2(1 + 3162.28)$$

$$C = 3100 \times \log_2(3163.28)$$

Using the change of base formula ($\log_2 x = \frac{\ln x}{\ln 2}$):

$$\log_2(3163.28) \approx \frac{8.059}{0.693} \approx 11.627 \text{ bits/Hz}$$

$$C = 3100 \times 11.627 \approx \mathbf{36,044.5} \text{ bps} \approx \mathbf{36.04} \text{ kbps}$$

The theoretical highest bit rate is approximately 36.04 **kbps.**

2. Calculating the Number of Amplitude Levels (L)
To find out how many discrete signal levels (L) are required to achieve this capacity over a noiseless channel of the same bandwidth, we equate the Shannon Capacity formula with the Nyquist Bit Rate formula.
Nyquist formula for maximum bit rate:

$$R_b = 2B\log_2(L)$$

Setting $R_b = C$:

$$2B\log_2(L) = B\log_2(1 + SNR)$$

Dividing both sides by B:

$$2\log_2(L) = \log_2(1 + SNR)$$

$$\log_2(L^2) = \log_2(1 + SNR)$$

$$L^2 = 1 + SNR$$

$$L = \sqrt{1 + SNR}$$

Now, substitute the linear SNR value into this simplified relationship:

$$L = \sqrt{1 + 3162.28}$$

$$L = \sqrt{3163.28}$$

$$L \approx 56.24$$

Since the number of amplitude levels must be an integer (and typically a power of 2 for binary systems to map bits evenly), a system could theoretically detect **56 distinct levels**. *(Note: In a practical binary-based system, one would choose $L = 32$ for reliability, or $L = 64$ accepting a slightly higher error rate, but the theoretical exact integer is 56).*

Digital Communication Systems

Solutions - Section A (Excluding 'b')

Question 1(a)

Consider a linear modulation scheme for which the modulated wave is defined by equation (1).

$$s(t) = b(t)c(t) \quad \ldots (1)$$

where $b(t)$ is the incoming binary wave and $c(t)$ is the sinusoidal carrier which has unit energy measured over one-bit duration. Under the band-pass assumption, show that the transmitted signal energy (on a per bit basis) is a scaled version of the energy in $b(t)$. (12)

Solution

Proof:

Let the bit duration be T_b. The transmitted modulated signal is given by:

$$s(t) = b(t)c(t)$$

The energy of the transmitted signal $s(t)$ over one bit duration is defined as:

$$E_s = \int_0^{T_b} s^2(t)dt = \int_0^{T_b} b^2(t)c^2(t)dt$$

Under the baseband signalling assumption for full-width pulses, the binary wave $b(t)$ remains constant over the duration of one bit ($0 \leq t \leq T_b$). Let this constant amplitude be V_k for the k-th bit. Since $b(t) = V_k$ is a constant during this integration interval, we can pull $b^2(t)$ out of the integral:

$$E_s = V_k^2 \int_0^{T_b} c^2(t)dt$$

The problem states that the carrier $c(t)$ has **unit energy** measured over one-bit duration. This means:

$$\int_0^{T_b} c^2(t)dt = 1$$

Substituting this back into the energy equation for the transmitted signal, we get:

$$E_s = V_k^2 \cdot 1 = V_k^2$$

Now, let's find the energy of the baseband binary wave $b(t)$ over the same one-bit duration, let's call it E_b:

$$E_b = \int_0^{T_b} b^2(t)dt = \int_0^{T_b} V_k^2 dt = V_k^2[t]_0^{T_b} = V_k^2 T_b$$

From this, we can express V_k^2 as:

$$V_k^2 = \frac{E_b}{T_b}$$

Since we already established that $E_s = V_k^2$, we can substitute the value of V_k^2:

$$\mathbf{E_s = \frac{E_b}{T_b}}$$

$$\mathbf{E_s = \left(\frac{1}{T_b}\right) E_b}$$

Conclusion: The transmitted signal energy E_s is equal to the baseband signal energy E_b multiplied by a scaling factor of $\frac{1}{T_b}$. Therefore, the transmitted signal energy (on a per bit basis) is indeed a scaled version of the energy in $b(t)$. *(Proved)*

Question 1(c)

Fig. 1(c) shows a digital data transmission scheme. The baseband signal generator uses full-width pulses and polar signalling. The data rate is 1 Mbit/s. (i) If the modulator generates a 16-PSK signal, calculate the baud rate and bandwidth of the modulated output. (ii) If the modulator generates a 4-FSK signal where the difference between the successive carrier frequencies is 100 kHz, determine the bandwidth of the modulated signal. For this 4-FSK case, assume that the transmission filter has a roll-off factor of 0.5. (10)

Solution

Given Parameters:

- Data Rate (Bit Rate), $R_b = 1$ Mbit/s $= 10^6$ bps
- Pulse shape: Full-width pulses (Rectangular pulses in time domain).

(i) For 16-PSK Signal

- **Number of Levels (M):** 16
- **Bits per symbol (n):** $n = \log_2(16) = 4$ bits/symbol

Baud Rate (Symbol Rate, R_s):

$$R_s = \frac{R_b}{n} = \frac{10^6 \text{ bps}}{4} = \mathbf{250} \text{ kbaud (or 250 ksps)}$$

Bandwidth of Modulated Output (B_T): For a baseband signal using full-width rectangular pulses, the spectrum follows a *sinc* function. The null-to-null bandwidth of the baseband signal is R_s. When modulated onto a carrier (like in PSK), it becomes a double-sideband signal, so the total **null-to-null transmission bandwidth** is twice the symbol rate:

$$B_T = 2 \times R_s = 2 \times 250 \text{ kHz} = \mathbf{500} \text{ kHz}$$

(Note: If only the absolute minimum Nyquist bandwidth is asked without considering the side-lobes of full-width pulses, it would just be $R_s = 250$ kHz. But for practical full-width pulses, null-to-null bandwidth is standardly $2R_s$).

(ii) For 4-FSK Signal

- **Number of Levels (M):** 4
- **Bits per symbol (n):** $n = \log_2(4) = 2$ bits/symbol
- **Symbol Rate (R_s):** $R_s = \frac{R_b}{n} = \frac{10^6}{2} = 500$ kbaud
- **Frequency Separation (Δf):** 100 kHz
- **Roll-off factor (r):** 0.5

Bandwidth of Modulated Output (B_T): For an M-ary FSK signal using pulse shaping with a roll-off factor r, the total transmission bandwidth is approximated by Carson's rule extended for M-ary FSK:

$$B_T = (M-1)\Delta f + (1+r)R_s$$

Substituting the given values:

$$\begin{aligned} B_T &= (4-1) \times 100 \text{ kHz} + (1+0.5) \times 500 \text{ kHz} \\ &= 3 \times 100 \text{ kHz} + 1.5 \times 500 \text{ kHz} \\ &= 300 \text{ kHz} + 750 \text{ kHz} \\ &= \mathbf{1050} \text{ kHz or } \mathbf{1.05} \text{ MHz} \end{aligned}$$

Answer: The bandwidth of the modulated 4-FSK signal is 1.05 MHz.

Digital Communication Systems

Solutions - Selected Questions

Question 1(a) - First Image

Explain ASK, FSK and PSK modulation schemes with their applications. (12)

Solution

1. ASK (Amplitude Shift Keying):
In ASK, the amplitude of the high-frequency carrier signal is varied in accordance with the baseband digital data (0s and 1s), while the frequency and phase remain constant. For example, binary '1' is represented by a high amplitude carrier (or simply the presence of the carrier, known as On-Off Keying or OOK), and binary '0' is represented by a lower amplitude (or absence of the carrier).

- **Applications:** Used in early telephone modem standards, optical fiber communications, infrared remote controls, and RFID tags due to its simplicity.

2. FSK (Frequency Shift Keying):
In FSK, the frequency of the carrier signal is varied according to the digital baseband data, while the amplitude and phase remain constant. For instance, binary '1' is transmitted using a higher frequency (f_1, called mark frequency), and binary '0' is transmitted using a lower frequency (f_2, called space frequency).

- **Applications:** Extensively used in low-speed digital data transmission like telemetry, caller ID, paging systems, and legacy dial-up modems.

3. PSK (Phase Shift Keying):
In PSK, the phase of the carrier signal is shifted to represent the digital data, keeping both amplitude and frequency constant. In BPSK (Binary PSK), binary '1' might be transmitted with a 0° phase shift, and binary '0' with a 180° phase shift. It is highly robust against noise.

- **Applications:** The foundation of modern high-speed communications. Used in Wi-Fi (IEEE 802.11), Bluetooth, satellite communications, biometric passports, and cellular networks.

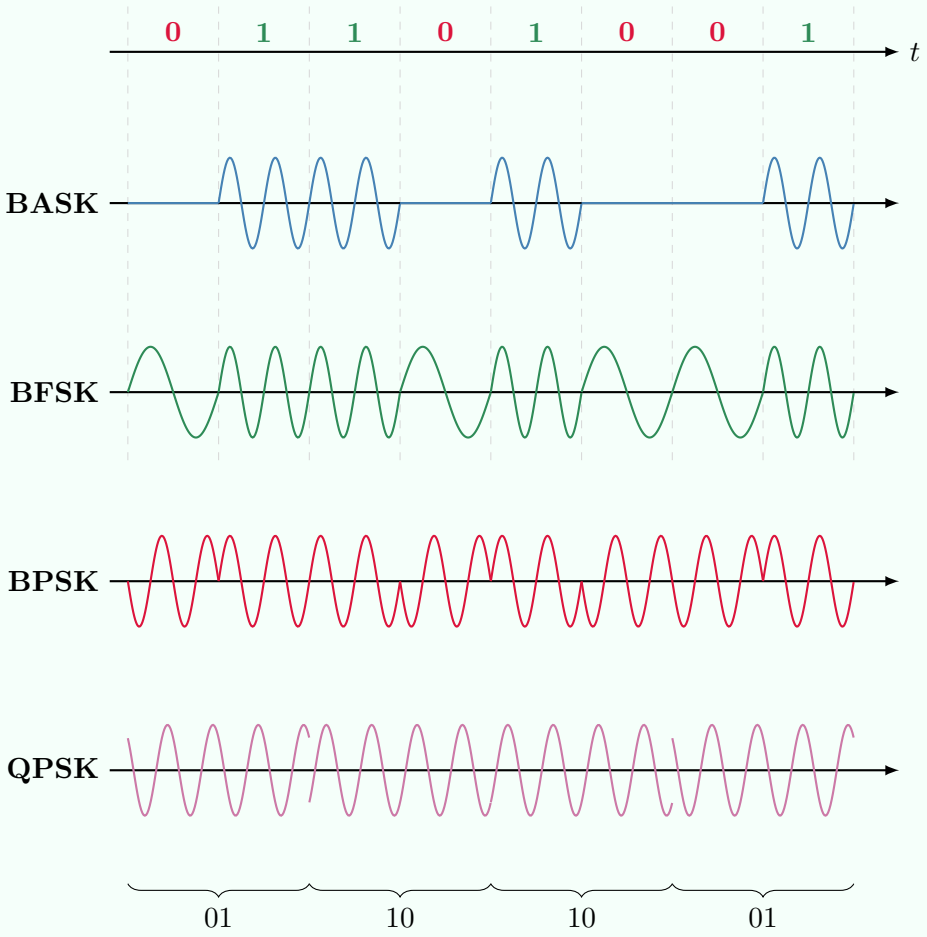
Question 1(c) - First Image

Why is coherent detection required for BPSK modulation? Draw the wave-shapes for modulated signal using BASK, BFSK, BPSK and QPSK modulation schemes to transmit a bit sequence '01101001'. (10)

Solution

1. Why Coherent Detection is required for BPSK:
In BPSK, the digital information is embedded entirely within the **phase** of the carrier signal (e.g., 0° for '1' and 180° for '0'). The amplitude remains perfectly constant. Non-coherent detectors (like envelope detectors) can only detect variations in amplitude and are completely blind to phase shifts. Therefore, to extract the phase information, the receiver must generate a local oscillator signal that is perfectly synchronized in both frequency and phase with the transmitter's carrier. This strict requirement makes **coherent detection** mandatory for BPSK.

2. Wave-shapes for the bit sequence '0 1 1 0 1 0 0 1':
For QPSK, the bits are paired (dibits). Sequence: **01, 10, 10, 01**. Each pair occupies $2T_b$ duration.

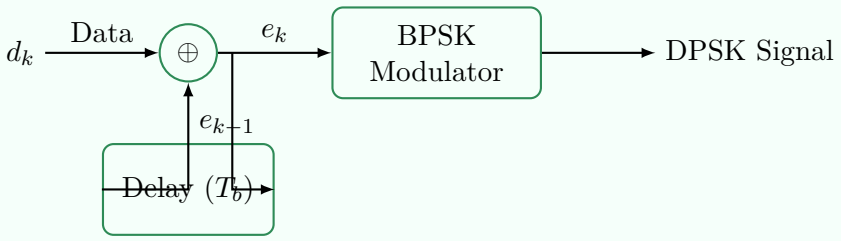


Question 2(a) - Second Image

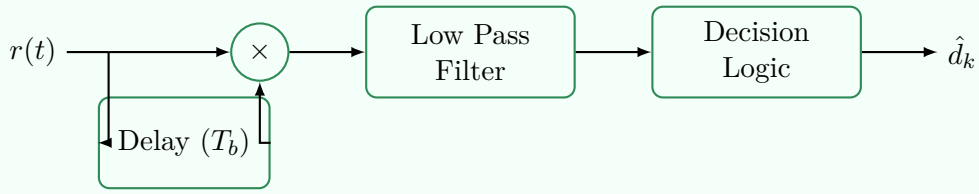
Briefly explain the generation and detection of differential PSK (DPSK) signal. (10)

Solution

1. Generation of DPSK (Transmitter):
DPSK eliminates the need for a coherent reference oscillator at the receiver by encoding information as the *difference* between successive phases rather than absolute phase. In the transmitter, the incoming binary data d_k is first differentially encoded using a logic gate (typically an XNOR or XOR gate) along with a 1-bit delay element (T_b). The encoded output sequence e_k is generated as: $e_k = d_k \oplus e_{k-1}$ (using XOR logic, meaning a phase shift occurs for '1' and no shift for '0'). This encoded bit stream e_k is then applied to a standard BPSK Modulator to produce the final DPSK signal.



2. Detection of DPSK (Receiver):
The receiver does not need a local carrier generator. Instead, it uses the received signal delayed by one bit period (T_b) as the phase reference for the current bit. The incoming signal $r(t)$ is multiplied by its own delayed version $r(t - T_b)$. If the phases are the same (representing a '0'), the multiplier output has a positive DC value. If the phases differ by 180° (representing a '1'), the output has a negative DC value. A Low Pass Filter (LPF) removes the high-frequency components, and a decision circuit (comparator) recovers the original data d_k.



Question 2(c) - Second Image

The binary sequence 01101000 is applied to a QPSK modulator. This sequence is divided in to two other sequences, consisting of odd and even numbered bits of the input sequence, which are basically representing the in-phase and quadrature components, respectively. Determine the final QPSK waveform. (13)

Solution

1. Data Splitting:
Input Sequence: **0 1 1 0 1 0 0 0** (Total 8 bits) According to the problem statement, the sequence is divided into:

- **Odd numbered bits (In-phase, I):** 1st, 3rd, 5th, 7th bits $\implies$ **0, 1, 1, 0**
- **Even numbered bits (Quadrature, Q):** 2nd, 4th, 6th, 8th bits $\implies$ **1, 0, 0, 0**

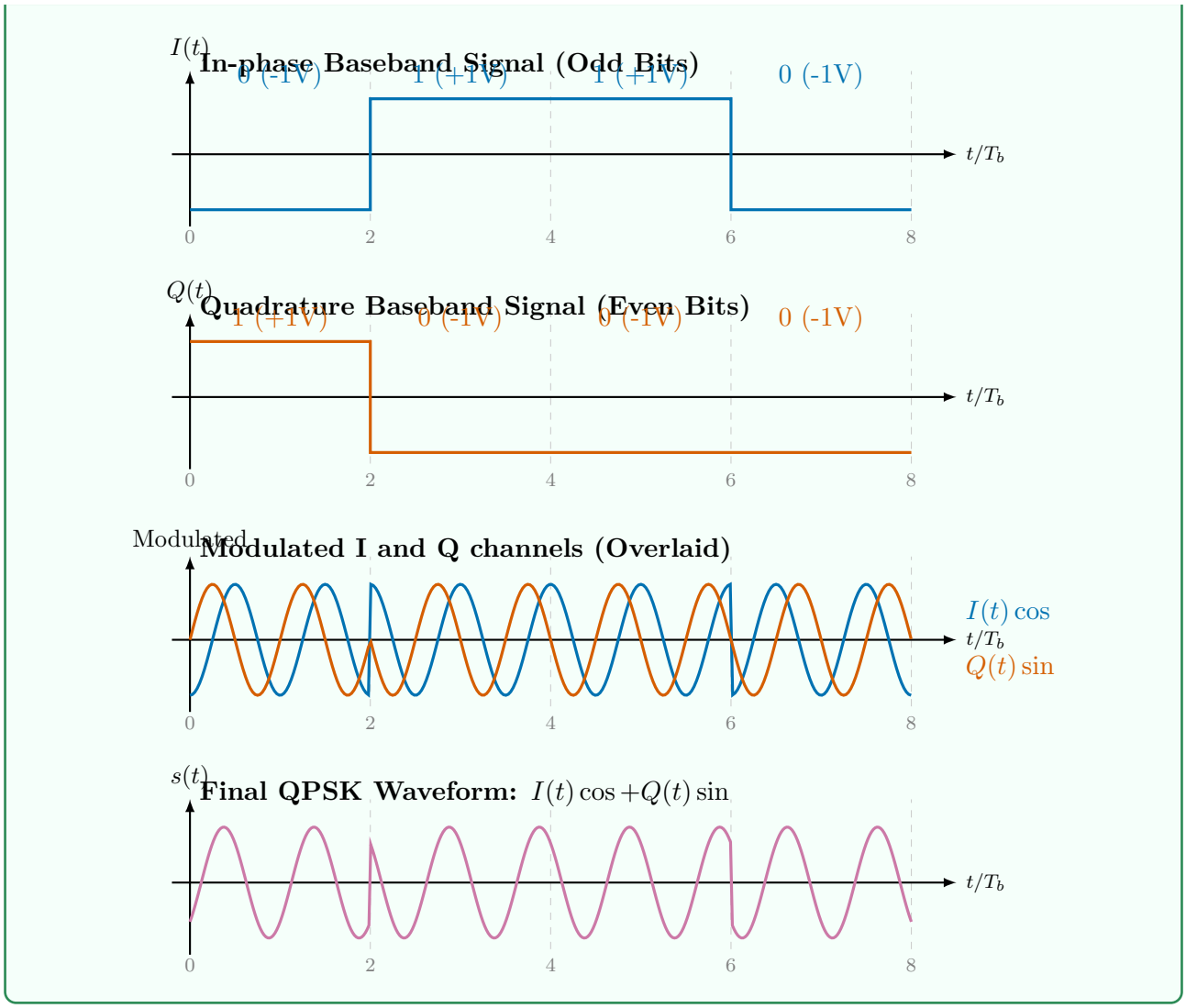
Using Polar NRZ mapping (0 → -1V, 1 → +1V):

- **I-Channel Sequence:** -1, +1, +1, -1
- **Q-Channel Sequence:** +1, -1, -1, -1

Each of these symbols in the I and Q channels will last for a duration of $2T_b$ (where T_b is the original bit duration).
2. Determination of the Final QPSK Waveform:
The final QPSK signal is the sum of the modulated I and Q components:

$$s(t) = I(t)\cos(2\pi f_c t) + Q(t)\sin(2\pi f_c t)$$

Below is the step-by-step waveform determination.


In-phase Baseband Signal (Odd Bits)
$I(t)$
0 (-1V)
1 (+1V)
1 (+1V)
0 (-1V)
t/T_b
0
2
4
6
8
Quadrature Baseband Signal (Even Bits)
$Q(t)$
1 (+1V)
0 (-1V)
0 (-1V)
0 (-1V)
t/T_b
0
2
4
6
8
Modulated I and Q channels (Overlaid)
Modulated
$I(t)\cos$
t/T_b
$Q(t)\sin$
0
2
4
6
8
Final QPSK Waveform: $I(t)\cos + Q(t)\sin$
$s(t)$
t/T_b
0
2
4
6
8

Digital Communication Systems

Solutions - Question 2 (Constellations & DPSK)

Question 2(a)

Derive the signal-space diagram of QPSK and 16-QAM signal.

Solution

1. Signal-Space Diagram of QPSK (Quadrature PSK)

In QPSK, $M = 4$. The transmitted signals can be expressed using two orthonormal basis functions:

$$\phi_1(t) = \sqrt{\frac{2}{T_s}}\cos(2\pi f_c t) \quad \text{and} \quad \phi_2(t) = -\sqrt{\frac{2}{T_s}}\sin(2\pi f_c t)$$

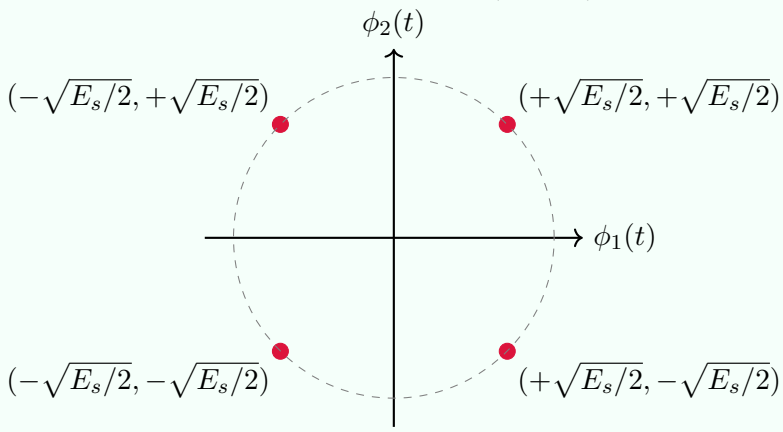
The transmitted signal for the i-th message is $s_i(t) = \sqrt{\frac{2E_s}{T_s}}\cos\left(2\pi f_c t + (2i-1)\frac{\pi}{4}\right)$ for $i = 1, 2, 3, 4$. Expanding this, the coordinates (s_{i1}, s_{i2}) in the signal space are:

$$s_{i1} = \sqrt{E_s}\cos\left((2i-1)\frac{\pi}{4}\right) = \pm\sqrt{\frac{E_s}{2}}, \quad s_{i2} = \sqrt{E_s}\sin\left((2i-1)\frac{\pi}{4}\right) = \pm\sqrt{\frac{E_s}{2}}$$

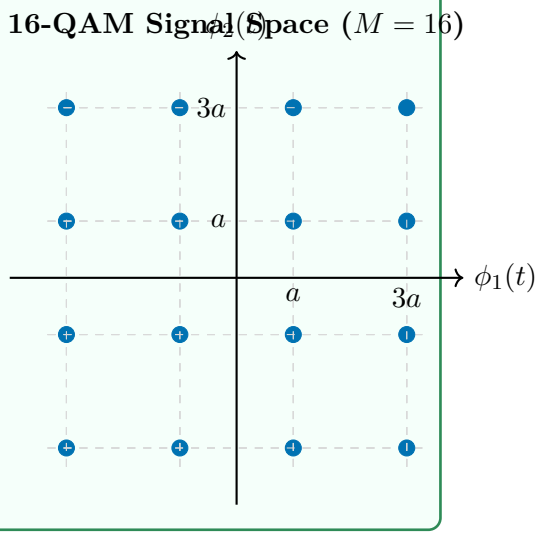
2. Signal-Space Diagram of 16-QAM

In 16-QAM, $M = 16$. Both amplitude and phase are varied. The signal points are generated by combining two independent 4-ASK signals on the orthogonal carriers $\phi_1(t)$ and $\phi_2(t)$. The coordinates are combinations of $\{-3a, -a, +a, +3a\}$ for both I (ϕ_1) and Q (ϕ_2) axes, where $2a$ is the distance between adjacent symbols.

QPSK Signal Space ($M = 4$)



16-QAM Signal Space ($M = 16$)



Question 2(b)

Describe QAM modulation process using the constellation diagram of 4-QAM.

Solution

QAM Modulation Process (using 4-QAM):

Quadrature Amplitude Modulation (QAM) is a technique where two separate carrier waves—one

cosine wave (In-phase, I) and one sine wave (Quadrature, Q)—are amplitude-modulated and then added together.
For **4-QAM**, the number of states is $M = 4$, which means each symbol carries $\log_2(4) = 2$ bits. Interestingly, the constellation of 4-QAM is mathematically equivalent to QPSK, as both have 4 points positioned in the four quadrants.
Modulation Steps:

1. **Splitting:** The incoming binary data stream is passed through a Serial-to-Parallel converter, which groups the bits into pairs (2 bits per symbol). The first bit goes to the I-channel and the second bit goes to the Q-channel.

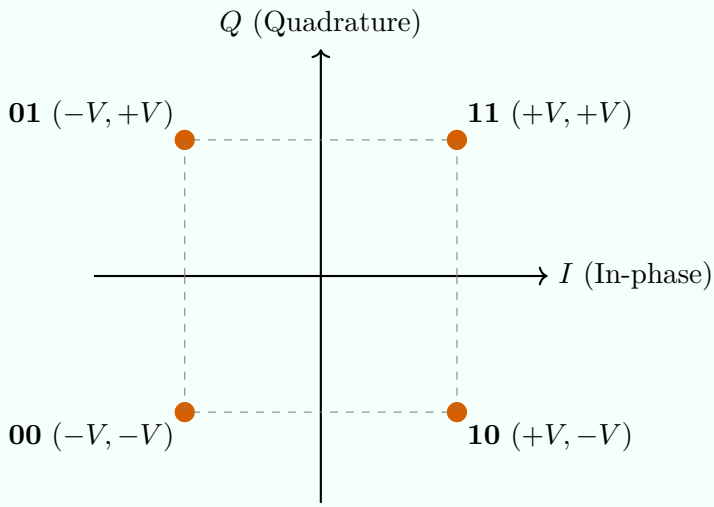
2. **Level Mapping:** Each bit is mapped to a discrete voltage level (e.g., bit '1' $\rightarrow +V$, bit '0' $\rightarrow -V$).

3. **Multiplication:**

   - The I-channel voltage multiplies the $\cos(2\pi f_c t)$ carrier.
   - The Q-channel voltage multiplies the $\sin(2\pi f_c t)$ carrier.

4. **Addition:** Both modulated components are added to produce the final 4-QAM signal.

4-QAM Constellation Diagram:
Assuming Gray coding (to minimize bit error rate), the mapping can be represented as follows:

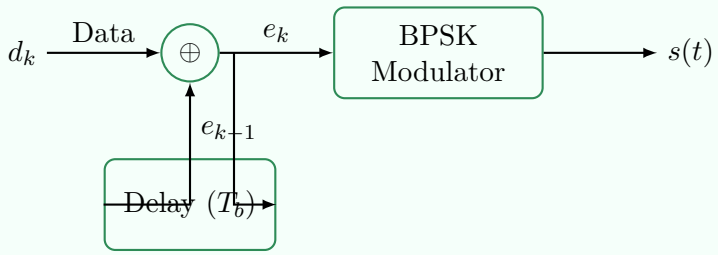


Question 2(c)

Draw the block diagram of DPSK transmitter and receiver and explain each block briefly.

Solution

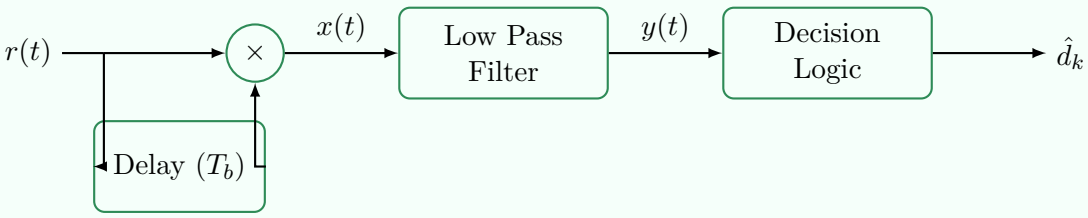
1. DPSK Transmitter



Explanation of Transmitter Blocks:

- **Logic Network (XNOR/XOR Gate):** It compares the current input data bit (d_k) with the previous encoded bit (e_{k-1}). Typically, an XNOR gate is used where '1' means no phase change and '0' means a phase change (or vice versa with XOR).
- **Delay (T_b):** This block stores the previous output bit for one bit duration (T_b) to feed it back to the logic network.
- **BPSK Modulator:** It takes the differentially encoded sequence (e_k) and modulates it using a standard BPSK scheme (shifting phase by 0° or 180°).

2. DPSK Receiver



Explanation of Receiver Blocks:

- **Delay (T_b):** Delays the received bandpass signal $r(t)$ by exactly one bit period. This acts as the "local reference" carrier for coherent detection.
- **Multiplier ($\times$):** Multiplies the current received signal with the delayed version. If the phases are the same, the output has a positive DC component. If they differ by 180°, it has a negative DC component.
- **Low Pass Filter (LPF):** Removes the high-frequency double-carrier components ($2f_c$) generated by the multiplier, leaving only the baseband signal (the DC component).
- **Decision Logic (Comparator):** Compares the LPF output against a zero threshold. Depending on whether the voltage is positive or negative, it decides if the original bit was a '1' or a '0'.

Digital Communication Systems

Solutions - Section A

Question 1(a)

Briefly explain the relative advantages and disadvantages of analog and digital communication system. (08)

Solution

1. Digital Communication Systems:

- **Advantages:**
  - **High Noise Immunity:** Digital signals (represented by discrete levels like 0 and 1) are highly resilient to channel noise and distortion compared to analog signals.
  - **Error Detection and Correction:** Advanced channel coding techniques can be applied to detect and automatically correct errors that occur during transmission.
  - **Security:** Digital data can be easily and highly encrypted, providing secure communication.
  - **Multiplexing:** Time Division Multiplexing (TDM) is much easier and more efficient to implement.
  - **Hardware:** Digital hardware is cheaper, more reliable, and allows data storage.
- **Disadvantages:**
  - **Higher Bandwidth:** Digital transmission generally requires a significantly larger bandwidth than analog transmission for the same information.
  - **Synchronization:** Strict timing and synchronization between the transmitter and receiver are absolutely necessary.
  - **Quantization Error:** Converting an analog signal to a digital format (A/D conversion) introduces permanent quantization noise.

2. Analog Communication Systems:

- **Advantages:**
  - **Lower Bandwidth:** Requires much less transmission bandwidth.
  - **Simple Circuitry:** Transmitter and receiver designs are generally simpler and historically cheaper.
  - **No Quantization:** Continuous signals do not suffer from quantization errors.
- **Disadvantages:**
  - **Poor Noise Immunity:** Any noise added to the signal directly degrades the output quality and cannot be separated easily.
  - **No Error Correction:** Once distorted, the exact original signal cannot be fully recovered.
  - **Lack of Security:** Very difficult to implement robust encryption on continuous analog waveforms.

Question 1(b)

Mention the advantages of coherent modulation scheme over non-coherent scheme. (08)

Solution

Advantages of Coherent Modulation Schemes:

1. **Superior Noise Performance:** Coherent detection provides a significantly lower Bit Error Rate (BER) for a given Signal-to-Noise Ratio (SNR) compared to non-coherent detection. It is optimal in the presence of Additive White Gaussian Noise (AWGN).

2. **Use of Phase Information:** Coherent schemes can perfectly track and utilize the phase of the carrier signal. This makes it possible to use highly bandwidth-efficient modulation techniques like PSK (Phase Shift Keying) and QAM (Quadrature Amplitude Modulation), which are impossible to detect non-coherently.

3. **Better Sensitivity:** Because it uses a synchronized local oscillator (LO) to mix down the signal, the receiver is much more sensitive and can detect weaker signals buried in noise.

4. **Rejection of Orthogonal Noise:** The correlator or multiplier in a coherent receiver effectively filters out the noise components that are orthogonal (out of phase) to the local carrier, improving the signal quality.

Note: The trade-off is that coherent systems are far more complex and expensive to build because they require carrier recovery circuits (like Phase-Locked Loops) to maintain exact phase and frequency synchronization.

Question 1(c)

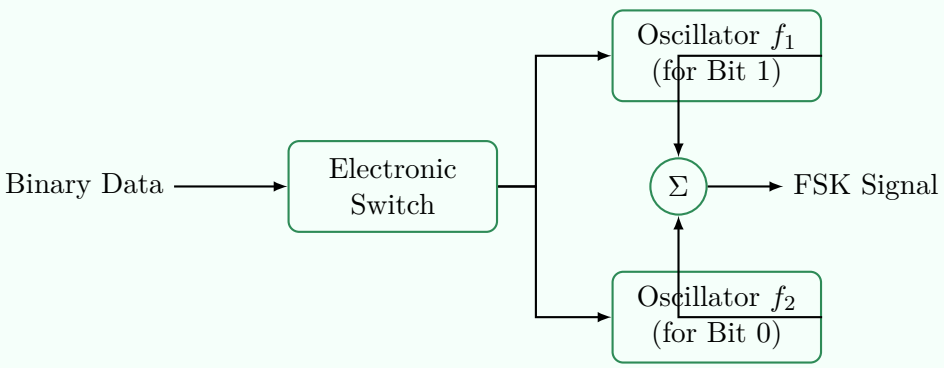
Describe the modulator and demodulator circuit of FSK modulation scheme. (14)

Solution

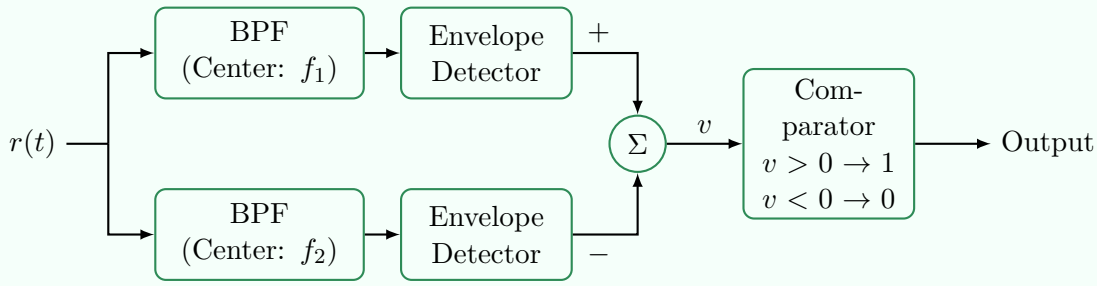
1. FSK Modulator Circuit:

Frequency Shift Keying (FSK) can be generated using a Voltage Controlled Oscillator (VCO) or by switching between two independent oscillators.

Description (Switching Method): The binary input data controls an electronic switch. When the input bit is '1', the switch connects to an oscillator generating a higher frequency f_1 (Mark frequency). When the input bit is '0', the switch connects to another oscillator generating a lower frequency f_2 (Space frequency). The outputs are combined to form the continuous FSK signal.



2. FSK Demodulator Circuit (Non-Coherent):
While FSK can be detected coherently, the non-coherent Envelope Detector method is standard and simpler.
Description: The received signal $r(t)$ is split into two parallel branches. The upper branch has a Bandpass Filter (BPF) tuned strictly to f_1, and the lower branch has a BPF tuned to f_2. If bit '1' was sent, the signal passes through the top filter, and the top Envelope Detector outputs a high voltage while the bottom one outputs near zero. A comparator subtracts the two envelope outputs. If the result is positive ($v > 0$), the decision is '1'. If negative ($v < 0$), the decision is '0'.



Question 1(d)

Why ASK is more affected by noise? (05)

Solution

Reason why ASK is highly susceptible to noise:
In any communication channel, noise (such as thermal noise or interference) is primarily an **additive amplitude disturbance**. It adds random voltage spikes to the transmitted signal.
Amplitude Shift Keying (ASK) conveys digital information purely by changing the **amplitude** (voltage level) of the carrier wave (e.g., a high amplitude for '1', and zero amplitude for '0'). Because noise directly alters the amplitude of the signal as it travels through the channel, it directly corrupts the very parameter that carries the information in ASK. A sudden spike in noise can make a '0' look like a '1', or attenuation can make a '1' look like a '0'.
In contrast, Frequency Shift Keying (FSK) and Phase Shift Keying (PSK) encode information in the frequency and phase of the carrier, respectively. These receivers can use "Amplitude Limiters" to clip off and ignore amplitude variations (noise) entirely without losing any of the encoded frequency or phase information, making them much more robust than ASK.

Digital Communication Systems

Solutions - Question 4

Question 4(a)

What is the number of bits per baud for the following techniques? (i) ASK with 8 different amplitudes (ii) FSK with 16 different frequencies (iii) PSK with different phases (iv) QAM with a constellation of 256 points.

Solution

Basic Formula:
The number of bits per baud (or bits per symbol), denoted by n, depends on the total number of distinct symbols or states (M) in the modulation scheme. The relationship is given by:

$$\mathbf{n} = \log_{\mathbf{2}}(\mathbf{M}) \text{ bits/baud}$$

(i) ASK with 8 different amplitudes
Here, $M = 8$.

$$n = \log_2(8) = \mathbf{3} \text{ bits/baud}$$

(ii) FSK with 16 different frequencies
Here, $M = 16$.

$$n = \log_2(16) = \mathbf{4} \text{ bits/baud}$$

(iii) PSK with different phases
(Note: The exact number of phases is missing in the question. Assuming a general M-ary PSK system with M different phases:)
If the system has M phases, the number of bits per baud is:

$$n = \log_{\mathbf{2}}(\mathbf{M}) \text{ bits/baud}$$

(For example, if it is QPSK with 4 phases, $n = 2$. If it is 8-PSK, $n = 3$.)
(iv) QAM with a constellation of 256 points
Here, $M = 256$.

$$n = \log_2(256) = \mathbf{8} \text{ bits/baud}$$

Question 4(b)

Draw the OQPSK signal components for the data "1100011001": (i) Modulating signal for in-phase component (ii) Modulated waveform of in-phase component (iii) Modulating signal for quadrature component (iv) Modulated waveform of quadrature component (v) OQPSK waveform.

Solution

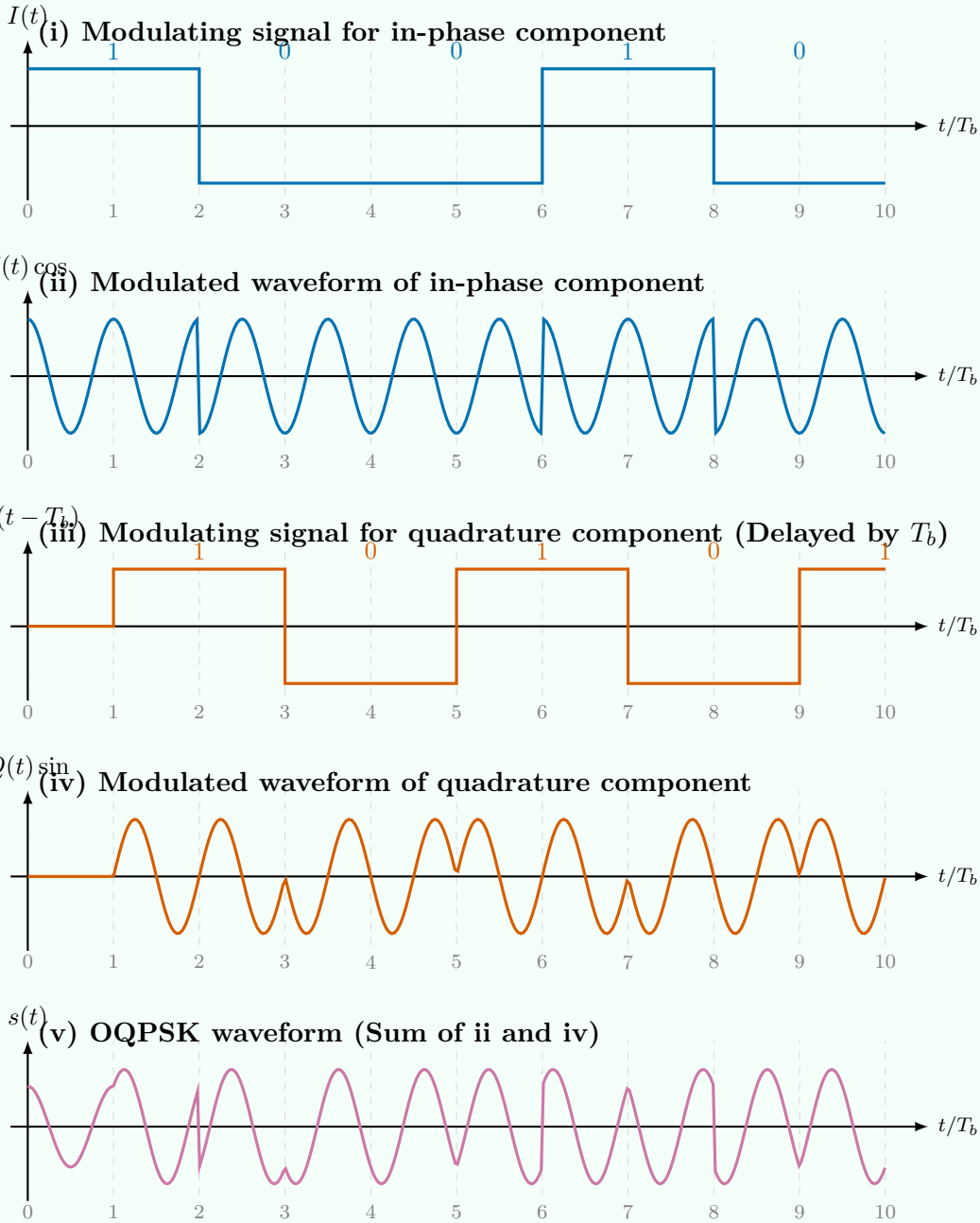
1. Data Splitting for OQPSK (Offset QPSK):
Data stream: $D = [1, 1, 0, 0, 0, 1, 1, 0, 0, 1]$. Length is 10 bits. In OQPSK, the input data is split into two streams: In-phase (I) for even bits, and Quadrature (Q) for odd bits. Each split bit stretches to a duration of $2T_b$.

- **I-channel (Even bits):** $d_0 = 1, d_2 = 0, d_4 = 0, d_6 = 1, d_8 = 0$. (Polar NRZ: $1 \to +1$, $0 \to -1$)

- **Q-channel (Odd bits):** $d_1 = 1, d_3 = 0, d_5 = 1, d_7 = 0, d_9 = 1$.

The Offset: In OQPSK, the Q-channel is delayed by exactly one bit period (T_b) relative to the I-channel. This prevents the I and Q channels from changing states at the exact same time, effectively avoiding 180° phase shifts.

2. OQPSK Waveform Plots:
For visual clarity, we use a carrier frequency $f_c = 1/T_b$ (1 cycle per bit period T_b).



Digital Communication Systems

Solutions - Question 2 (Bandwidth & BFSK)

Question 2(a)

The telephone line has 8kHz bandwidth. What is the maximum number of bits we can send using each of the following techniques? Let $d = 0$. (i) BASK (ii) QPSK (iii) 64-QAM (iv) 128-QAM (10)

Solution

Given Parameters:

- Bandwidth, $B = 8$ kHz $= 8000$ Hz.
- Roll-off factor, $d = 0$ (Ideal Nyquist filter).

For bandpass digital modulation schemes (like ASK, PSK, QAM), the transmission bandwidth is related to the symbol rate (baud rate, R_s) by the equation:

$$B = R_s(1 + d)$$

Since $d = 0$, we have $B = R_s$. Therefore, the maximum symbol rate we can transmit is:

$$R_s = 8 \text{ kbaud} = 8000 \text{ symbols/sec}$$

The maximum bit rate (data rate), R_b, depends on the number of bits per symbol, $n = \log_2(M)$, where M is the number of levels/states.

$$R_b = n \times R_s = \log_2(M) \times B$$

(i) BASK (Binary ASK):
$M = 2 \implies n = \log_2(2) = 1$ bit/symbol.

$$R_b = 1 \times 8 \text{ kbaud} = \mathbf{8} \text{ kbps}$$

(ii) QPSK (Quadrature PSK):
$M = 4 \implies n = \log_2(4) = 2$ bits/symbol.

$$R_b = 2 \times 8 \text{ kbaud} = \mathbf{16} \text{ kbps}$$

(iii) 64-QAM:
$M = 64 \implies n = \log_2(64) = 6$ bits/symbol.

$$R_b = 6 \times 8 \text{ kbaud} = \mathbf{48} \text{ kbps}$$

(iv) 128-QAM:
$M = 128 \implies n = \log_2(128) = 7$ bits/symbol.

$$R_b = 7 \times 8 \text{ kbaud} = \mathbf{56} \text{ kbps}$$

Question 2(b)

A BFSK signal with discontinuous phase is generated... (i) Show that this BFSK signal may be expressed as the sum of two independent BASK signals. (ii) For θ_1 = (last two digits of your student ID)° and θ_2 = (middle two digits of your student ID)°, plot the BFSK waveform for the binary data stream "01101001" and compare your result to continuous-phase BFSK. (15)

Solution

(i) Expressing BFSK as the sum of two independent BASK signals

Let the incoming binary data stream be represented by an unipolar switching function $d(t)$, where:

$$d(t) = \begin{cases} 1, & \text{for symbol 1} \\ 0, & \text{for symbol 0} \end{cases}$$

The inverse of this sequence can be represented as $1 - d(t)$, which is:

$$1 - d(t) = \begin{cases} 0, & \text{for symbol 1} \\ 1, & \text{for symbol 0} \end{cases}$$

The given BFSK signal $s(t)$ transmits frequency f_1 for symbol 1 and frequency f_2 for symbol 0. Using the switching function $d(t)$, we can combine the two equations into a single expression for $s(t)$:

$$s(t) = d(t)\sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_1 t + \theta_1) + [1 - d(t)]\sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_2 t + \theta_2)$$

Let's analyze the terms:

- $s_1(t) = d(t)\sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_1 t + \theta_1)$ is a **BASK (OOK)** signal at frequency f_1. It is 'ON' when data is 1 and 'OFF' when data is 0.
- $s_2(t) = [1 - d(t)]\sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_2 t + \theta_2)$ is another **BASK (OOK)** signal at frequency f_2. It is 'ON' when data is 0 and 'OFF' when data is 1.

Therefore, $s(t) = s_1(t) + s_2(t)$, which clearly shows that a discontinuous-phase BFSK signal can be expressed as the sum of two independent, non-overlapping BASK signals operating at different frequencies. (Proved).

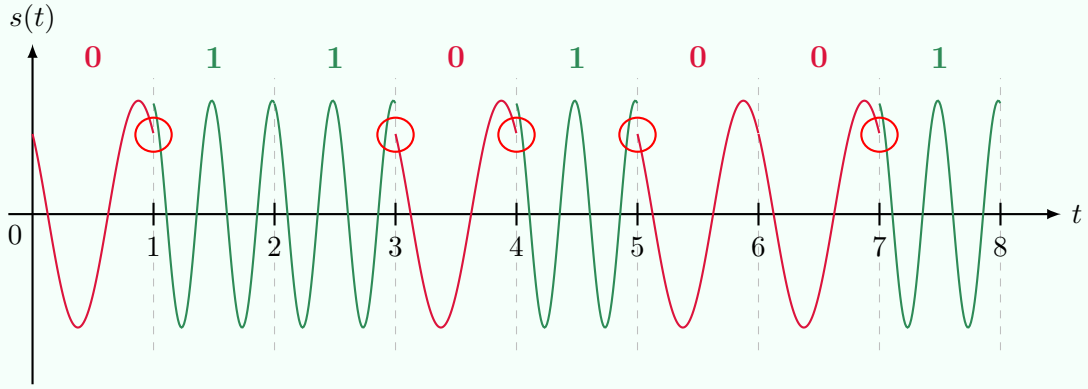
(ii) Plotting the BFSK waveform

Assumptions based on Student ID:

- Last two digits = **13** $\implies \theta_1 = 13°$
- Middle two digits (Assumed) = **45** $\implies \theta_2 = 45°$

Data Stream: **0 1 1 0 1 0 0 1**

For visualization, we will assume $f_1 = 2$ Hz (higher frequency for bit 1) and $f_2 = 1$ Hz (lower frequency for bit 0) with a bit duration of 1 s.



Comparison to Continuous-Phase BFSK (CPFSK):

- **Discontinuous Phase (Plotted Above):** Because two separate oscillators are switched abruptly, the phase of the signal does not align at the bit boundaries ($t = 1, 3, 4, 5, 7$ as circled in red). This creates sharp edges or jumps in the waveform. These sudden changes cause "spectral leakage," meaning the signal requires a much larger bandwidth and generates high-frequency noise (side-lobes).
- **Continuous-Phase BFSK (CPFSK):** In CPFSK (such as MSK), a single Voltage Controlled Oscillator (VCO) is used, ensuring that the phase transitions smoothly without any jumps at the bit boundaries. This makes the waveform visually continuous and significantly reduces the required bandwidth, making it much more spectrally efficient.

Digital Communication Systems

Solutions - Section A (New)

Question 1(a)

We have a baseband channel with a 2-MHz bandwidth. What is the data rate for this channel if we use one of the following line coding schemes? (i) NRZ-L, (ii) Manchester, (iii) Differential Manchester, (iv) NRZ-I (12)

Solution

Given: Baseband Channel Bandwidth, $B = 2$ MHz.
The relationship between the minimum required bandwidth B and the data rate R_b depends on the line coding scheme. According to Nyquist's theorem for baseband transmission, the fundamental relationship is $B = c \times R_b$, where c depends on the line coding technique.

- **(i) NRZ-L (Non-Return to Zero - Level):**
  For NRZ schemes, the minimum bandwidth (or Nyquist bandwidth) required is half the data rate, i.e., $B = \frac{R_b}{2}$.

$$R_b = 2 \times B = 2 \times 2 \text{ MHz} = \mathbf{4} \text{ Mbps}$$

- **(ii) Manchester:**
  Manchester encoding ensures a transition at the middle of every bit period. Because the maximum fundamental frequency is twice that of NRZ, the required bandwidth is equal to the data rate, i.e., $B = R_b$.

$$R_b = B = \mathbf{2} \text{ Mbps}$$

- **(iii) Differential Manchester:**
  Similar to standard Manchester encoding, Differential Manchester also has transitions in the middle of every bit. Thus, its bandwidth requirement is the same as Manchester, i.e., $B = R_b$.

$$R_b = B = \mathbf{2} \text{ Mbps}$$

- **(iv) NRZ-I (Non-Return to Zero - Invert):**
  Like NRZ-L, the NRZ-I scheme does not return to zero and has the same basic frequency characteristics. The minimum bandwidth is $B = \frac{R_b}{2}$.

$$R_b = 2 \times B = 2 \times 2 \text{ MHz} = \mathbf{4} \text{ Mbps}$$

Conclusion: For a 2-MHz channel, NRZ schemes can support up to 4 Mbps, while Manchester schemes can support up to 2 Mbps due to their higher bandwidth requirements.

Question 1(b)

The binary sequence "last two digits of your student ID" is applied to an BASK modulator. The bit duration is 1 μs, and sinusoidal carrier wave used to represent symbol 1 has a frequency equal to "last two digits of your student ID in decimal" MHz. (i) Find the transmission bandwidth of the transmitted signal. (ii) Plot the waveform of the transmitted BASK signal. (13)

Solution

Given Parameters based on Student ID:

- Last two digits of Student ID = **13**.
- Binary sequence: The decimal number 13 converted to binary is **1101**.
- Bit duration, $T_b = 1\ \mu s = 10^{-6}$ s.
- Carrier Frequency, $f_c = \mathbf{13}$ **MHz**.

From the bit duration, we can calculate the data rate (bit rate), R_b:

$$R_b = \frac{1}{T_b} = \frac{1}{10^{-6}} = 10^6 \text{ bps} = \mathbf{1} \text{ Mbps}$$

(i) Transmission Bandwidth

For a Binary Amplitude Shift Keying (BASK / OOK) signal, the transmission bandwidth (B_T) is twice the baseband signal bandwidth. Assuming the first null bandwidth is used for the baseband signal ($B_{baseband} = R_b$):

$$B_T = 2 \times B_{baseband} = 2 \times R_b$$

$$B_T = 2 \times 1 \text{ Mbps} = \mathbf{2} \text{ MHz}$$

(Note: Depending on the specific roll-off factor r of the pulse shaping filter, the bandwidth is generally expressed as $B_T = (1 + r)R_b$. Without any filter specified, the fundamental required bandwidth is $2R_b$).

(ii) Plot of the transmitted BASK signal waveform

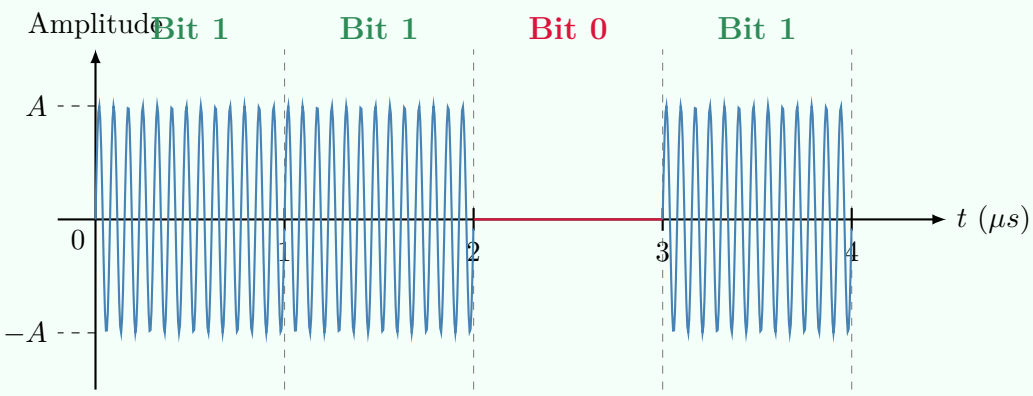
The binary sequence is **1 1 0 1**. For BASK (also called On-Off Keying):

- For bit '1': The carrier $A\cos(2\pi f_c t)$ is transmitted.
- For bit '0': No signal (0 amplitude) is transmitted.

The number of carrier cycles per bit duration is:

$$\text{Cycles per bit} = f_c \times T_b = 13 \times 10^6 \times 10^{-6} = \mathbf{13} \text{ cycles}$$

Below is the plotted waveform. For bits '1', exactly 13 cycles of the carrier wave are shown per bit interval (1 μs).



Note: In the plot above, the high-frequency sine waves accurately represent $f_c = 13$ MHz, showing exactly 13 full sinusoidal cycles within each 1 μs interval for the bit '1'.

Digital Communication Systems

Solutions - Question 2

Question 2(a)

Mention the advantages of M-ary modulation scheme over the binary modulation. (08)

Solution

Advantages of M-ary Modulation over Binary Modulation:

1. **Higher Bandwidth Efficiency:** In M-ary modulation, each symbol represents $n = \log_2(M)$ bits. This allows the transmission of multiple bits per signal element, significantly increasing the bit rate for a given bandwidth compared to binary modulation (where 1 symbol = 1 bit).

2. **Reduced Bandwidth Requirement:** For a fixed data rate (R_b), the symbol rate or baud rate (R_s) is reduced by a factor of $\log_2(M)$ ($R_s = R_b/\log_2(M)$). This means the required transmission bandwidth is proportionately reduced (especially in M-ary PSK and QAM).

3. **Better Power Efficiency (in M-ary FSK):** While M-ary PSK/QAM improves bandwidth efficiency at the cost of power, orthogonal M-ary FSK improves power efficiency (requires less E_b/N_0 for a given Bit Error Rate) at the expense of requiring more bandwidth.

4. **High-Speed Data Transmission:** It makes technologies like 4G, 5G, Wi-Fi, and digital microwave links feasible, where transmitting high-speed data over a limited frequency spectrum is crucial.

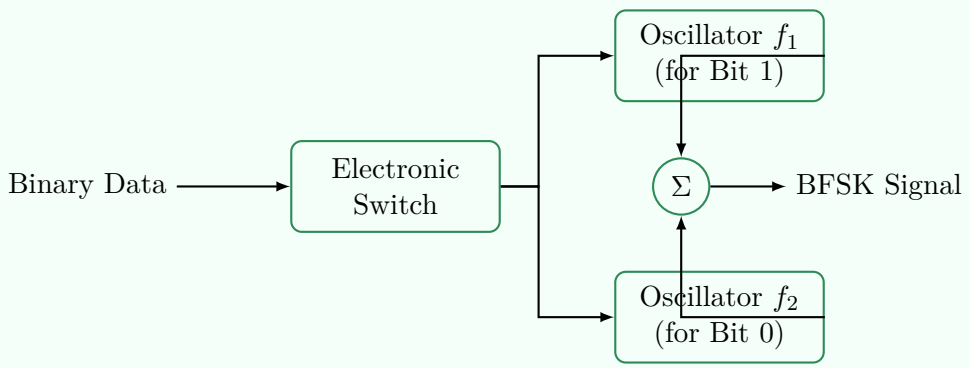
Question 2(b)

Explain the generation and both the coherent & non-coherent detection system of BFSK modulation scheme. (15)

Solution

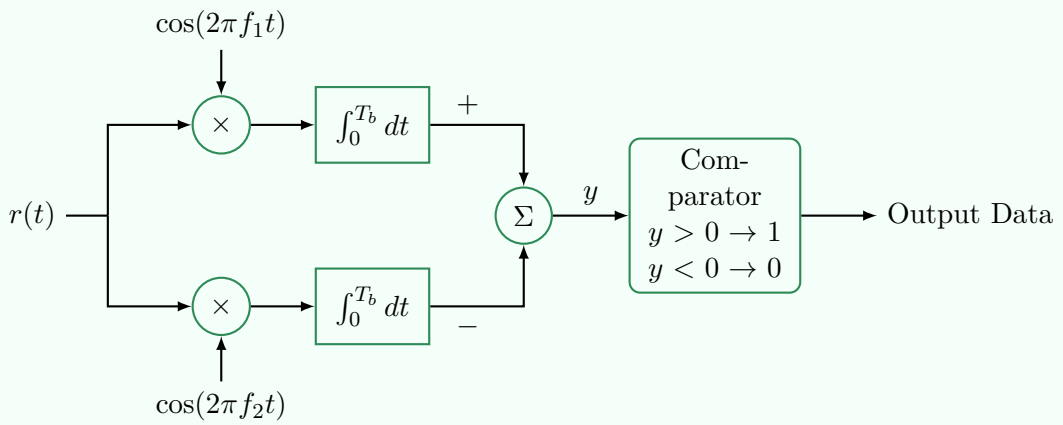
1. Generation of BFSK Signal

A Binary Frequency Shift Keying (BFSK) signal uses two different frequencies, f_1 and f_2, to represent binary '1' and '0' respectively. It can be generated by using the input binary data to switch between two independent oscillators, or by using a Voltage Controlled Oscillator (VCO).



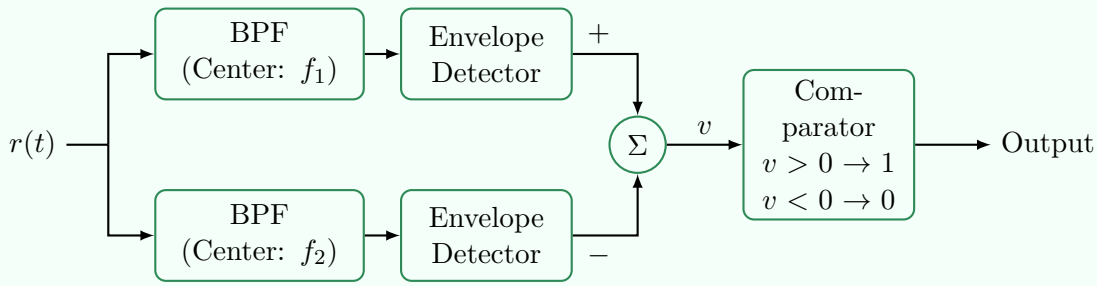
2. Coherent Detection of BFSK

In coherent detection, the receiver uses local oscillators that are exactly synchronized in both frequency and phase with the transmitter. The received signal is correlated with both f_1 and f_2 separately, integrated, and then compared.



3. Non-Coherent Detection of BFSK

Non-coherent detection does not require phase synchronization. It uses two Bandpass Filters (BPF) tuned to f_1 and f_2, followed by Envelope Detectors to measure the signal amplitude in each branch. The larger envelope determines the bit.



Question 2(c)

Bangladesh Satellite Company Limited (BSCL) uses an AWGN channel with 6 kHz bandwidth and the noise power spectral density $n_0/2 = 10^{-12}$ W/Hz. The signal power is required at the receiver is 0.2 mW. Find the capacity of the channel. If the above channel is to be used to transmit 100 kbps of data determine the minimum BW required of the channel of same noise power spectral density characteristics. (12)

Solution

Part 1: Finding the Capacity of the Channel

Given parameters:

- Bandwidth, $B = 6$ kHz $= 6 \times 10^3$ Hz
- Noise Power Spectral Density, $\frac{N_0}{2} = 10^{-12}$ W/Hz $\implies N_0 = 2 \times 10^{-12}$ W/Hz
- Signal Power, $S = 0.2$ mW $= 0.2 \times 10^{-3}$ W $= 2 \times 10^{-4}$ W

Total Noise Power, $N = N_0 \times B$:

$$N = (2 \times 10^{-12}) \times (6 \times 10^3) = 12 \times 10^{-9} \text{ W}$$

Signal-to-Noise Ratio (SNR):

$$\text{SNR} = \frac{S}{N} = \frac{2 \times 10^{-4}}{12 \times 10^{-9}} = \frac{10^5}{6} \approx 16666.67$$

Shannon-Hartley Channel Capacity (C):

$$\begin{aligned}
C &= B \log_2 \left(1 + \frac{S}{N}\right) \\
C &= 6000 \times \log_2(1 + 16666.67) \\
C &= 6000 \times \log_2(16667.67) \\
C &\approx 6000 \times 14.0247 \\
C &\approx 84,148.3 \text{ bps} \approx \mathbf{84.15} \text{ kbps}
\end{aligned}$$

The capacity of the channel is approx 84.15 kbps.

Part 2: Finding the Minimum Bandwidth for 100 kbps
We want a new capacity $C_{new} = 100$ kbps $= 100,000$ bps. Assuming the signal power (S) and noise PSD (N_0) remain the same. We need to find the new bandwidth B_{new}.

$$C_{new} = B_{new} \log_2 \left(1 + \frac{S}{N_0 \cdot B_{new}}\right)$$

$$100000 = B_{new} \log_2 \left(1 + \frac{2 \times 10^{-4}}{2 \times 10^{-12} \cdot B_{new}}\right)$$

$$100000 = B_{new} \log_2 \left(1 + \frac{10^8}{B_{new}}\right)$$

This is a transcendental equation and cannot be solved algebraically. We can solve it using numerical methods or trial and error: Let $f(B) = B \log_2 \left(1 + \frac{10^8}{B}\right)$

- If $B = 7000$: $f(7000) = 7000 \log_2(1 + 14285.7) = 7000 \times 13.802 = 96,614$ bps
- If $B = 7500$: $f(7500) = 7500 \log_2(1 + 13333.3) = 7500 \times 13.702 = 102,765$ bps
- If $B = 7275$: $f(7275) = 7275 \log_2(1 + 13745.7) = 7275 \times 13.746 = 99,998$ bps $\approx 100,000$ bps

The minimum required bandwidth is approximately 7275 **Hz or 7.275 kHz.**

Digital Communication Systems

Solutions - Section A (Question 1)

Question 1(a)

Briefly explain the relative advantages and disadvantages of analog and digital communication systems. (10)

Solution

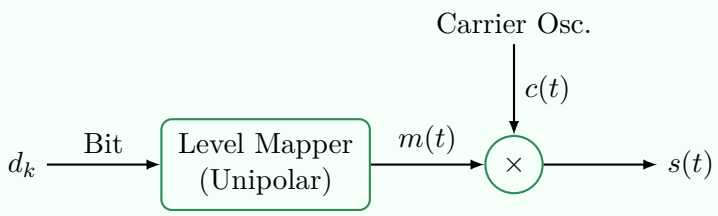
1. **Digital Communication Systems:**
   - **Advantages:**
     - **Noise Immunity:** Digital signals are highly robust against channel noise and interference because receivers only need to distinguish between discrete levels (e.g., 0 and 1), not exact continuous values.
     - **Error Detection & Correction:** Channel coding techniques can be applied to detect and correct errors introduced during transmission.
     - **Security:** Digital data can be easily encrypted, ensuring high privacy and security.
     - **Multiplexing:** Time Division Multiplexing (TDM) is easier and more efficient to implement with digital signals than Frequency Division Multiplexing (FDM) in analog.
     - **Hardware:** Digital hardware (VLSI technology) is cheaper, more reliable, and allows for data storage.
   - **Disadvantages:**
     - **High Bandwidth:** Transmitting digital pulses requires significantly more bandwidth than the original analog baseband signal.
     - **Synchronization:** Strict timing and synchronization between the transmitter and receiver are mandatory, adding complexity to the system.
     - **Quantization Error:** The process of converting an analog signal to digital (A/D conversion) introduces permanent quantization noise.
2. **Analog Communication Systems:**
   - **Advantages:**
     - **Lower Bandwidth:** Requires much less bandwidth compared to digital systems for transmitting the same information.
     - **Simpler Circuits:** Transmitter and receiver designs (e.g., AM/FM radios) are relatively simple and inexpensive.
     - **No Quantization Noise:** Since the signal is continuous, there is no error introduced by sampling or quantization.
   - **Disadvantages:**
     - **Poor Noise Immunity:** Any noise added to the signal during transmission directly degrades the signal quality and cannot be easily separated.
     - **No Error Correction:** Once the signal is distorted, it is almost impossible to perfectly reconstruct the original signal.
     - **Security limitations:** Analog signals are extremely difficult to encrypt effectively.

Question 1(b)

Consider an ASK-based digital communication system that transmits two different signals, $A_0 \cos(\omega_c t)$ for bit 0 and $A_1 \cos(\omega_c t)$ for bit 1. Draw a block diagram of the transmitter and receiver of the system, and provide the expressions for the signals at each point of the transmitter and receiver. (13)

Solution

1. Transmitter Block Diagram and Expressions:



Expressions at Transmitter:

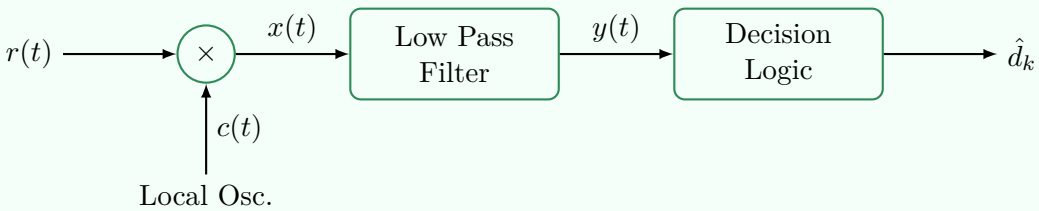
- **Input:** Binary data sequence $d_k \in \{0, 1\}$.
- **Baseband Signal $m(t)$:** The Level Mapper converts the binary bit to a discrete amplitude level.

$$m(t) = \begin{cases} A_0, & \text{if bit is 0} \\ A_1, & \text{if bit is 1} \end{cases}$$

- **Carrier Signal $c(t)$:** $c(t) = \cos(\omega_c t)$
- **Transmitted Signal $s(t)$:**

$$s(t) = m(t) \cdot c(t) = \begin{cases} A_0 \cos(\omega_c t), & \text{for bit 0} \\ A_1 \cos(\omega_c t), & \text{for bit 1} \end{cases}$$

2. Receiver Block Diagram and Expressions (Coherent Detection):



Expressions at Receiver:

- **Received Signal $r(t)$:** Assuming ideal channel (no noise for simplicity of expression), $r(t) = s(t) = m(t)\cos(\omega_c t)$.
- **Local Carrier:** Synchronized local oscillator generates $c(t) = \cos(\omega_c t)$.
- **Multiplier Output $x(t)$:**

$$\begin{aligned} x(t) &= r(t) \cdot \cos(\omega_c t) = m(t)\cos^2(\omega_c t) \\ &= \frac{m(t)}{2}[1 + \cos(2\omega_c t)] = \frac{m(t)}{2} + \frac{m(t)}{2}\cos(2\omega_c t) \end{aligned}$$

- **LPF Output $y(t)$:** The Low Pass Filter removes the high-frequency $2\omega_c$ component.

$$y(t) = \frac{m(t)}{2} = \begin{cases} \frac{A_0}{2}, & \text{for bit 0} \\ \frac{A_1}{2}, & \text{for bit 1} \end{cases}$$

- **Decision Logic Output $\hat{d}_k$:** The comparator compares $y(t)$ with a threshold $V_{th} = \frac{(A_0/2+A_1/2)}{2} = \frac{A_0+A_1}{4}$.

$$\hat{d}_k = \begin{cases} 1, & \text{if } y(t) > V_{th} \\ 0, & \text{if } y(t) < V_{th} \end{cases}$$

(Note: An Envelope Detector can also be used non-coherently, which directly extracts $m(t)$ from the envelope of $r(t)$ and compares it with a threshold of $(A_0 + A_1)/2$.)

Question 1(c)

What is meant by MSK modulation scheme? Demonstrate that the minimum tone spacing for noncoherently detected orthogonal FSK signaling, where symbols are equiprobable, is given by: $f_1 - f_2 = \frac{1}{T}$ Where symbols have their usual meaning. (12)

Solution

1. What is MSK Modulation Scheme?

MSK (Minimum Shift Keying) is a special type of Continuous Phase Frequency Shift Keying (CPFSK). In MSK, the modulation index is precisely $h = 0.5$. It is called "Minimum" Shift Keying because it uses the minimum possible frequency spacing ($\Delta f = \frac{1}{2T}$) that allows the two FSK signals to be coherently orthogonal. The most significant feature of MSK is that there are no abrupt phase changes at the symbol boundaries (phase is continuous), which results in a highly compact power spectrum with very low sidelobes, conserving bandwidth.

2. Proof of Minimum Tone Spacing for Non-coherent Orthogonal FSK:

Let the two FSK signals for a symbol duration T be represented as:

$$s_1(t) = A\cos(2\pi f_1 t + \phi_1)$$
$$s_2(t) = A\cos(2\pi f_2 t + \phi_2)$$

where ϕ_1 and ϕ_2 are arbitrary, unknown carrier phases (which is the case for non-coherent detection).

For these two signals to be **orthogonal for non-coherent detection**, their cross-correlation must be zero regardless of the values of ϕ_1 and ϕ_2. This condition is mathematically equivalent to stating that the inner product of their complex equivalent baseband signals over the period T must be zero.

Let the complex exponential equivalents of the signals be: $u_1(t) = e^{j2\pi f_1 t}$ and $u_2(t) = e^{j2\pi f_2 t}$.

The orthogonality condition requires:

$$\int_0^T u_1(t)u_2^*(t)dt = 0$$

Substituting the expressions:

$$\int_0^T e^{j2\pi f_1 t} \cdot e^{-j2\pi f_2 t} dt = 0$$

$$\int_0^T e^{j2\pi(f_1-f_2)t}dt = 0$$

Now, let's evaluate this integral. Let $\Delta f = f_1 - f_2$:

$$\left[\frac{e^{j2\pi\Delta ft}}{j2\pi\Delta f}\right]_0^T = 0$$

$$\frac{e^{j2\pi\Delta fT} - 1}{j2\pi\Delta f} = 0$$

For this fraction to be zero, the numerator must be zero (and the denominator non-zero):

$$e^{j2\pi\Delta fT} = 1$$

From Euler's identity, we know that $e^{j\theta} = 1$ is true only when θ is an integer multiple of 2π. Therefore:

$$2\pi\Delta fT = n(2\pi), \quad \text{where } n = 1, 2, 3, \ldots$$

$$\Delta fT = n$$

$$\Delta f = f_1 - f_2 = \frac{n}{T}$$

To find the **minimum** tone spacing, we must take the smallest non-zero positive integer value for n, which is $n = 1$.
Therefore, the minimum tone spacing for noncoherently detected orthogonal FSK is:

$$\mathbf{f_1 - f_2 = \frac{1}{T}}$$

(Proved).

Digital Communication Systems

Solutions - Question 2

Question 2(a)

Describe the process of generating signals in an 8-PSK modulation scheme. (12)

Solution

Process of Generating an 8-PSK Signal:
In 8-PSK (Phase Shift Keying), the phase of the carrier signal can take one of 8 distinct values, meaning each symbol represents $\log_2(8) = 3$ bits.

1. Serial-to-Parallel Conversion: The incoming binary data stream (R_b bps) is passed through a serial-to-parallel converter. This divides the data into groups of 3 bits (Tribits). The symbol rate becomes $R_s = R_b/3$. Let the 3 bits be b_1, b_2, b_3.

2. Signal Mapping (Level Generation): The 3-bit sequences are mapped to specific discrete voltage levels for the In-phase (I) and Quadrature (Q) channels. The phases are given by $\theta_i = \frac{2\pi i}{8}$ for $i = 0, 1, \ldots, 7$. The amplitudes are proportional to:

$$I\text{-level} = \cos(\theta_i), \quad Q\text{-level} = \sin(\theta_i)$$

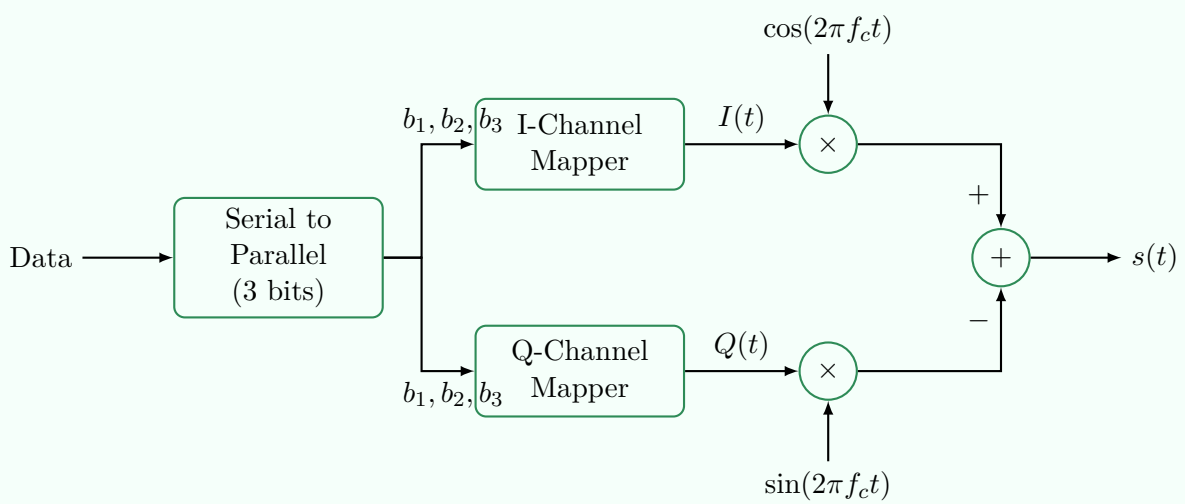
3. Modulation (Multiplication):

- The I-level signal is multiplied by a cosine carrier wave, $\cos(2\pi f_c t)$.
- The Q-level signal is multiplied by a sine carrier wave (which is orthogonal), $\sin(2\pi f_c t)$.

4. Summation: Finally, the modulated I and Q signals are added or subtracted together in a linear summer to produce the final 8-PSK bandpass signal $s(t)$.

$$s(t) = \sqrt{\frac{2E_s}{T_s}} \cos(2\pi f_c t + \theta_i) = \underbrace{I \cdot \cos(2\pi f_c t)}_{\text{In-phase}} - \underbrace{Q \cdot \sin(2\pi f_c t)}_{\text{Quadrature}}$$

Block Diagram:



Question 2(b)

Derive the expression for the average probability of bit error for a coherent binary BPSK transmission system. (13)

Solution

Derivation of Average Probability of Bit Error for BPSK:

1. Signal Representation:

In BPSK, the transmitted signals for bit '1' and bit '0' are:

$$s_1(t) = \sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_c t) \quad \text{for bit 1}$$
$$s_2(t) = -\sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_c t) \quad \text{for bit 0}$$

Where E_b is the bit energy and T_b is the bit duration. The basis function is $\phi_1(t) = \sqrt{\frac{2}{T_b}}\cos(2\pi f_c t)$. Thus, the signal points in the constellation diagram are $s_1 = +\sqrt{E_b}$ and $s_2 = -\sqrt{E_b}$.

2. Received Signal and Noise:

The received signal is $r(t) = s_i(t) + n(t)$, where $n(t)$ is Additive White Gaussian Noise (AWGN) with zero mean and a power spectral density of $N_0/2$. The output of the correlator is $r = s_i + n$, where the noise component n is a Gaussian random variable with zero mean and variance $\sigma^2 = N_0/2$.

3. Decision Rule and Error Probability:

The optimum decision threshold for equiprobable bits is exactly in the middle of s_1 and s_2, which is 0.

- Error given '1' is sent ($P(e|1)$): Happens if $r < 0$.
- Error given '0' is sent ($P(e|0)$): Happens if $r > 0$.

Due to symmetry, $P(e|1) = P(e|0)$. The average probability of error P_e is:

$$P_e = \frac{1}{2}P(e|1) + \frac{1}{2}P(e|0) = P(e|1)$$

4. Calculating $P(e|1)$:

Given s_1 is sent, r is a Gaussian variable with mean $+\sqrt{E_b}$ and variance $N_0/2$. The probability density function (PDF) is:

$$f(r|s_1) = \frac{1}{\sqrt{\pi N_0}}\exp\left(-\frac{(r-\sqrt{E_b})^2}{N_0}\right)$$

The probability of error is the area under the tail from $-\infty$ to 0:

$$P_e = \int_{-\infty}^{0}\frac{1}{\sqrt{\pi N_0}}\exp\left(-\frac{(r-\sqrt{E_b})^2}{N_0}\right)dr$$

Let us change variables: $z = \frac{\sqrt{E_b}-r}{\sqrt{N_0/2}}$. When $r = 0$, $z = \frac{\sqrt{E_b}}{\sqrt{N_0/2}} = \sqrt{\frac{2E_b}{N_0}}$. When $r \to -\infty$, $z \to \infty$. $dr = -\sqrt{N_0/2}dz$.

$$P_e = \int_{\sqrt{2E_b/N_0}}^{\infty}\frac{1}{\sqrt{\pi N_0}}\exp(-z^2)(\sqrt{N_0/2})dz$$

$$P_e = \frac{1}{\sqrt{2\pi}} \int_{\sqrt{2E_b/N_0}}^{\infty} \exp\left(-\frac{z^2}{2}\right) dz$$

This integral is exactly the definition of the standard Q-function, $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-z^2/2} dz$. Therefore, the final expression for the probability of error is:

$$\mathbf{P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)}$$

(Alternatively, using the complementary error function: $P_e = \frac{1}{2} erfc\left(\sqrt{\frac{E_b}{N_0}}\right)$)

Question 2(c)

Assume that a 64-QAM signal is used to send data at a rate of 100 Mbits/s over a satellite transponder. The transponder has a bandwidth of 30 MHz. If the satellite signal is equalized to have an equivalent raised cosine filter characteristic, what is the rolloff factor r required? Could a rolloff factor r be found so that a 180 Mbit/s data rate could be supported? (10)

Solution

Part 1: Finding the required rolloff factor r for 100 Mbits/s

- Data Rate, $R_b = 100$ Mbits/s $= 100 \times 10^6$ bps
- Modulation: 64-QAM. Number of levels $M = 64$.
- Bits per symbol, $n = \log_2(M) = \log_2(64) = 6$ bits/symbol
- Available Bandwidth, $B = 30$ MHz $= 30 \times 10^6$ Hz

First, we calculate the symbol rate (Baud rate), R_s:

$$R_s = \frac{R_b}{n} = \frac{100 \text{ Mbits/s}}{6 \text{ bits/symbol}} = 16.667 \text{ Mbaud}$$

The transmission bandwidth for a signal using a raised cosine filter (for double sideband modulation like QAM) is given by:

$$B = R_s(1 + r)$$

Where r is the roll-off factor. We need to find r:

$$\begin{aligned} 30 \text{ MHz} &= 16.667 \text{ Mbaud} \times (1 + r) \\ 1 + r &= \frac{30}{16.667} \\ 1 + r &= 1.8 \\ r &= 1.8 - 1 = \mathbf{0.8} \end{aligned}$$

The required roll-off factor is 0.8 **(or 80%).**

Part 2: Could a 180 Mbit/s data rate be supported?

Let's check if $R_b = 180$ Mbits/s is possible. New symbol rate, $R_{s(\text{new})} = \frac{180 \text{ Mbits/s}}{6} = 30$ Mbaud.

Using the bandwidth equation again:

$$
\begin{aligned}
B &= R_{s(\text{new})}(1+r) \\
30 \text{ MHz} &= 30 \text{ Mbaud} \times (1+r) \\
1+r &= \frac{30}{30} = 1 \\
r &= \mathbf{0}
\end{aligned}
$$

Conclusion: Yes, mathematically, a data rate of 180 Mbits/s can be supported if the roll-off factor is exactly $r = 0$. This corresponds to an ideal Nyquist (brick-wall) filter. However, practically, designing a filter with $r = 0$ is impossible because it requires infinite time delay and is highly sensitive to timing jitter. So, while theoretically possible, it is not practically feasible.

Digital Communication Systems

Solutions - Question 2

Question 2(a)

Describe the process of generating signals in an 8-PSK modulation scheme. (12)

Solution

Process of Generating an 8-PSK Signal:
In 8-PSK (Phase Shift Keying), the phase of the carrier signal can take one of 8 distinct values, meaning each symbol represents $\log_2(8) = 3$ bits.

1. Serial-to-Parallel Conversion: The incoming binary data stream (R_b bps) is passed through a serial-to-parallel converter. This divides the data into groups of 3 bits (Tribits). The symbol rate becomes $R_s = R_b/3$. Let the 3 bits be b_1, b_2, b_3.

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4. Summation: Finally, the modulated I and Q signals are added or subtracted together in a linear summer to produce the final 8-PSK bandpass signal $s(t)$.

$$s(t) = \sqrt{\frac{2E_s}{T_s}} \cos(2\pi f_c t + \theta_i) = \underbrace{I \cdot \cos(2\pi f_c t)}_{\text{In-phase}} - \underbrace{Q \cdot \sin(2\pi f_c t)}_{\text{Quadrature}}$$

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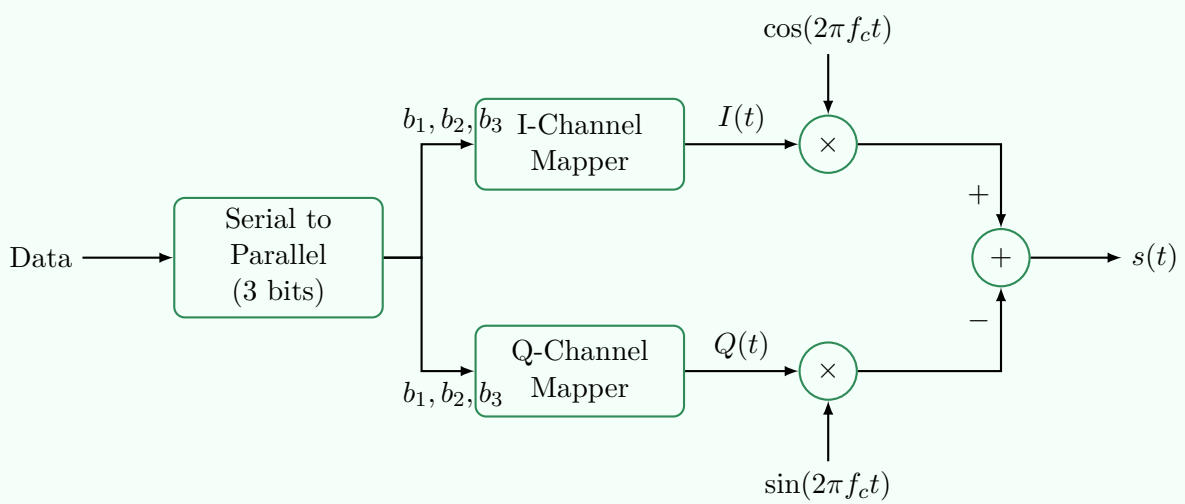


Question 2(b)

Derive the expression for the average probability of bit error for a coherent binary BPSK transmission system. (13)

Solution

Derivation of Average Probability of Bit Error for BPSK:

1. Signal Representation:

In BPSK, the transmitted signals for bit '1' and bit '0' are:

$$s_1(t) = \sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_c t) \quad \text{for bit 1}$$
$$s_2(t) = -\sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_c t) \quad \text{for bit 0}$$

Where E_b is the bit energy and T_b is the bit duration. The basis function is $\phi_1(t) = \sqrt{\frac{2}{T_b}}\cos(2\pi f_c t)$. Thus, the signal points in the constellation diagram are $s_1 = +\sqrt{E_b}$ and $s_2 = -\sqrt{E_b}$.

2. Received Signal and Noise:

The received signal is $r(t) = s_i(t) + n(t)$, where $n(t)$ is Additive White Gaussian Noise (AWGN) with zero mean and a power spectral density of $N_0/2$. The output of the correlator is $r = s_i + n$, where the noise component n is a Gaussian random variable with zero mean and variance $\sigma^2 = N_0/2$.

3. Decision Rule and Error Probability:

The optimum decision threshold for equiprobable bits is exactly in the middle of s_1 and s_2, which is 0.

- Error given '1' is sent ($P(e|1)$): Happens if $r < 0$.
- Error given '0' is sent ($P(e|0)$): Happens if $r > 0$.

Due to symmetry, $P(e|1) = P(e|0)$. The average probability of error P_e is:

$$P_e = \frac{1}{2}P(e|1) + \frac{1}{2}P(e|0) = P(e|1)$$

4. Calculating $P(e|1)$:

Given s_1 is sent, r is a Gaussian variable with mean $+\sqrt{E_b}$ and variance $N_0/2$. The probability density function (PDF) is:

$$f(r|s_1) = \frac{1}{\sqrt{\pi N_0}}\exp\left(-\frac{(r-\sqrt{E_b})^2}{N_0}\right)$$

The probability of error is the area under the tail from $-\infty$ to 0:

$$P_e = \int_{-\infty}^{0}\frac{1}{\sqrt{\pi N_0}}\exp\left(-\frac{(r-\sqrt{E_b})^2}{N_0}\right)dr$$

Let us change variables: $z = \frac{\sqrt{E_b}-r}{\sqrt{N_0/2}}$. When $r = 0$, $z = \frac{\sqrt{E_b}}{\sqrt{N_0/2}} = \sqrt{\frac{2E_b}{N_0}}$. When $r \to -\infty$, $z \to \infty$. $dr = -\sqrt{N_0/2}dz$.

$$P_e = \int_{\sqrt{2E_b/N_0}}^{\infty}\frac{1}{\sqrt{\pi N_0}}\exp(-z^2)(\sqrt{N_0/2})dz$$

$$P_e = \frac{1}{\sqrt{2\pi}} \int_{\sqrt{2E_b/N_0}}^{\infty} \exp\left(-\frac{z^2}{2}\right) dz$$

This integral is exactly the definition of the standard Q-function, $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-z^2/2} dz$. Therefore, the final expression for the probability of error is:

$$\mathbf{P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)}$$

(Alternatively, using the complementary error function: $P_e = \frac{1}{2} erfc\left(\sqrt{\frac{E_b}{N_0}}\right)$)

Question 2(c)

Assume that a 64-QAM signal is used to send data at a rate of 100 Mbits/s over a satellite transponder. The transponder has a bandwidth of 30 MHz. If the satellite signal is equalized to have an equivalent raised cosine filter characteristic, what is the rolloff factor r required? Could a rolloff factor r be found so that a 180 Mbit/s data rate could be supported? (10)

Solution

Part 1: Finding the required rolloff factor r for 100 Mbits/s

- Data Rate, $R_b = 100$ Mbits/s $= 100 \times 10^6$ bps
- Modulation: 64-QAM. Number of levels $M = 64$.
- Bits per symbol, $n = \log_2(M) = \log_2(64) = 6$ bits/symbol
- Available Bandwidth, $B = 30$ MHz $= 30 \times 10^6$ Hz

First, we calculate the symbol rate (Baud rate), R_s:

$$R_s = \frac{R_b}{n} = \frac{100 \text{ Mbits/s}}{6 \text{ bits/symbol}} = 16.667 \text{ Mbaud}$$

The transmission bandwidth for a signal using a raised cosine filter (for double sideband modulation like QAM) is given by:

$$B = R_s(1 + r)$$

Where r is the roll-off factor. We need to find r:

$$\begin{aligned} 30 \text{ MHz} &= 16.667 \text{ Mbaud} \times (1 + r) \\ 1 + r &= \frac{30}{16.667} \\ 1 + r &= 1.8 \\ r &= 1.8 - 1 = \mathbf{0.8} \end{aligned}$$

The required roll-off factor is 0.8 **(or 80%).**

Part 2: Could a 180 Mbit/s data rate be supported?

Let's check if $R_b = 180$ Mbits/s is possible. New symbol rate, $R_{s(\text{new})} = \frac{180 \text{ Mbits/s}}{6} = 30$ Mbaud.

Using the bandwidth equation again:

$$
\begin{aligned}
B &= R_{s(\text{new})}(1+r) \\
30 \text{ MHz} &= 30 \text{ Mbaud} \times (1+r) \\
1+r &= \frac{30}{30} = 1 \\
r &= \mathbf{0}
\end{aligned}
$$

Conclusion: Yes, mathematically, a data rate of 180 Mbits/s can be supported if the roll-off factor is exactly $r = 0$. This corresponds to an ideal Nyquist (brick-wall) filter. However, practically, designing a filter with $r = 0$ is impossible because it requires infinite time delay and is highly sensitive to timing jitter. So, while theoretically possible, it is not practically feasible.

Digital Communication Systems

Section A Solutions

Question 1(a)

Explain the process of generating a BPSK signal. Can such a signal be detected using non-coherent techniques? With the appropriate expressions and waveforms, illustrate each stage of the detection process. (12)

Solution

1. Generation of BPSK Signal:
In Binary Phase Shift Keying (BPSK), the phase of a high-frequency carrier signal is shifted by 180° (π radians) based on the input digital data. To generate it, the binary sequence is first converted into a polar Non-Return-to-Zero (NRZ) baseband signal, $m(t)$, where bit '1' is represented by $+V$ and bit '0' by $-V$. This $m(t)$ is then multiplied with a sinusoidal carrier signal $c(t) = A\cos(2\pi f_c t)$ using a balanced modulator (multiplier). The resulting BPSK signal is: $s(t) = m(t) \cdot A\cos(2\pi f_c t)$.

2. Can BPSK be detected non-coherently?
No. BPSK cannot be detected using non-coherent techniques (like envelope detection). This is because all the information in BPSK is contained entirely within the *phase* of the signal, not the amplitude. A non-coherent detector ignores phase information, making it impossible to distinguish between a '1' (phase 0°) and a '0' (phase 180°). **Coherent detection** is mandatory.

3. Stages of Coherent Detection (with Expressions):
In coherent detection, the receiver generates a local carrier that is perfectly phase-locked with the transmitter's carrier.

- **Stage 1: Multiplication.** The received signal $s(t) = \pm A\cos(2\pi f_c t)$ is multiplied by the local carrier $\cos(2\pi f_c t)$.

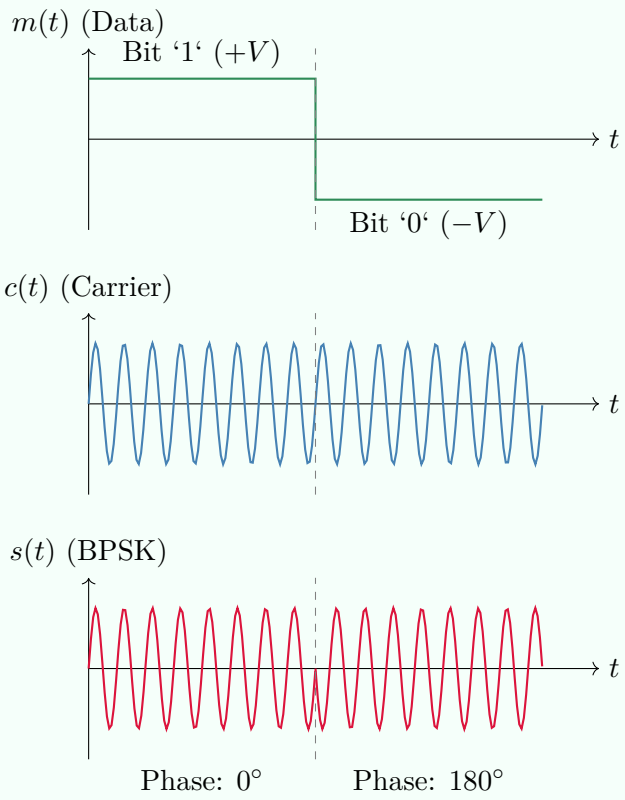
$$x(t) = \pm A\cos(2\pi f_c t) \cdot \cos(2\pi f_c t) = \pm\frac{A}{2}[1 + \cos(4\pi f_c t)]$$

- **Stage 2: Integration/LPF.** The signal passes through an integrator or Low Pass Filter (LPF) which removes the high-frequency double-carrier term ($\cos(4\pi f_c t)$).

$$y = \int_0^{T_b} x(t)dt \approx \pm\frac{A}{2}T_b$$

- **Stage 3: Decision.** A threshold logic compares y with 0V. If $y > 0$, it decides Bit '1'. If $y < 0$, it decides Bit '0'.

4. Waveforms representing the Generation and Detection process:

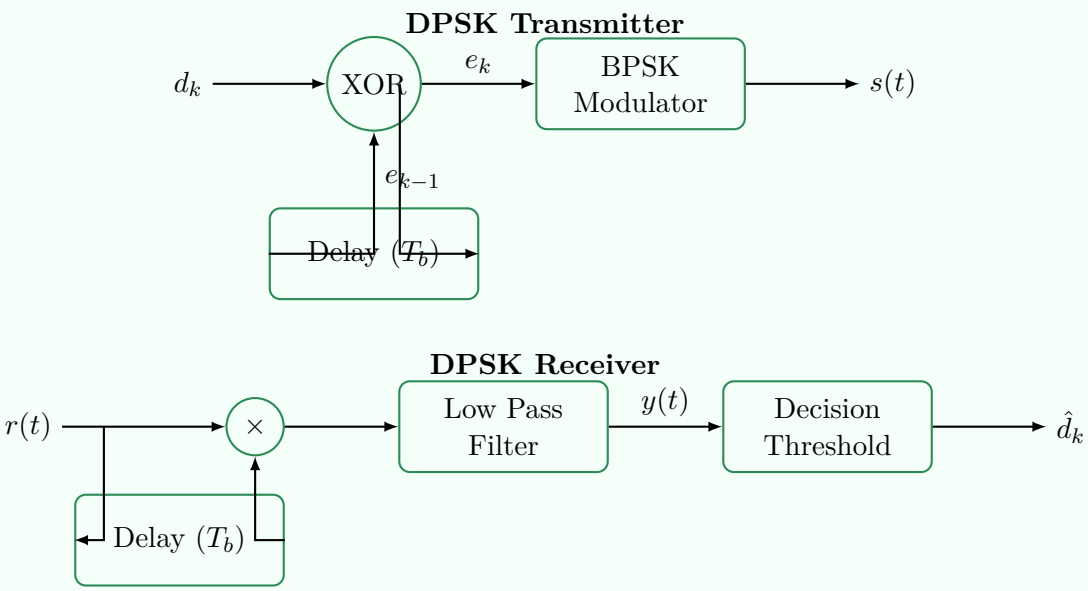


Question 1(b)

Design and draw a DPSK transmitter and receiver, where a '1' indicates a transition. Justify your design by specifying the appropriate logic gate at the transmitter and the comparator logic at the receiver. Additionally, for the original data stream [1 0 1 0 0 1 1 0], determine (i) differentially encoded bits, (ii) transmitted phases, and (iii) detected bits. (13)

Solution

1. Transmitter and Receiver Design



2. Justification of Logic Gates:

- **Transmitter (XOR Gate):** The condition states "a '1' indicates a transition". If the input bit $d_k = 1$, the encoded bit e_k must differ from the previous bit e_{k-1}. If $d_k = 0$, e_k remains the same as e_{k-1}. The Boolean equation is $e_k = d_k \oplus e_{k-1}$. Thus, an **XOR gate** perfectly satisfies this condition.

- **Receiver (Comparator Logic):** The multiplier multiplies the current signal with its 1-bit delayed version. The LPF outputs a positive voltage ($+y$) if the phases were the same, and a negative voltage ($-y$) if the phases were different (transition). Since a transition means $d_k = 1$, the comparator logic is: **If $y(t) < 0$ (phase changed), decide $\hat{d}_k = 1$. If $y(t) > 0$ (phase unchanged), decide $\hat{d}_k = 0$.**

3. Data Stream Analysis:
Given $d_k = [1, 0, 1, 0, 0, 1, 1, 0]$. Let the reference encoded bit be $e_{-1} = 1$. Phase Mapping: $0 \to 0$ rad, $1 \to \pi$ rad.

Index k	**Ref**	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Original Data (d_k)		1	0	1	0	0	1	1	0
(i) Encoded Bits (e_k)	**1**	0	0	1	1	1	0	1	1
(ii) Transmitted Phases (ϕ_k)	π	0	0	π	π	π	0	π	π
Phase Diff ($\Delta\phi = \phi_k - \phi_{k-1}$)		$-\pi \equiv \pi$	0	π	0	0	$-\pi \equiv \pi$	π	0
(iii) Detected Bits ($\hat{d}_k$)		**1**	**0**	**1**	**0**	**0**	**1**	**1**	**0**

Question 1(c)

Digital information is to be transmitted by carrier modulation through an additive gaussian noise channel with bandwidth of 100 kHz and $N_0 = 10^{-10} W/Hz$. Determine the maximum rate that can be transmitted through the channel for four phase PSK, binary FSK and four-frequency orthogonal FSK, which is detected noncoherently. (10)

Solution

Assumption: The problem asks for the maximum rate for specific modulation schemes within a strictly band-limited channel ($B = 100$ kHz). Since the Signal Power (S) is not provided, we cannot calculate the Shannon capacity constraint using S/N_0. Therefore, we will determine the **Nyquist maximum signaling rate** limited by the modulation bandwidth.
Given Bandwidth $B = 100$ kHz.

1. Four Phase PSK (QPSK)

For an M-ary PSK system, the theoretical minimum passband bandwidth (assuming ideal Nyquist filtering) is equal to the symbol rate (R_s).

$$B = R_s \implies R_s = 100 \text{ kbaud}$$

Since QPSK uses $M = 4$, the number of bits per symbol is $n = \log_2(4) = 2$.

$$\text{Maximum Bit Rate, } R_b = n \times R_s = 2 \times 100 \text{ kbaud} = \mathbf{200} \text{ kbps}$$

2. Binary FSK (BFSK) - Detected Noncoherently

For non-coherent orthogonal FSK, the minimum frequency separation between carriers is $\Delta f =$

$\frac{1}{T_s} = R_s$. The approximate transmission bandwidth for M-ary FSK is given by $B = M \cdot \Delta f = M \cdot R_s$. For Binary FSK, $M = 2$, and $R_s = R_b$.

$$B = 2 \times R_b \implies 100 \text{ kHz} = 2 \times R_b$$

$$\text{Maximum Bit Rate, } R_b = \frac{100 \text{ kHz}}{2} = \mathbf{50} \text{ kbps}$$

3. Four-Frequency Orthogonal FSK (4-FSK) - Detected Noncoherently

Using the same non-coherent orthogonal bandwidth formula ($B = M \cdot R_s$), with $M = 4$:

$$B = 4 \times R_s \implies 100 \text{ kHz} = 4 \times R_s$$

$$\text{Symbol Rate, } R_s = \frac{100 \text{ kHz}}{4} = 25 \text{ kbaud}$$

Since $M = 4$, each symbol carries $n = \log_2(4) = 2$ bits.

$$\text{Maximum Bit Rate, } R_b = n \times R_s = 2 \times 25 \text{ kbaud} = \mathbf{50} \text{ kbps}$$

Digital Communication Systems

Assignment / Exam Solutions (Part 2)

Question 3(a)

How does the M-ary modulation scheme differ from the binary modulation scheme? What is the significance of M-ary modulation schemes in digital communication systems? (10)

Solution

1. Differences between M-ary and Binary Modulation Schemes:

- **Number of Symbols:** In a binary modulation scheme, only two distinct signals or symbols are used to represent data (e.g., bit '0' and bit '1'). In an M-ary modulation scheme, there are M distinct symbols available for transmission, where $M > 2$ (typically $M = 2^n$).
- **Bits per Symbol:** Binary schemes transmit exactly 1 bit per symbol. M-ary schemes transmit $n = \log_2(M)$ bits per symbol. For example, a 16-QAM system transmits $\log_2(16) = 4$ bits per single symbol.
- **Symbol Rate (Baud Rate):** For a given data bit rate (R_b), the symbol rate (R_s) in a binary system is equal to the bit rate ($R_s = R_b$). In an M-ary system, the symbol rate is reduced by a factor of $\log_2(M)$, meaning $R_s = R_b/\log_2(M)$.

2. Significance of M-ary Modulation Schemes:

- **Bandwidth Efficiency:** The primary significance of M-ary modulation is the conservation of bandwidth. Since multiple bits are packed into a single symbol, the required transmission bandwidth for a given bit rate is significantly reduced compared to binary modulation. Bandwidth efficiency (measured in bits/s/Hz) increases as M increases.
- **High Data Rate Transmission:** It allows for very high data rates over limited bandwidth channels (like telephone lines or wireless spectrum), making technologies like 4G/5G, Wi-Fi, and digital television possible.
- **Trade-off (The Cost):** The trade-off for this higher bandwidth efficiency is power efficiency. As M increases, the symbols become closer to each other in the signal constellation. This makes the system more susceptible to noise, requiring a higher Signal-to-Noise Ratio (SNR) to maintain the same Bit Error Rate (BER).

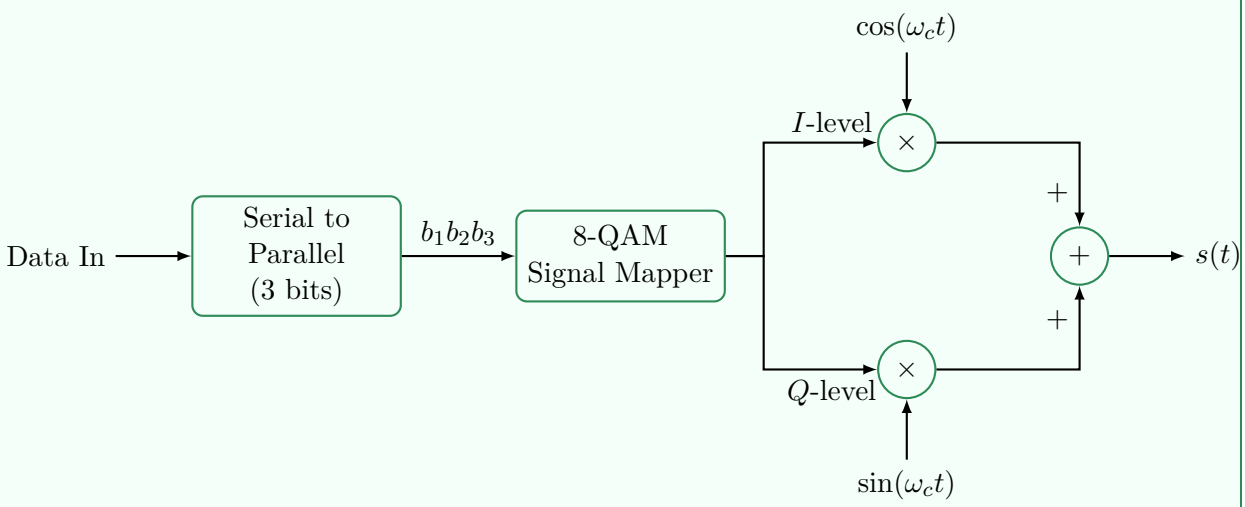
Question 3(b)

Draw the block diagram of an 8-QAM transmitter and receiver. Explain how the modulation and demodulation work with a specific input bit sequence. (15)

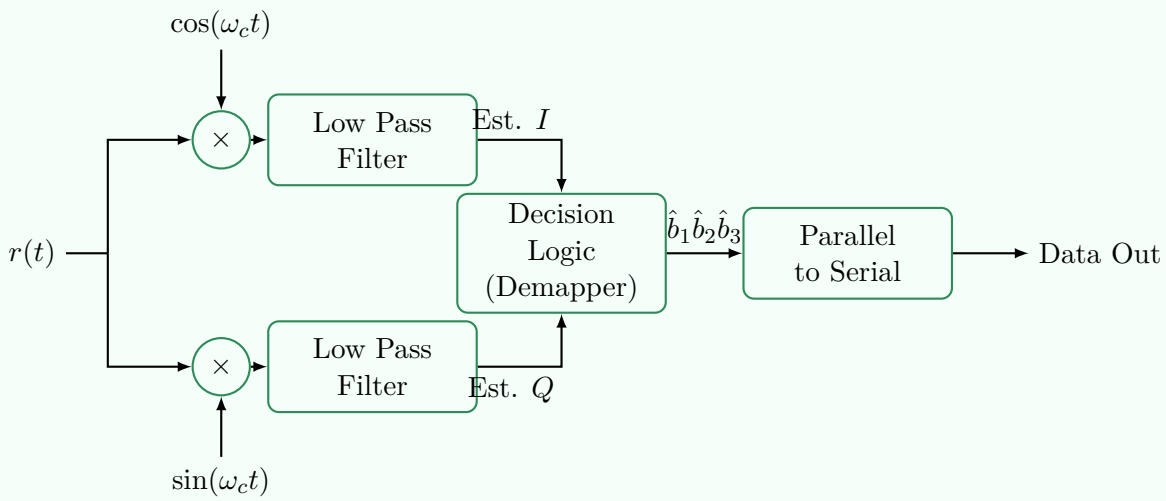
Solution

1. Block Diagrams for 8-QAM

a) 8-QAM Transmitter:



b) 8-QAM Receiver:



2. Explanation with a Specific Input Sequence:
Let's assume the specific input bit sequence is **'101'**.

- **Modulation (Transmitter):**
  1. The Serial-to-Parallel converter takes the stream and groups it into 3 bits: **101**.
  2. The 8-QAM signal mapper receives **101**. Based on its internal constellation diagram mapping, it assigns specific voltage levels to the I (In-phase) and Q (Quadrature) channels. Let's assume the mapping for '101' is $I = -1V$ and $Q = +3V$.
  3. The I level (-1V) multiplies the $\cos(\omega_c t)$ carrier, and the Q level (+3V) multiplies the $\sin(\omega_c t)$ carrier.
  4. The transmitted signal becomes: $s(t) = -1 \cdot \cos(\omega_c t) + 3 \cdot \sin(\omega_c t)$.
- **Demodulation (Receiver):**
  1. The receiver gets $r(t) = -1 \cdot \cos(\omega_c t) + 3 \cdot \sin(\omega_c t)$ (ignoring noise for simplicity).
  2. The upper branch multiplies $r(t)$ by locally generated $\cos(\omega_c t)$: $r(t)\cos(\omega_c t) = -1\cos^2(\omega_c t) + 3\sin(\omega_c t)\cos(\omega_c t) = -0.5(1 + \cos(2\omega_c t)) + 1.5\sin(2\omega_c t)$.

3. The Low Pass Filter (LPF) removes the high-frequency $2\omega_c t$ terms, leaving a DC level proportional to -0.5. Scaling restores this to the estimated $I = -1V$.
4. Similarly, the lower branch multiplies by $\sin(\omega_c t)$ and filters it, recovering the estimated $Q = +3V$.
5. The Demapper receives $(I = -1, Q = +3)$, looks up its constellation table, and decides the bits were **101**.
6. The Parallel-to-Serial converter outputs the bit stream '101'.

Question 3(c)

For an 8-PSK system, operating with an information bit rate of 24 kbps, determine: (i) baud, (ii) minimum bandwidth, and (iii) bandwidth efficiency. (10)

Solution

Given Parameters:

- Bit rate, $R_b = 24$ kbps $= 24,000$ bits/second
- Modulation Scheme = 8-PSK
- Number of levels, $M = 8$

The number of bits per symbol is $n = \log_2(M) = \log_2(8) = 3$ bits/symbol.

(i) Baud (Symbol Rate, R_s)

The baud rate is the number of symbols transmitted per second.

$$R_s = \frac{R_b}{n} = \frac{24 \text{ kbps}}{3} = \mathbf{8} \text{ kbaud (or 8000 symbols/sec)}$$

(ii) Minimum Bandwidth (B_{min})

For an M-ary bandpass modulation scheme like 8-PSK, the minimum theoretical passband bandwidth (assuming ideal Nyquist filtering where roll-off factor $\alpha = 0$) is equal to the symbol rate.

$$B_{\text{min}} = R_s = \mathbf{8} \text{ kHz}$$

(iii) Bandwidth Efficiency (η)

Bandwidth efficiency is defined as the ratio of the bit rate to the minimum required bandwidth.

$$\eta = \frac{R_b}{B_{\text{min}}} = \frac{24 \text{ kbps}}{8 \text{ kHz}} = \mathbf{3} \text{ bits/s/Hz}$$

(Alternatively, for ideal 8-PSK, bandwidth efficiency is simply $\log_2(M) = \log_2(8) = 3$ *bits/s/Hz).*

Question 4(a)

What is thermal noise in a communication system? Derive the equation of equivalent thermal noise voltage and current with necessary sketches. (10)

Solution

1. What is Thermal Noise?

Thermal noise (also known as Johnson-Nyquist noise) is the electrical noise generated by the random thermal motion of electrons (charge carriers) inside an electrical conductor, regardless of any applied voltage. It occurs in all resistive electronic components and is dependent on temperature. Because its power spectral density is nearly flat across the frequency spectrum used in communications, it is often modeled as White Noise.

2. Derivation of Equivalent Thermal Noise Voltage and Current

The power spectral density (PSD) of thermal noise voltage across a resistor R is given by physics principles as:

$$S_v(f) = 2kTR \quad (\text{in Volts}^2/\text{Hz})$$

Where:

- k = Boltzmann's constant (1.38×10^{-23} J/K)
- T = Absolute temperature in Kelvin (K)
- R = Resistance in Ohms (Ω)

Because this is a double-sided PSD, the total mean square noise voltage $\overline{v_n^2}$ measured over a specific system bandwidth B (in Hz) is calculated by integrating over positive frequencies (hence multiplying by 2):

$$\overline{v_n^2} = 2 \times S_v(f) \times B = 2 \times (2kTR) \times B = 4kTRB$$

Therefore, the **root-mean-square (RMS) thermal noise voltage (V_n)** is:

$$\mathbf{V_n = \sqrt{4kTRB}}$$

To find the **equivalent thermal noise current (I_n)**, we use Norton's equivalent circuit (Ohm's law, $I = V/R$):

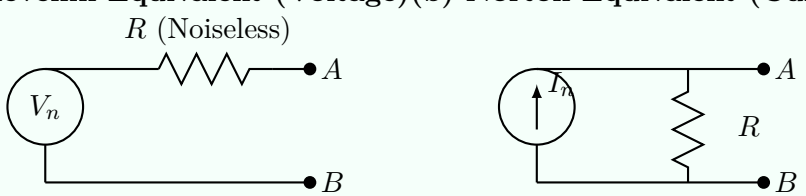
$$I_n = \frac{V_n}{R} = \frac{\sqrt{4kTRB}}{R} = \sqrt{\frac{4kTRB}{R^2}}$$

$$\mathbf{I_n = \sqrt{\frac{4kTB}{R}}}$$

3. Necessary Sketches (Equivalent Circuits)

A noisy resistor can be modeled either as a noiseless resistor in series with a voltage source (Thévenin) or in parallel with a current source (Norton).

(a) Thévenin Equivalent (Voltage) **(b) Norton Equivalent (Current)**



Digital Communication Systems

Assignment / Exam Solutions

Question 2(a)

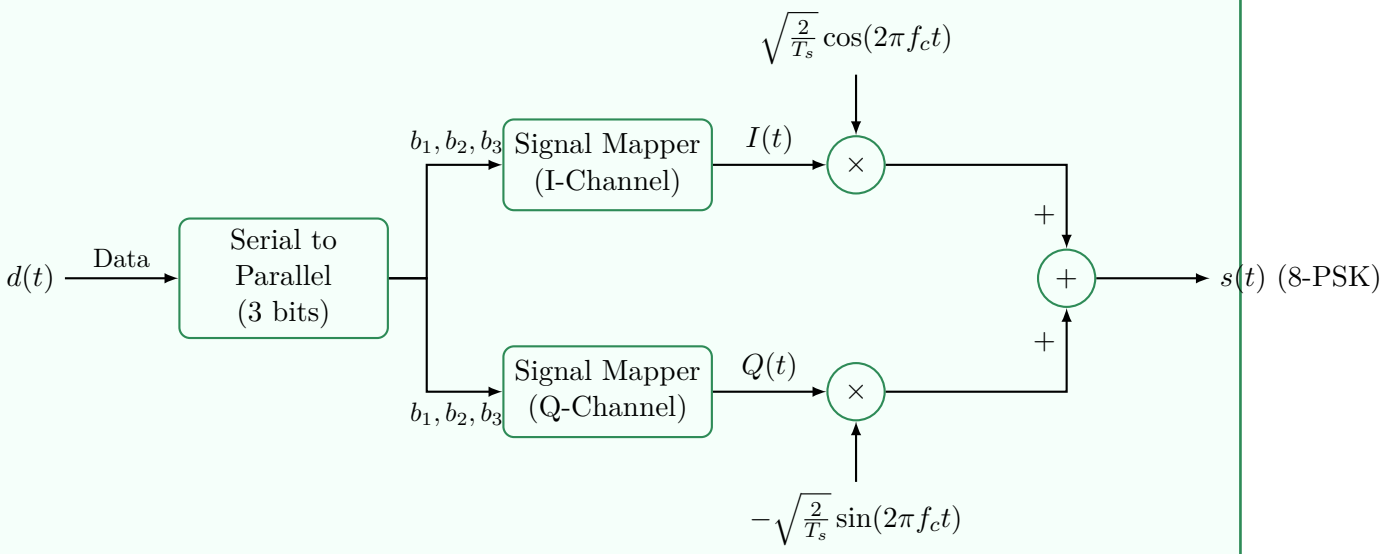
Design a PSK transmitter capable of transmitting three bits per signal element. Also calculate the amplitudes and phases of the resulting PSK modulated signal for all possible input bit combinations. (12)

Solution

1. Transmitter Design (8-PSK)

Since the transmitter is capable of transmitting 3 bits per signal element, the number of levels M is $2^3 = 8$. This corresponds to an **8-PSK (Phase Shift Keying)** modulation scheme.

In an 8-PSK transmitter, the incoming binary data stream is divided into groups of 3 bits using a serial-to-parallel converter. These 3 bits act as an index to a signal mapper that determines the in-phase (I) and quadrature (Q) amplitudes. These I and Q levels are then multiplied by the carrier waves and added together.



2. Amplitudes and Phases Calculation

The general equation for an M-ary PSK signal is:

$$s_i(t) = \sqrt{\frac{2E_s}{T_s}} \cos\left(2\pi f_c t + \theta_i\right)$$

Where $\theta_i = \frac{2\pi i}{M}$ for $i = 0, 1, 2, \ldots, 7$.

By expanding using trigonometric identities:

$$s_i(t) = \underbrace{\sqrt{\frac{2E_s}{T_s}} \cos(\theta_i)}_{I-\text{amplitude}} \cos(2\pi f_c t) - \underbrace{\sqrt{\frac{2E_s}{T_s}} \sin(\theta_i)}_{Q-\text{amplitude}} \sin(2\pi f_c t)$$

Let the peak carrier amplitude be $A = \sqrt{2E_s/T_s}$. For simplicity, let $A = 1$ to calculate normalized I and Q amplitudes.

Input Bits ($b_1b_2b_3$)	**Phase** (θ_i)	**I-Amplitude** ($\cos\theta_i$)	**Q-Amplitude** ($\sin\theta_i$)
000	$0°$	1	0
001	$45°$ ($\pi/4$)	0.707 ($1/\sqrt{2}$)	0.707 ($1/\sqrt{2}$)
010	$90°$ ($\pi/2$)	0	1
011	$135°$ ($3\pi/4$)	-0.707 ($-1/\sqrt{2}$)	0.707 ($1/\sqrt{2}$)
100	$180°$ (π)	-1	0
101	$225°$ ($5\pi/4$)	-0.707 ($-1/\sqrt{2}$)	-0.707 ($-1/\sqrt{2}$)
110	$270°$ ($3\pi/2$)	0	-1
111	$315°$ ($7\pi/4$)	0.707 ($1/\sqrt{2}$)	-0.707 ($-1/\sqrt{2}$)

Note: The exact bit-to-phase mapping can follow Gray coding to minimize bit error rate, but the above illustrates the direct sequential mapping principle.

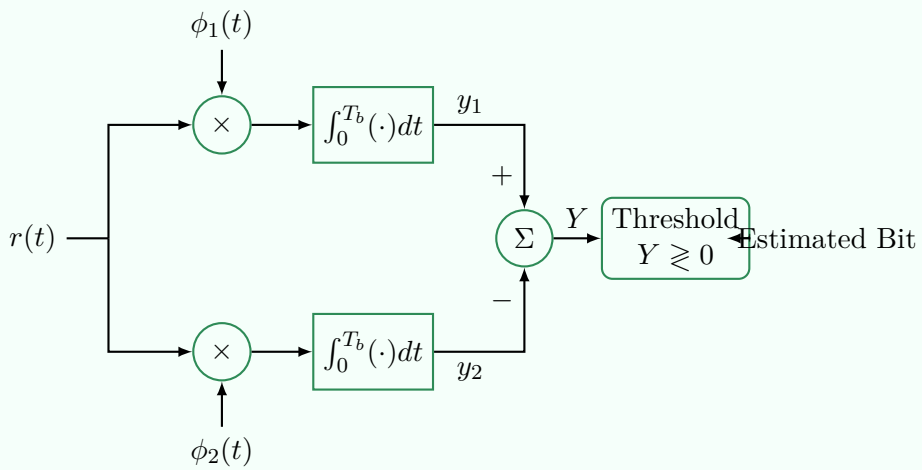
Question 2(b)

Sketch and explain the coherent detection of a BFSK signal, providing the necessary mathematical derivation. (13)

Solution

1. Sketch of Coherent BFSK Detector

Coherent detection requires the receiver to have exact knowledge of the carrier frequencies and phases. The receiver consists of two correlators (or matched filters) corresponding to the two possible signal frequencies f_1 (for bit 1) and f_2 (for bit 0).



2. Mathematical Derivation

The transmitted BFSK signals for symbol duration T_b are:

$$s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_1 t) \quad \text{for bit '1'}$$
$$s_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_2 t) \quad \text{for bit '0'}$$

The receiver generates two orthonormal basis functions using local oscillators:

$$\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_1 t)$$
$$\phi_2(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_2 t)$$

Assuming Additive White Gaussian Noise (AWGN), the received signal is $r(t) = s_i(t) + w(t)$.
Correlator Outputs: The received signal is multiplied by $\phi_1(t)$ and $\phi_2(t)$ and integrated over T_b:

$$y_1 = \int_0^{T_b} r(t)\phi_1(t)dt$$
$$y_2 = \int_0^{T_b} r(t)\phi_2(t)dt$$

Case Analysis (Assuming $s_1(t)$ was transmitted):
If bit '1' was sent, $r(t) = s_1(t) + w(t)$.

$$y_1 = \int_0^{T_b} \left[\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_1 t) + w(t)\right] \sqrt{\frac{2}{T_b}} \cos(2\pi f_1 t)dt$$
$$= \sqrt{E_b} \int_0^{T_b} \frac{2}{T_b} \cos^2(2\pi f_1 t)dt + n_1 = \sqrt{E_b} + n_1$$

Because BFSK signals are chosen to be orthogonal, $\int_0^{T_b} s_1(t)\phi_2(t)dt = 0$.

$$y_2 = \int_0^{T_b} [s_1(t) + w(t)]\, \phi_2(t)dt = 0 + n_2$$

Decision Rule: The test statistic is $Y = y_1 - y_2$.

- If $Y > 0$ (i.e., $y_1 > y_2$), the receiver decides that **Bit '1'** was transmitted.
- If $Y < 0$ (i.e., $y_1 < y_2$), the receiver decides that **Bit '0'** was transmitted.

This coherent detection yields optimal performance in the presence of AWGN.

Question 2(c)

A binary baseband signal is passed through a raised cosine roll-off filter with a 40% roll-off factor and is then modulated onto a carrier. The data rate is 128 kbit/s. Evaluate:

(i) the absolute bandwidth of the resulting OOK signal.

(ii) the approximate bandwidth of the resulting FSK signal when the mark frequency is 250

kHz and the space frequency is 260 kHz.

(10)

Solution

Given Parameters:

- Data rate, $R_b = 128 \text{ kbit/s} = 128 \times 10^3 \text{ bps}$
- Roll-off factor, $\alpha = 40\% = 0.4$

First, let's find the required absolute bandwidth for the **baseband** signal passed through the raised cosine filter.

$$B_{\text{baseband}} = \frac{R_b}{2}(1+\alpha)$$

$$B_{\text{baseband}} = \frac{128 \text{ kHz}}{2}(1+0.4) = 64 \text{ kHz} \times 1.4 = 89.6 \text{ kHz}$$

(i) Absolute bandwidth of the resulting OOK signal

On-Off Keying (OOK) is a form of Amplitude Shift Keying (ASK). When a baseband signal is amplitude-modulated onto a carrier, it creates a double-sideband signal. Therefore, the transmission bandwidth is twice the baseband bandwidth.

$$\begin{aligned} B_{\text{OOK}} &= 2 \times B_{\text{baseband}} \\ &= R_b(1+\alpha) \\ &= 128 \text{ kHz} \times (1+0.4) \\ &= \mathbf{179.2} \text{ kHz} \end{aligned}$$

(ii) Approximate bandwidth of the resulting FSK signal

Given:

- Mark frequency, $f_m = 250$ kHz
- Space frequency, $f_s = 260$ kHz

The frequency shift (separation between the two carrier frequencies) is:

$$2\Delta f = |f_m - f_s| = |250 \text{ kHz} - 260 \text{ kHz}| = 10 \text{ kHz}$$

Using Carson's Rule adapted for digital signals with pulse shaping, the approximate bandwidth of an FSK signal is the sum of the frequency separation and the bandwidth of the corresponding amplitude-modulated signal (which we found in part i):

$$\begin{aligned} B_{\text{FSK}} &\approx 2\Delta f + B_{\text{OOK}} \\ &= 10 \text{ kHz} + 179.2 \text{ kHz} \\ &= \mathbf{189.2} \text{ kHz} \end{aligned}$$

(Note: If treating the baseband bandwidth formally as $2\Delta f + 2B_{baseband}$, the calculation yields the exact same 189.2 kHz.)

Telecommunication-II (EEE-4203)

Complete Question Bank (2017 - 2024)

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1. Fundamentals & Information Theory

Theoretical Questions

- Briefly explain the relative advantages and disadvantages of analog and digital communication systems.
- What is thermal noise in a communication system? Derive the equation of equivalent thermal noise voltage and current with necessary sketches.

Mathematical Problems

- **Packet Error:** An information packet contains 200 bits. This packet is transmitted over a communications channel where the probability of error for each bit is 10^{-3}. Suppose the packet includes an error-correcting code that can correct up to three errors. What is the probability that a particular packet is received in error?
- **Channel Capacity:** An AWGN channel with 6 kHz bandwidth and noise power spectral density $N_0/2 = 10^{-12}$ W/Hz. Signal power required is 0.2 mW. Find the capacity. If used to transmit 100 kbps, determine the minimum BW required for the same noise characteristics.
- **Channel Rate:** Digital info transmitted through AWGN channel with BW=100 kHz and $N_0 = 10^{-10}$ W/Hz. Determine maximum rate for 4-PSK, BFSK, and 4-FSK (non-coherent).

2. Line Coding & Baseband Transmission

- What is line coding? Why is it important to minimize DC components in line coding, and which schemes help achieve this?
- **Data Rate:** A baseband channel has a 2-MHz bandwidth. What is the data rate if we use: (i) NRZ-L, (ii) Manchester, (iii) Differential Manchester, (iv) NRZ-I?
- **Multiplexing (TDM/FDM):** 20 sources multiplexed in each TDM block. Source BW=4 kHz. Nyquist sampling, 8 bits/sample. Interleaved unit is 8 bits + 1 bit sync. FDM multiplexes 4 TDM outputs with ASK carrier and 2 kHz guard band. Determine: (i) TDM output data rate, (ii) FDM signal BW.
- **Decoding Task:** The waveform below represents MLT-3 encoding. Decode the data stream.

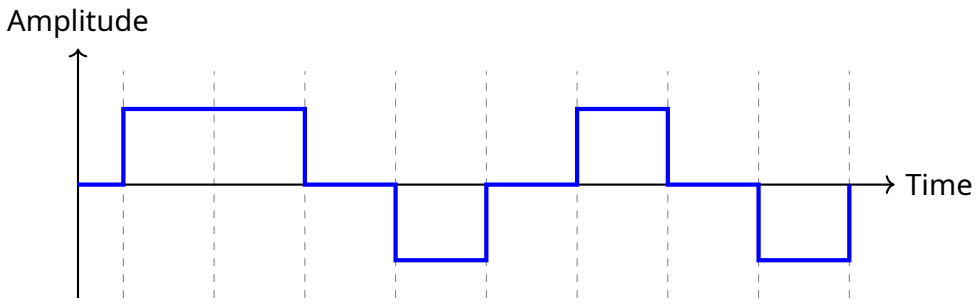

Figure: MLT-3 Waveform for Decoding

3. Digital Modulation Techniques

Theoretical & Diagram Questions

- Explain ASK, FSK, and PSK modulation schemes with their applications. Why is ASK more affected by noise?
- **ASK/OOK:** Draw transmitter/receiver block diagram for ASK ($A_0 \cos \omega_c t$ for 0, $A_1 \cos \omega_c t$ for 1).
- **BPSK:** Explain BPSK generation. Can it be detected non-coherently? Why is coherent detection required? Derive average probability of bit error for coherent BPSK.
- **BFSK:** Explain generation, coherent, and non-coherent detection of BFSK. Derive average probability of symbol error for coherent BFSK. Show BFSK with discontinuous phase is the sum of two independent BASK signals.
- **DPSK:** Briefly explain generation and detection. Design a DPSK Tx and Rx (specify logic gates).
- **QPSK & OQPSK:** Explain QPSK Tx and Rx with block diagrams and constellations. Draw OQPSK signal components (modulating/modulated I & Q, final waveform) for "1100011001".
- **MSK:** What is MSK? Show minimum tone spacing $f_1 - f_2 = 1/T$ for orthogonal non-coherent FSK. Derive phase expression for MSK.
- **M-ary, QAM & 8-PSK:** Advantages of M-ary. Explain 8-QAM Tx/Rx with block diagrams. Derive signal-space diagram of QPSK, 8-PSK, 16-QAM. Explain QAM using 4-QAM constellation. Explain the advantage of Gray encoding in multilevel modulation.

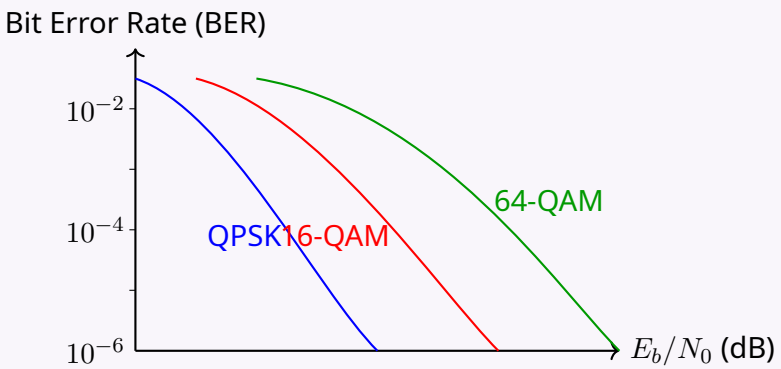

Figure: Concept of BER vs E_b/N_0 for different M-ary QAM schemes

Mathematical Problems

- **Baud/Bits:** What is the number of bits per baud for: ASK (8 amp), FSK (16 freq), PSK (different phases), 256-QAM? Telephone line (8kHz), max bits for BASK, QPSK, 64-QAM, 128-QAM ($d = 0$)?
- **8-PSK Math:** Information bit rate 24 kbps. Determine: baud, min BW, BW efficiency.
- **64-QAM Satellite:** Data rate 100 Mbits/s, Transponder BW 30 MHz. Find required rolloff factor r. Can it support 180 Mbit/s?
- **Baseband filtering:** Binary baseband through raised cosine (40% rolloff), 128 kbps. Evaluate absolute BW of OOK, and approx BW of FSK ($f_m = 250$ kHz, $f_s = 260$ kHz).

4. Networking & Data Communication

- What is meant by protocol? Define data communication.
- Explain simplex, half-duplex, and full-duplex data transmission schemes with examples.
- State features of LANs, WANs, and MANs. Describe various LAN topologies (Bus, Ring, Mesh, Star) with diagrams.
- Briefly explain the frame transmission technique in a ring LAN.
- Discuss consequences if a connection fails: (i) 6-device bus, (ii) 4-device ring, (iii) 5-device mesh, (iv) 7-device star.
- What is the purpose of a reference model? Differentiate OSI and TCP/IP models. Mention responsibilities of each OSI/TCP layer.
- **Network Routing Problem:** 4 computers, 2 routers, 5 LANs. Router 1 connected to LAN 1, 2, 4. Router 2 connected to LAN 2, 3, 5. Comp 1, 2, 3, 4 connected to LAN 1, 4, 5, 3 respectively. Send packet from Comp 2 to Comp 4. Draw connection showing physical and logical addresses.

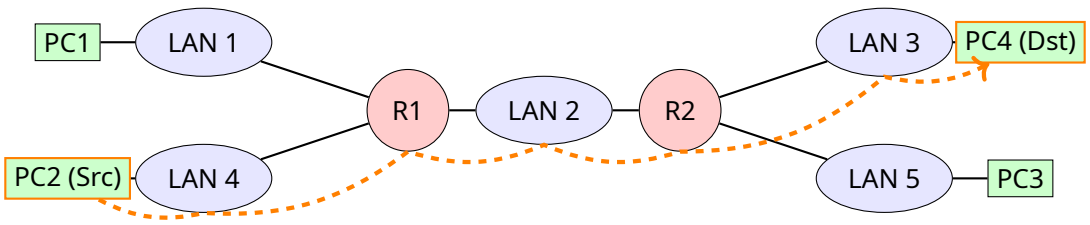

Figure: Solution Diagram for Network Routing Problem

5. Satellite Communication

- Describe the need for satellite launching methods. Describe one method of launching (including geo-stationary).
- What is the main function of a transponder? Explain using detailed diagrams. Illustrate single and double frequency conversion transponders (e.g., 6/4 GHz and 14/11 GHz

bands).

- What are the main tasks of an earth station? Explain different sub-stations. Describe with a block diagram how terrestrial components communicate through a satellite.
- Why is the uplink frequency different from the downlink frequency?

6. Radar Systems

- Define Radar. Briefly explain working principle and applications. Discuss types of radar. How is radar utilized in Air Traffic Control (ATC)?
- Explain the operation and applications of Doppler Radar. Explain working principle of Continuous-Wave (C.W.) radar with necessary diagram.
- **Radar Range Eq:** Derive free-space radar range equation. Comment on the relationship between max range (R_{max}) and operating wavelength (λ) / frequency.
- **Math 1:** $f = 12$ GHz, $P_t = 300$ kW, $A_e = 5\ m^2$, RCS $= 25\ m^2$, $P_{\text{min}} = 1$ nW. Calculate max range.
- **Math 2:** $P_t = 250$ kW, $P_{\text{min}} = 10$ pW, RCS $= 20\ m^2$, Gain $= 4000$, $A_e = 4.5\ m^2$. Find max range.
- **Math 3:** $P_t = 400$ kW, $f = 10$ GHz, $A_e = 5\ m^2$, RCS $= 30\ m^2$, $P_{\text{min}} = 10^{-10}$ W. Find max range.

7. Optical Fiber Communication

Theory & Principles

- Define Acceptance angle, Numerical Aperture (NA), Critical propagation angle, and TIR. Show NA depends on core/cladding difference.
- Explain construction of optical fiber. Draw block diagram of fiber-optic communication system. List advantages and disadvantages of optical fiber.
- Define attenuation. List major causes of attenuation and explain their mechanisms.
- Define dispersion. Explain chromatic dispersion, modal dispersion, and polarization mode dispersion (PMD). Express pulse-spreading time caused by modal dispersion in terms of relative refractive index difference. How can such dispersions be compensated?

Mathematical Problems

- **Angle/NA Calculations:** Given n_1 and n_2 (e.g., 1.52/1.50, 1.52/1.48, 1.50/1.47, 1.78/1.76, 1.4513/1.4468). Determine critical propagation angle, critical angle at interface, acceptance angle, and NA.
- **Velocity relation:** Velocity in core $= 2.05 \times 10^8$ m/s, critical angle $= 70°$. Determine NA, acceptance angle, critical propagation angle.
- **Pulse Spread:** $n_1 = 1.78, n_2 = 1.76$. How much will a pulse spread over 6 km?
- **Max Link Length:** $P_{in} = 1.55$ mW (or 1.6 mW), attenuation $= 0.25$ dB/km (or 0.26 dB/km), $P_{\text{min}} = 2.55$ μW (or 2.5 μW). Find maximum link length without repeater.
- **Splice Loss:** $P_{in} = 125$ μW over 8km, $P_{out} = 4$ μW. Find overall loss, loss/km, overall loss for 10km link with splices at 1km intervals (1dB each), and numerical I/O ratio.

8. Optical Sources & Detectors

- Compare characteristics of LED and LD. Main reasons to use LD for long-haul links. Draw stimulated emission, light amplification, positive feedback for LD.
- Define responsivity and quantum efficiency. Derive $R = \frac{\eta e \lambda}{hc}$. Show that at 1248 nm, R and η are equal.
- **Photodiode Math:** Given η (e.g., 75%, 70%, 60%) and λ (e.g., 0.80, 0.87, 0.85, 0.9 μm). Calculate Responsivity, Received optical power if I_p is given (20 μA, 16 μA, 15 μA, 1 μA), and corresponding number of received photons.

9. Cellular & Mobile Communication

- Define cellular concept, cell, and cluster. Why is the hexagonal shape widely used?
- Explain frequency reuse. Show for hexagonal geometry, co-channel reuse ratio $Q = \sqrt{3N}$ where $N = i^2 + ij + j^2$.
- Define cell splitting and cell sectoring. How do they enhance capacity?
- **Capacity Math:** Total BW (30, 35, 33 MHz), FDD, 50kHz (or 55kHz) simplex channels. Compute channels/cell for $N = 4, 7, 12$. If 1 MHz dedicated to control, distribute control and voice channels equitably.
- Explain GSM network architecture and components' roles.
- Mention classification of handover. Briefly explain handover process in GSM.
- Differentiate among TDMA, FDMA, and CDMA.
- Describe and compare various generations: 2G, 3G, 4G, and 5G. Evaluate data services evolution.

10. Advanced Communication Topics

- **Adaptive Equalizer:** Draw block diagram of adaptive equalizer in the receiver and briefly describe components.
- **Diversity:** What are diversity techniques? Describe different types.
- **Speech Coding:** What is speech coding? What are its objectives?
- **Short Notes:** JPGG (JPEG) and MPGA (MPEG).

Advanced Communication Systems

Comprehensive Question Bank & Study Guide

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1. Communication Fundamentals

- Relative advantages and disadvantages of analog and digital communication systems.
- What is thermal noise in a communication system? Derive the equation for equivalent thermal noise voltage and current with necessary sketches.
- **Information theory calculation:** Determine the probability of a packet being received in error for a 200-bit packet over a channel with a 10^{-3} bit error probability, assuming an error-correcting code can fix up to three errors.

2. Line Coding and Baseband Transmission

- What is line coding? Why is it important to minimize DC components, and which schemes help achieve this?
- **Decoding task:** Decode a data stream from a given waveform representing Multiline Transmission (MLT-3) encoding.
- **Data rate calculations:** Determine the data rate for a 2-MHz bandwidth baseband channel using NRZ-L, Manchester, Differential Manchester, and NRZ-I coding.

3. Digital Modulation Techniques

- **Basic Shift Keying:** Define and explain ASK, FSK, and PSK modulation schemes with their applications.
- **BASK:** Find the transmission bandwidth and plot the waveform for a given binary sequence.
- **BFSK:**
  - Sketch and explain the coherent detection of a BFSK signal with mathematical derivation.
  - Explain the generation and both coherent and non-coherent detection systems of BFSK.
  - Show that a BFSK signal can be expressed as the sum of two independent BASK signals.
  - Derive the expression for the average probability of symbol error for coherent binary FSK.
- **BPSK:**
  - Explain the generation process and determine if it can be detected non-coherently.

- Derive the average probability of bit error for a coherent binary BPSK system.
- Explain why coherent detection is required for BPSK.

- **DPSK:** Design and draw a DPSK transmitter and receiver (specifying logic gates) and determine coded bits and transmitted phases for a bit sequence. Briefly explain the generation and detection of a Differential PSK (DPSK) signal.
- **QPSK:** Explain the operation of a QPSK transmitter and coherent QPSK receiver with block diagrams and constellation diagrams. Draw the OQPSK signal components for a specific data sequence (e.g., "1100011001").
- **MSK:** What is meant by MSK modulation? Demonstrate the minimum tone spacing for non-coherently detected orthogonal FSK ($f_1 - f_2 = 1/T$) or derive the phase expression for MSK.
- **M-ary Modulation and QAM:** How does the M-ary modulation scheme differ from binary? What is its significance/advantage in digital systems?
- **8-PSK & QAM:** Describe the signal generation process. Calculate baud rate, minimum bandwidth, and efficiency for an 8-PSK system operating at 24 kbps. Draw the block diagram of an 8-QAM transmitter and receiver; explain the process with a bit sequence. Describe the modulation process using a constellation diagram. Calculate the required rolloff factor for a 64-QAM signal sent at 100 Mbits/s through a 30 MHz transponder.
- **Gray Encoding:** Explain its advantage in multilevel modulation with an example.

4. Networking and Data Communication

- **OSI and TCP/IP Models:** Describe the function of each layer in the OSI model or TCP/IP model. How is the OSI reference model different from the TCP/IP model?
- **LAN Topologies:** Define data communication and describe various LAN topologies (Bus, Ring, Mesh, Star) with diagrams.
- **Protocols and Transmission:** What is meant by a protocol in networking? Explain simplex, half-duplex, and full-duplex data transmission schemes. Explain the frame transmission technique in a Ring LAN.
- **Network Routing:** Draw a network connection diagram for a system of 4 computers, 2 routers, and 5 LANs, showing physical and logical addresses at each hop.

5. Satellite Communication

- **Launching:** Describe satellite launching methods, including those for geo-stationary orbits.
- **Transponders:** Explain the function of a satellite transponder using a detailed diagram. Illustrate block diagrams for single- and double-frequency conversion transponders.
- **Earth Stations:** What are the main tasks of an earth station? Describe with a block diagram how terrestrial components communicate through a satellite. Explain the different sub-stations of a satellite earth station.
- **Frequency:** Why is the uplink frequency different from the downlink frequency?

6. Radar Systems

- **General & Types:** Define Radar and explain its working principle and applications. Discuss different types of radar systems and explain how radar is utilized in Air Traffic Control.
- **Doppler & C.W. Radar:** Explain the operation and applications of Doppler Radar. Explain the working principle of continuous-wave radar with a diagram.
- **Radar Range Equation & Calculations:** Derive the radar range equation and comment on the relationship between maximum range (R_{max}) and operating wavelength (λ). Calculate the maximum range of a radar system given parameters like peak power, antenna gain, effective aperture, and target radar cross section (RCS).

7. Optical Fiber Communication

- **Fiber Structure and Propagation:** Define acceptance angle and numerical aperture (NA). Show that NA depends on the difference between core and cladding refractive indices. Explain the construction of an optical fiber and the block diagram of a fiber-optic communication system. Define critical propagation angle and total internal reflection (TIR).
- **Calculations:** Determine NA, critical propagation angles, and acceptance angles for various silica fibers based on refractive indices or light velocity.
- **Attenuation:** Define attenuation and list its major causes. Calculate maximum possible link length without a repeater given launch power, attenuation (dB/km), and minimum required detector power.
- **Dispersion:** Define and briefly explain chromatic dispersion, modal dispersion, and polarization mode dispersion (PMD). Express the compensation techniques for these types of dispersion.

8. Light Sources and Photodetectors

- **Light Sources (LED vs. LD):** Compare the characteristics of LED and Laser Diode (LD) as optical sources. Describe the main reasons to use LD instead of LED for long-haul links. Draw and explain stimulated emission, light amplification, and the positive feedback process in an LD.
- **Photodetectors (Photodiodes):** Define responsivity and quantum efficiency of a photodiode and derive their relationship $\left(R = \frac{\eta e \lambda}{hc}\right)$.
- **Calculations:** Calculate responsivity, received optical power, or the number of received photons for a p-n photodiode.

9. Mobile and Cellular Communication

- **Cellular Concepts:** Define the cellular concept, "cell," and "cluster." Why is the hexagonal shape used? Explain frequency reuse and show that for hexagonal geometry, the co-channel reuse ratio is $Q = \sqrt{3N}$. Explain how capacity is enhanced by "cell-splitting" and "sectoring."

- **Handovers & GSM:** Define the classification of handovers and explain the handover process in a cellular/GSM network. Describe the architecture and structure of a GSM network, detailing each component's role.
- **Generations & Access:** Compare/describe the various generations of cellular systems (2G, 3G, 4G, and 5G). Differentiate among TDMA, FDMA, and CDMA in wireless communication.

10. Advanced Communication Topics

- **Adaptive Equalizer:** Draw the block diagram of an adaptive equalizer in a receiver and briefly describe its components.
- **Speech Coding:** What is speech coding, and what are its objectives?
- **Multiplexing:** Joint TDM and FDM system calculations involving Nyquist sampling, bit conversion, and carrier modulation.
- **Diversity Techniques:** What are diversity techniques in communication systems? Describe different types.
- **Short Notes:** Provide short notes on JPGG (JPEG) and MPGA (MPEG).